Analysis

Spring 2026

1 Product Measures and Fubini's Theorem

1.1 Product Measure

Definition 1. *If X is a set of points and $\mathcal{M}$ is a σ-algebra of sets in X, we say that $\mu : \mathcal{M} \to [0, \infty]$ is a* ***measure*** *when*

1. $\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j)$ *when $E_j \in \mathcal{M}$ and are pairwise disjoint*
2. $\mu(\varnothing) = 0$

We construct measures by starting with an **outer measure** $\mu^* : \mathcal{P}(X) \to [0, \infty]$, which has

1. $\mu^*(\varnothing) = 0$
2. $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$
3. $\mu^*\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \mu^*(E_j)$ for all sets $\{E_j\}_{j=1}^{\infty}$ (not necessarily disjoint)

Outer measures generate measures via the Caratheodory construction by restricting domain. We say a set is E **Caratheodory measurable** when for any set $K \subseteq X$, $\mu^*(K) = \mu^*(K \cap E) + \mu^*(K \cap E^c)$.

Given some measure spaces $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ we wish to build a measure on $X \times Y$ which can be reasonably understood as the product of the two given measure spaces.

For our space, we take $X \times Y$ = Cartesian product of X and Y.

For our σ-algebra, we cannot take $\mathcal{M} \times \mathcal{N}$ because this is not a σ-algebra. Instead, we take $\mathcal{M} \otimes \mathcal{N} :=$ smallest σ-algebra containing $\mathcal{M} \times \mathcal{N}$.

For our measure, we define $\pi : \mathcal{M} \otimes \mathcal{N} \to [0, \infty]$ in the following way:

If $Z \in \mathcal{M} \otimes \mathcal{N}$ has the form $E \times F$ for $E \in \mathcal{M}$ and $F \in \mathcal{N}$, we would like to have $\pi(Z) = \mu(E)\nu(F)$.

We will assume both of our measures μ, ν are σ-finite. We can define an outer measure π^* on $\mathcal{P}(X \times Y)$ in the following way: Given $Z \subset X \times Y$,

$$\pi^*(Z) = \inf \left\{ \sum_{i=1}^{\infty} \mu(E_i)\nu(F_i) \mid \{E_i\}_{i=1}^{\infty} \subseteq X, \{F_i\}_{i=1}^{\infty} \subseteq Y, \bigcup_{i=1}^{\infty} E_i \times F_i \supseteq Z \right\}$$

Claim: If you take the collection of sets that are product sets and then form the algebra (not σ-algebra), any set in the algebra can be written as a disjoint union of product sets.
Proof:
Want to show that the set of disjoint finite unions of product sets form an algebra. That is, show that given two sets Z_1, Z_2 which are both finite disjoint product sets, show that $Z_1 \cup Z_2, Z_1 \cap Z_2$, and Z_1^c all also have the same form. First observe:

$$(E_1 \times F_1) \cap (E_2 \times F_2) = (E_1 \cap E_2) \times (F_1 \cap F_2) \text{ and}$$

$$(E_1 \times F_1)^c = (E_1^c \times F_1) \cup (E_1 \times F_1^c) \cup (E_1^c \times F_1^c)$$

Thus, we have:

1. Let $Z_1 = \bigcup_{i=1}^{n} R_i$ and $Z_2 = \bigcup_{j=1}^{m} S_j$ with $\{R_i\}$ and $\{S_j\}$ both collections of pairwise disjoint product sets. Then:

$$Z_1 \cap Z_2 = \left(\bigcup_{i=1}^{n} R_i \right) \cap \left(\bigcup_{j=1}^{m} S_j \right) = \bigcup_{i,j} (R_i \cap S_j)$$

   Since $R_i \cap S_j = (E_i \cap E_j) \times (F_i \cap F_j)$ and the pairs (i, j) yield mutually disjoint sets, $Z_1 \cap Z_2$ is a disjoint union of product sets.

2. If $Z = \bigcup_{i=1}^{n} R_i$, then $Z^c = \bigcap_{i=1}^{n} R_i^c$. Since we observed that each R_i^c is a disjoint union of three product sets and we already showed that the set of disjoint unions of product sets is closed under finite intersections, it follows that Z^c is a disjoint union of product sets.

3. Since the set of disjoint unions of product sets is closed under complements and intersections, it is closed under unions by De Morgan's Law: $Z_1 \cup Z_2 = (Z_1^c \cap Z_2^c)^c$.

Therefore, the set of disjoint unions of product sets is an algebra.
Conclusion: for finite unions of product sets, we may always manipulate or rewrite pieces so that they are pairwise disjoint.

Now suppose $Z \subseteq X \times Y$ and we have a covering $Z \subseteq \bigcup_{i=1}^{\infty} E_i \times F_i$.

Let $W_N = \bigcup_{i=1}^N E_i \times F_i$. By above, there is a way of writing W_N as a finite union of disjoint product sets.

$W_N \setminus W_{N-1}$ is in the algebra (of disjoint unions of product sets) so it is also equal to a disjoint union of product sets.

$$\bigcup_{i=1}^{\infty} E_i \times F_i = W_1 \cup \bigcup_{i=2}^{\infty} W_i \setminus W_{i-1} = \text{ countable union of disjoint product sets}$$

So we may also rewrite countable unions of product sets as disjoint countable unions of product sets. Changing the cover in this way is fine because it does not change the sum of the products of the measures.

Related/ more precise statement: $\pi^*(E \times F) = \mu(E)\nu(F)$:
Proof:
$\pi^*(E \times F) \leq \mu(E)\nu(F)$ by covering with itself. For the reverse inequality,

$$\chi_{E\times F}(x,y) \leq \sum_{i=1}^{\infty} \chi_{E_i \times F_i}(x,y) \text{ when } \{E_i \times F_i\}_{i=1}^{\infty} \text{ covers } E \times F.$$

Integrate with respect to y (measure ν):

$$\int \chi_{E\times F}(x,y) d\nu(y) \leq \int \sum_{i=1}^{\infty} \chi_{E_i \times F_i}(x,y) d\nu(y) \implies$$

$$\chi_E(x)\nu(F) \leq \sum_{i=1}^{\infty} \chi_{E_i}(x)\nu(F_i)$$

Now integrate in x (measure μ):

$$\mu(E)\nu(F) \leq \sum_{i=1}^{\infty} \mu(E_i)\nu(F_i)$$

The inf of the stuff on the right-hand-side is how we get outer measure. So $\mu(E)\nu(F) \leq \pi^*(E \times F)$.

Corollary 1. *All product sets are measurable in the Caratheodory sense.*

Proof. We need to show $\pi^*(Z) = \pi^*(Z \cap (E \times F)) + \pi^*(Z \cap (E \times F)^c)$.

The $\leq$ direction is immediate from properties of outer measure, so it suffices to show $\geq$.

Let $\{E_i \times F_i\}_{i=1}^{\infty}$ be a covering of Z. We can rewrite this covering as disjoint product sets. Then $E_i \times F_i =$
(collection of pieces intersecting $E\times F$)$\cup$(collection of pieces intersecting $(E\times F)^c$)

$$= ((E_i \times F_i) \cap (E \times F)) \cup ((E_i \times F_i) \cap (E \times F)^c)$$

Taking the first term over all i forms a cover of $Z \cap (E \times F)$. Taking the second term over all i forms a cover of $Z \cap (E \times F)^c$.

Key fact: when we do this splitting, the sum of the products of measures does not change. So we get covers of $Z \cap (E \times F)$ and $Z \cap (E \times F)^c$ such that the sum of the sizes of the first cover + the sum of the sizes of the second cover $\leq \sum_{i=1}^{\infty} \mu(E_i)\nu(F_i)$. $\square$

We know: the outer measure π^* has $E \times F$ Caratheodory measurable for every $E \in \mathcal{M}$ and $F \in \mathcal{N}$, and $\pi^*(E \times F) = \mu(E)\nu(F)$. That is,

$$\int \int \chi_{E \times F}(x, y) d\nu(y) d\mu(x) = \int \chi_{E \times F}(x, y) d\pi(x, y) = \int \int \chi_{E \times F}(x, y) d\mu(x) d\nu(y).$$

Claim: If μ, ν are both σ-finite, the property $\pi(E \times F) = \mu(E)\nu(F)$ uniquely specifies π on $\mathcal{M} \otimes \mathcal{N}$.
Proof:
First, observe that given any set $Z \subseteq X \times Y$ and any $\epsilon > 0$, there exists a sequence of disjoint measurable rectangles $\{E_i \times F_i\}_{i=1}^{\infty}$ (where $E_i \in \mathcal{M}$ and $F_i \in \mathcal{N}$) such that

$$\pi^*(Z) \leq \sum_{i=1}^{\infty} \pi(E_i \times F_i) < \pi^*(Z) + \epsilon.$$

Suppose $Z \in \mathcal{M} \otimes \mathcal{N}$. We will show that if π' is a measure on $\mathcal{M} \otimes \mathcal{N}$ which factors on product sets, then $\pi' = \pi$ on all of $\mathcal{M} \otimes \mathcal{N}$:

First observe that $\pi'(Z) \leq \pi^*(Z)$, since we can take Z, cover by $\{E_i \times F_i\}$, and then:

$$\pi'(Z) \leq \sum_{i=1}^{\infty} \pi'(E_i \times F_i),$$

and since the inf of the right-hand-side is the outer measure, $\pi'(Z) \leq \pi^*(Z)$.

For the reverse direction, if $Z \in \mathcal{M} \otimes \mathcal{N}$, then take Z' a countable union of disjoint product sets such that $Z \subseteq Z'$ and $\pi(Z' \setminus Z) < \varepsilon$. Then

$$\pi^*(Z) \leq \pi^*(Z') + \varepsilon = \pi'(Z') + \varepsilon = \pi'(Z) + \pi'(Z' \setminus Z) + \varepsilon \leq \pi'(Z) + \pi^*(Z' \setminus Z) + \varepsilon \leq \pi'(Z) + 2\varepsilon$$

and since ε is arbitrary, $\pi^*(Z) = \pi'(Z)$.

1.2 Fubini's Theorem

At this point, we know that for $Z = E \times F \in \mathcal{M} \times \mathcal{N}$:

- $\chi_Z(x, y)$ is a $(\mathcal{M} \otimes \mathcal{N})$-measurable function

- For each x, $y \mapsto \chi_Z(x,y)$ is $\mathcal{N}$-measurable
- $x \mapsto \int \chi_Z(x,y)d\nu(y)$ is $\mathcal{M}$-measurable
- $(*)\quad \int \left[\int \chi_Z(x,y)d\nu(y)\right] d\mu(x) = \int \chi_Z(x,y)d\pi(x,y)$, since $\chi_Z(x,y) = \chi_E(x)\chi_F(y)$.

At this point, we would like to show that (*) holds for all $Z \in \mathcal{M} \otimes \mathcal{N}$. In order to do this, we will study the class of sets Z satisfying (*) and argue that containing product sets forces it to contain all of $\mathcal{M} \otimes \mathcal{N}$.

Definition 2. *A* ***monotone class*** $\mathcal{C}$ *is a collection of sets that is closed under countable increasing unions and countable decreasing intersections.*

We will show that (1) sets satisfying (*) are a monotone class and (2) the initial class of sets we have is rich enough so that monotone class $\implies$ σ-algebra, in this particular case.

Let $\mathcal{Z}$ be the collection of sets satisfying (*) and associated measurability requirements. We'll show $\mathcal{Z}$ is a monotone class containing an algebra $\mathcal{A}$ that contains product sets. Notice:

1. $E \times F \in \mathcal{Z}$ when $E \in \mathcal{M}$, $F \in \mathcal{N}$ by the definition of the product measure.
2. If Z is a countable disjoint union of product sets, then $Z \in \mathcal{Z}$, because $\chi_Z(x,y) = \sum_{j=1}^{\infty} \chi_{E_j \times F_j}(x,y)$ and we can apply the Monotone Convergence Theorem to both sides of $(*)$.
3. If $Z_1 \subseteq Z_2 \subseteq \cdots \in \mathcal{Z}$ then $\lim_{j\to\infty} \chi_{Z_j}(x,y) = \chi_{\bigcup_{j=1}^{\infty} Z_j}(x,y)$ with pointwise monotone convergence. By applying MCT to both the inner and outer integrals, we see $\bigcup Z_j \in \mathcal{Z}$.
4. If X, Y are both finite measure and $Z_1 \supseteq Z_2 \supseteq Z_3 \supseteq \cdots$ all in $X \times Y$ are in $\mathcal{Z}$, then since $\chi_{Z_j}(x,y) \leq 1$ and $1 \in L^1(\mu \times \nu)$ for finite spaces, we apply dominated convergence:

   $$\lim_{j\to\infty} \int \left[\int \chi_{Z_j}(x,y)d\nu(y)\right] d\mu(x) = \int \left[\lim_{j\to\infty} \int \chi_{Z_j}(x,y)d\nu(y)\right] d\mu(x)$$

   provided the limit exists for all x. Fix x; since $\chi_{Z_j}(x,y)$ is dominated by the integrable function $\chi_Y(y) = 1$, by dominated convergence:

   $$\lim_{j\to\infty} \int \chi_{Z_j}(x,y)d\nu(y) = \int \chi_{\bigcap_{j=1}^{\infty} Z_j}(x,y)d\nu(y)$$

   because $\lim_{j\to\infty} \chi_{Z_j}(x,y) = \chi_{\bigcap_{j=1}^{\infty} Z_j}(x,y)$. So if X and Y are finite measure, $\mathcal{Z}$ is closed under countable decreasing intersections and is thus a monotone class.

   For the σ-finite case, let X_k, Y_k be nested, increasing sets of finite measure such that $\bigcup_k X_k = X$ and $\bigcup_k Y_k = Y$.

By the Monotone Class Theorem on the finite rectangles $X_k \times Y_k$, we know $(*)$ holds for all $Z \in \mathcal{M} \otimes \mathcal{N}$ with μ restricted to X_k and ν restricted to Y_k. Thus:

$$\chi_{Z_j}(x,y)\chi_{X_k \times Y_k}(x,y) = \chi_{Z_j \cap (X_k \times Y_k)}.$$

As $j \to \infty$, for fixed k:

$$\pi(Z_j \cap (X_k \times Y_k)) \to \pi\left(\left(\bigcap_{j=1}^{\infty} Z_j\right) \cap (X_k \times Y_k)\right)$$

hence

$$\lim_{j\to\infty} \int_{X_k} \left[\int_{Y_k} \chi_{Z_j}(x,y)d\nu(y)\right] d\mu(x) = \int_{X_k} \left[\int_{Y_k} \chi_{(\bigcap_{j=1}^{\infty} Z_j)}(x,y)d\nu(y)\right] d\mu(x)$$

Now take the limit as $k \to \infty$. By the Monotone Convergence Theorem (since the sets $X_k \times Y_k$ are increasing):

$$\lim_{k\to\infty} \int_{X} \left[\int_{Y} \chi_{Z\cap(X_k \times Y_k)}(x,y)d\nu(y)\right] d\mu(x) = \int_{X} \left[\int_{Y} \chi_{Z}(x,y)d\nu(y)\right] d\mu(x)$$

This shows the property holds for the full σ-finite space.

Theorem 1 (Monotone Class Theorem)**.** *If $\mathcal{A}$ is an algebra, then $\mathcal{C}(\mathcal{A})$ coincides with the σ-algebra generated by $\mathcal{A}$.*

Proof. Every σ-algebra is a monotone class. We know the σ-algebra generated by $\mathcal{A}$ contains $\mathcal{C}(\mathcal{A})$. So it suffices to show $\mathcal{C}(\mathcal{A})$ contains the σ-algebra generated by $\mathcal{A}$.

Let $\mathcal{C}_E := \{F \in \mathcal{C}(\mathcal{A}) \mid F \setminus E, E \setminus F, E \cap F \in \mathcal{C}(\mathcal{A})\}$.

We want to show that $\mathcal{C}_E$ is "large."

1. If $E \in \mathcal{A}$, then $A \in \mathcal{C}_E$ (because $\mathcal{A}$ is an algebra and $\mathcal{A} \subseteq \mathcal{C}(\mathcal{A})$)
2. For every E, $\mathcal{C}_E$ is itself a monotone class.
3. If $E \in \mathcal{A}$, then $\mathcal{C}_E$ is a monotone class containing $\mathcal{A} \implies \mathcal{C}_E = \mathcal{C}(\mathcal{A})$.
4. By symmetry, if $F \subseteq \mathcal{C}(\mathcal{A})$, then $E \in \mathcal{C}_F$.
5. This means $\mathcal{A} \subseteq \mathcal{C}_F \implies \mathcal{C}_F = \mathcal{C}(\mathcal{A})$.
6. As a consequence, $\mathcal{C}(\mathcal{A})$ is closed under differences and intersections.
7. $X \in \mathcal{A}$, so $X \setminus F \in \mathcal{C}(\mathcal{A})$ for all $F \in \mathcal{C}(\mathcal{A}) \implies \mathcal{C}(\mathcal{A})$ is closed under complements.

8. If we can show $\mathcal{C}(\mathcal{A})$ is closed under countable unions (not nested), we're done.

   Let $\{Z_j\}_{j=1}^{\infty} \subseteq \mathcal{C}(\mathcal{A})$ be arbitrary.

   Let
$$F_j = \bigcup_{k=1}^{j} Z_k$$
   The F_j's are increasing and have $\bigcup_{j=1}^{\infty} F_j = \bigcup_{j=1}^{\infty} Z_j$.

   Each F_j is constructed via operations that $\mathcal{C}(\mathcal{A})$ is known to be closed under. So $F_j \in \mathcal{C}(\mathcal{A})$.

□

Corollary 2. *If μ, ν are σ-finite and $Z \in \mathcal{M} \otimes \mathcal{N}$,*

1. *For each x, $y \mapsto \chi_Z(x,y)$ is ν-measurable*
2. *$x \mapsto \int \chi_Z(x,y) d\nu(y)$ is μ-measurable*
3. *$\int \left[\int \chi_Z(x,y) d\nu(y)\right] d\mu(x) = \int \chi_Z(x,y) d\pi(x,y)$*
4. *Same for reverse ordering.*

Once you know this fact,

Theorem 2. *If μ, ν are σ-finite, f is π-measurable on $X \times Y$ and f is nonnegative or $f \in L^1(\pi)$, then*

1. *For all x, $f(x,y)$ is ν-measurable*
2. *$x \mapsto \int_Y f(x,y) d\nu(y)$ is defined and μ-measurable*
3. *$\int_X \left[\int_Y f(x,y) d\nu(y)\right] d\mu(x) = \int_{X \times Y} f(x,y) d\pi(x,y)$*

In practice, the measurability requirement can be a bit tedious because Lebesgue measure is in general not the same as product measure.

Theorem 3. *If $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ are σ-finite complete measure spaces, then every $Z \subseteq X \times Y$ which is Caratheodory measurable has the property that*

1. *$y \mapsto \chi_Z(x,y)$ is ν-measurable for μ a.e. x*
2. *$x \mapsto \int \chi_Z(x,y) d\nu(y)$ is μ-measurable*
3. *$\int \left[\int \chi_Z(x,y) d\nu(y)\right] d\mu(x) = \int \chi_Z(x,y) d\pi(x,y)$*
4. *And same for reverse ordering.*

Proof. Z is Caratheodory measurable if and only if $Z_1 \subseteq Z \subseteq Z_2$ for some $Z_1, Z_2 \in \mathcal{M} \otimes \mathcal{N}$ with $\pi^*(Z_2 \setminus Z_1) = 0$.

Apply the first version of Fubini to sets Z_1 and Z_2.

1. For all x, $y \mapsto \chi_{Z_1}(x, y)$ and $y \mapsto \chi_{Z_2}(x, y)$ are ν-measurable
2. Both can be integrated in x
3. $\int \left[\int \chi_{Z_1}(x, y) d\nu(y)\right] d\mu(x) = \int \chi_{Z_1}(x, y) d\pi(x, y)$ and
   $\int \left[\int \chi_{Z_2}(x, y) d\nu(y)\right] d\mu(x) = \int \chi_{Z_2}(x, y) d\pi(x, y)$

Now consider $Z_2 \setminus Z_1$:
By (3),

$$\int \left[\int \chi_{Z_2 \setminus Z_1}(x, y) d\nu(y)\right] d\mu(x) = \int \chi_{Z_2 \setminus Z_1}(x, y) d\pi(x, y) = 0 \implies$$

$\int \chi_{Z_2 \setminus Z_1}(x, y) d\nu(y) = 0$ for μ a.e. x
$\implies$ for μ a.e. $x, \chi_{Z_2 \setminus Z_1}(x, y) = 0$ for ν a.e. y.

But $Z = Z_1 \cup Z^*$ where $Z^* \subseteq Z \setminus Z_1$. So:

1. For μ a.e. x, $\chi_{Z^*}(x, y) = 0$ for ν a.e.y
2. For μ a.e. x, $y \mapsto \chi_{Z^*}(x, y)$ is ν-measurable
3. $$\int \chi_{Z_1}(x, y) d\nu(y) \leq \int \chi_{Z^*}(x, y) d\nu(y) \leq \int \chi_{Z_2}(x, y) d\nu(y)$$

□

1.3 Lebesgue measure on Euclidean space

Definition 3. *Lebesgue measure on $\mathbb{R}^n$ is the n-fold product measure of 1-dimensional Lebesgue measure, on the σ-algebra that is the completion of the product σ-algebra of the Borel σ-algebra in 1 dimension.*

Alternatively, Lebesgue outer measure m^* is given by

$$m^*(E) = \inf \left\{ \sum_{j=1}^{\infty} |R_j| \mid E \subseteq \bigcup_{j=1}^{\infty} R_j \right\}$$

where R_j = rectangle = product of closed bounded intervals, and
the Lebesgue σ-algebra = the Caratheodory σ-algebra for this outer measure.

Let $\pi^*(E)$ = product outer measure derived by coverings using product sets.

Claim: $\pi^* = m^*$.
Proof:
$m^*(E) \geq \pi^*(E)$ because coverings by rectangles are product coverings, but not vice versa.

For $m^*(E) \leq \pi^*(E)$, take an arbitrary set E and cover it by products of 1-dimensional measurable sets:

$$E \subseteq \bigcup_{i=1}^{\infty} A_{1_i} \times \cdots \times A_{n_i}$$

where each A_{j_i} = 1-dimensional Lebesgue measurable set. Then

$$\pi^*(E) = \inf \left\{ \sum |A_{1_i}| \cdots |A_{n_i}| \mid E \subseteq \bigcup_{i=1}^{\infty} A_{1_i} \times \cdots \times A_{n_i} \right\}$$

Since

$$|A_{j_i}| = \inf \left\{ \sum_{k=1}^{\infty} |I_k| \mid A_{j_i} \subseteq \bigcup_{k=1}^{\infty} I_k \right\}$$

and

$$A_{j_i} \subseteq \bigcup_{n=1}^{\infty} I_{j_i}^{(k)}, \text{ where } I_{j_i}^{(k)} \text{ are closed bounded intervals (or empty),}$$

write

$$R^{(i)}_{k_1,\dots,k_n} = I^{(k_1)}_{1_i} \times \cdots \times I^{(k_n)}_{n_i} \implies |R^{(i)}_{k_1,\dots,k_n}| = |I^{(k_1)}_{1_i}| \cdots |I^{(k_n)}_{n_i}|.$$

Then

$$E \subseteq \bigcup_{i=1}^{\infty} A_{1_i} \times \cdots \times A_{n_i} \subseteq \bigcup_{i=1}^{\infty} \bigcup_{k_1,\dots,k_n} R^{(i)}_{k_1,\dots,k_n}$$

and

$$\sum_{i,k_1,\dots,k_n} |R^{(i)}_{k_1,\dots,k_n}| = \sum_i \prod_j \sum_k |I^{(k_j)}_{j_i}|.$$

Now choose each covering of A_{j_i} so that $\sum_k |I^{(k_j)}_{j_i}| = |A_{j_i}| + \varepsilon_{j_i}$ (where ε_{j_i} is to be determined in a moment). Then

$$\sum_{i,k_1,\dots,k_n} |R^{(i)}_{k_1,\dots,k_n}| = \sum_i \prod_j (|A_{j_i}| + \varepsilon_{j_i})$$

Since the ε_{j_i} are arbitrary, make then small enough that the product makes sense and then let them tend to zero. Thus

$$\inf \sum_{i,k_1,\dots,k_n} |R^{(i)}_{k_1,\dots,k_n}| \leq \inf \sum_i \prod_j |A_{j_i}|$$

Since inf of the right-hand-side is $\pi^*(E)$, $m^*(E) \leq \pi^*(E)$.

Some observations:

1. Products of measurable things are measurable

2. All open sets are measurable, because all open sets can be written as the union of all boxes contained in the set with rational endpoints in each direction (= countable union of product rectangles)

   Dyadic cubes: $[2^{-j}n_1, 2^{-j}(n_1+1)] \times \cdots \times [2^{-j}n_d, 2^{-j}(n_d+1)] \subseteq \mathbb{R}^d$

   For any two dyadic cubes, they are either nonoverlapping (i.e. disjoint interiors) or one is contained in the other.

   We can show that open sets can be approximated by dyadic cubes: take the set of all dyadic cubes contained in your set and then keep only the maximal ones.

3. **Definition:**
   Outer regular: for all $\varepsilon > 0$, for all measurable sets E, there exists an open set U so that $E \subseteq U$ and $|U \setminus E| < \varepsilon$.
   Inner regular: for all measurable sets E, there exists a sequence of compact sets $\{K_j\}_{j=1}^\infty$ contained in E so that $\lim_{j\to\infty} |K_j| = |E|$.

   Lebesgue measure on $\mathbb{R}^n$ is inner and outer regular:
   Outer regularity:
   Take your set E, decompose it into sets $E_j = E_{(j_1,\ldots,j_n)}$ where $E_j = E \cap [j_1, j_1+1] \times \cdots \times [j_n, j_n+1]$.
   Cover E_j by rectangles R_i so that $\sum_{i=1}^\infty |R_i| \le |E_j| + \varepsilon_j$.
   Let $U_j = \bigcup_{i=1}^\infty R_i^*$, where R_i^* is a bigger open rectangle containing R_i.
   Then $\sum_{i=1}^\infty |R_i^*| \le |E_j| + 2\varepsilon_j \implies |U_j| \le |E_j| + 2\varepsilon_j$ and $E_j \subset U_j$ so $|U_j \setminus E_j| \le 2\varepsilon_j$.
   Now take $U = \bigcup_j U_j \implies E \subseteq U$ and $|U \setminus E| \le \sum_{j=1}^\infty 2\varepsilon_j$. Choose $2\varepsilon_j$ so that $\sum_{j=1}^\infty 2\varepsilon_j \le \varepsilon$.

   Inner regularity:
   Outer regularity of $E^c \implies$
   $\exists$ closed set C so that $C \subseteq E$ and $|E \setminus C| < \varepsilon$.
   Restrict C to $[-N, N]^n$.
   If we take N large enough, we can make $|C \cap [-N, N]^n|$ close to $|C|$
   (ε close to $|C|$ if $|C| < \infty$ and arbitrarily large if $|C| = \infty$).

Corollary 3. *A set $E \subseteq \mathbb{R}^n$ is Lebesgue measurable if and only if there exists $A_1 \subseteq E \subseteq A_2$ so that A_1 is F_σ, A_2 is G_δ, and $|A_2 \setminus A_1| = 0$.*

Proof. ($\implies$): Take a sequence of ε's going to zero. Use outer and inner regularity to find $C_\varepsilon \subseteq E \subseteq O_\varepsilon$ so that C_ε is closed, O_ε is open, and $|O_\varepsilon \setminus E| < \varepsilon$, $|E \setminus C_\varepsilon| < \varepsilon$. Take $A_1 = \bigcup_\varepsilon C_\varepsilon$ and $A_2 = \bigcap_\varepsilon O_\varepsilon$.

($\impliedby$): We know A_1 and A_2 have to be Lebesgue measurable because open sets are Lebesgue measurable. So $E = A_1 \cup E^*$ where E^* is a subset of a Lebesgue null set $A_2 \setminus A_1$. Because the Lebesgue measure is complete, E^* is measurable $\implies A_1 \cup E^*$ is measurable. $\square$

Proposition 1. *$E \subseteq \mathbb{R}^n$ is Lebesgue measurable if and only if for all $\varepsilon > 0, \exists$ a countable collection of disjoint closed boxes $\{R_i\}_i$ so that $m^*(E \triangle \bigcup_i R_i) < \varepsilon$. If $m^*(E) < \infty$ then the same statement holds with "countable" replaced by "finite."*

Proof. ($\implies$):
Pick $\varepsilon > 0$, use outer regularity to get an open set U so that $|U \setminus E| < \frac{\varepsilon}{2}$.
This U = a nonoverlapping union of closed dyadic cubes.
If $U = \bigcup_k Q_k$, replace Q_k by Q_k^* so that Q_k^* is a smaller box with the same center and $|Q_k \setminus Q_k^*| < \varepsilon_k$ (choose ε_k so that $\sum_k \varepsilon_k < \frac{\varepsilon}{2}$).

($\impliedby$):
Suppose $R = \bigcup_{i=1}^{\infty} R_i$ = union of disjoint closed boxes so that $m^*(E \triangle R) < \varepsilon$. (for $m^*(E) < \infty$ take $R = \bigcup_{i=1}^{N} R_i$ where $\left|\bigcup_{i=N+1}^{\infty} R_i\right| < \frac{\varepsilon}{3}$)

Now take an arbitrary set A.

$$m^*(A \cap E) \leq m^*(A \cap R) + m^*(E \setminus R) \leq m^*(A \cap R) + \varepsilon$$

and

$$m^*(A \cap E^c) \leq m^*(A \cap R^c) + m^*(R \setminus E) \leq m^*(A \cap R^c) + \varepsilon$$

so

$$m^*(A \cap E) + m^*(A \cap E^c) \leq m^*(A \cap R) + m^*(A \cap R^c) + 2\varepsilon = m^*(A) + 2\varepsilon.$$

ε is arbitrary so $m^*(A \cap E) + m^*(A \cap E^c) \leq m^*(A)$. $\square$

As a consequence, we can use this result to find dense classes of functions in $L^1(\mu)$. For example, finite linear combinations of boxes are dense in L^1 (and by smoothly approximating, we get that C^∞ functions of compact support are dense too).

2 Signed Measures

Recall: A **measure** μ on $(X, \mathcal{M})$ is a map $\mu : \mathcal{M} \to [0, \infty]$ such that:

1. $\mu(\emptyset) = 0$
2. For measurable and pairwise disjoint $\{E_j\}$, $\mu(\bigcup E_j) = \sum \mu(E_j)$

Definition 4. *A **signed measure** μ on $(X, \mathcal{M})$ is a map $\mu : \mathcal{M} \to [-\infty, \infty]$ such that:*

1. $\mu(\emptyset) = 0$

2. *μ assumes at most one of the values $+\infty$ or $-\infty$*

3. *For pairwise disjoint measurable $\{E_j\}$, $\mu(\bigcup E_j) = \sum \mu(E_j)$ with the sum always being well-defined and being absolutely convergent when $\mu(\bigcup E_j)$ is finite.*

Example 1. *Take a positive measure μ on $\mathcal{M}$. For $f \in L^1(\mu)$ and g nonnegative and measurable, define $\nu(E) = \int_E f d\mu + \int_E g d\mu$. Then ν is a signed measure.*

Proposition 2. *If $\{E_j\}_{j=1}^{\infty}$ is an increasing sequence of measurable sets, then*

$$\mu(\bigcup_{j=1}^{\infty} E_j) = \lim_{j\to\infty} \mu(E_j)$$

and if $\{F_j\}_{j=1}^{\infty}$ is an decreasing sequence of measurable sets and at least one of F_j has finite measure, then

$$\mu(\bigcap_{j=1}^{\infty} F_j) = \lim_{j\to\infty} \mu(F_j)$$

Proof. Let $E_0 := \varnothing$ and for $j \geq 1$ let $B_j := E_j \setminus E_{j-1}$.
Then$\{B_j\}_{j=1}^{\infty}$ are pairwise disjoint because the E_j's are nested and increasing. Also $\bigcup_{j=1}^{\infty} B_j = \bigcup_{j=1}^{\infty} E_j$ and $\bigcup_{j=1}^{N} B_j = E_N$.
By definition:

$$\mu(\bigcup_{j=1}^{\infty} B_j) = \sum_{j=1}^{\infty} \mu(B_j) = \lim_{N\to\infty} \sum_{j=1}^{N} \mu(B_j) = \lim_{N\to\infty} \mu(E_N).$$

If $\{F_j\}_{j=1}^{\infty}$ is an decreasing sequence of measurable sets, then let $A_j := F_j \setminus F_{j+1}$. Then $\{A_j\}_{j=1}^{\infty}$ are pairwise disjoint,

$$F_N = \left(\bigcup_{j=N}^{\infty} A_j\right) \cup \left(\bigcap_{k=1}^{\infty} F_k\right), \quad \text{and}$$

$$\mu(F_N) = \mu(\bigcap_{k=1}^{\infty} F_k) + \sum_{j=N}^{\infty} \mu(A_j)$$

Suppose $\mu(F_N)$ is finite for some N. This implies that $\sum_{j=N}^{\infty} \mu(A_j)$ must be finite, as it is a tail of a convergent series. So $\lim_{N\to\infty} \sum_{j=N}^{\infty} \mu(A_j) = 0$. Then

$$\lim_{N\to\infty} \mu(F_N) = \lim_{N\to\infty} \left[\mu(\bigcap_{k=1}^{\infty} F_k) + \sum_{j=N}^{\infty} \mu(A_j)\right] = \mu(\bigcap_{k=1}^{\infty} F_k).$$

$\square$

2.1 Hahn Decomposition and Jordan Decomposition

Definition 5. *Given a signed measure space $(X, \mathcal{M}, \mu)$:*

1. *$E \in \mathcal{M}$ is **positive** when $\mu(E') \geq 0$ for all $E' \subseteq E$ measurable*
2. *$E \in \mathcal{M}$ is **negative** when $\mu(E') \leq 0$ for all $E' \subseteq E$ measurable*
3. *$E \in \mathcal{M}$ is **null** when $\mu(E') = 0$ for all $E' \subseteq E$ measurable*

Proposition 3. *Countable unions of positive sets are positive (same for negative, null).*

Proof. Let $\{P_j\}_{j=1}^{\infty}$ be a countable collection of positive sets. Let $P = \bigcup_{j=1}^{\infty} P_j$. Let $P' \subseteq P$ be any measurable set. Need to show $\mu(P') \geq 0$. Let

$$B_j := P_j \setminus \left(\bigcup_{k=1}^{j-1} P_k \right)$$

These are pairwise disjoint and $B_j \subseteq P_j$ so B_j is positive.

We can write $P' = \bigcup_{j=1}^{\infty} (P' \cap B_j)$. We note that $\{P' \cap B_j\}$ are pairwise disjoint. Also $P' \cap B_j \subseteq P_j$ so $\mu(P' \cap B_j) \geq 0$. Then $\mu(P') = \sum_{j=1}^{\infty} \mu(P' \cap B_j)$ and all these terms in the sum are nonnegative $\implies \mu(P') \geq 0$. $\square$

Theorem 4 (Hahn Decomposition Theorem)**.** *Given a signed measure $(X, \mathcal{M}, \mu)$, there exist measurable sets P, N such that*

$$P \cup N = X \quad \textit{and} \quad P \cap N = \varnothing.$$

and P is positive for μ and N is negative for μ. And if P', N' is any other such decomposition, $P \triangle P'$, $N \triangle N'$ are null sets.

Proof. Assume without loss of generality that μ is never equal to $+\infty$.
Let $S = \{\text{positive sets } P' \subseteq X\}$.
$\varnothing \in S$ so S is nonempty.
Let P_j be a sequence of positive sets so that $\mu(P_j) \to \sup_{P' \in S} \mu(P')$ as $j \to \infty$.
Without loss of generality, we may assume the P_j are nested. Then

$$\mu(\bigcup_{j=1}^{\infty} P_j) = \lim_{j \to \infty} \mu(P_j) = \sup_{P' \in S} \mu(P').$$

So the sup is attained on $\bigcup_{j=1}^{\infty} P_j$ because it is a positive set too.

By assumption, $\mu(\bigcup_{j=1}^{\infty} P_j)$ is nonnegative and finite because it cannot be $+\infty$.

Let $P = \bigcup_{j=1}^{\infty} P_j$ and let $N = X \setminus P$.

If N is a negative set, we're done.

If N is not negative, there exists $E \subseteq N$ so that $\mu(E) > 0$.

The set E cannot be positive, since $E \cup P$ is positive and
$\mu(E \cup P) = \mu(E) + \mu(P) > \mu(P) = \sup_{P' \in S} \mu(P')$.
E cannot have any positive subsets either (other than null).

Now construct a sequence $\{E_j\}_{j=1}^{\infty}$ so that $E_1 = E$, $E_j \supseteq E_{j+1}$ for each j, and $\mu(E_{j+1})$ is as large as possible relative to E_j in the following way:

Given E_j, if there exists a subset $E' \subseteq E_j$ so that $\mu(E') \geq \mu(E_j) + 1$, let $E_{j+1} := E'$.
If not, choose E_{j+1} so that $\mu(E_{j+1}) \geq \frac{1}{2}\left(\mu(E_j) + \sup_{E' \subseteq E_j} \mu(E')\right)$.

Now the E_j's are nested, so the measures are increasing (and all of the measures are finite).Then $\mu(\bigcap_{j=1}^{\infty} E_j) = \lim_{j\to\infty} \mu(E_j)$.
The left hand side cannot be $+\infty$. And the right hand side is positive, so is a finite positive limit. So $\lim_{j\to\infty} \mu(E_j)$ is finite $\implies$

$$\mu(E_{j+1}) - \mu(E_j) \to 0 \text{ as } j \to \infty.$$

Now since we chose each E_j so that

$$\mu(E_{j+1}) \geq \frac{1}{2}\left(\mu(E_j) + \sup_{E' \subseteq E_j} \mu(E')\right),$$

we have

$$\sup_{E' \subseteq E_j} \mu(E') \leq \mu(E_{j+1}) + [\mu(E_{j+1}) - \mu(E_j)].$$

We claim that this combination of information forces $\bigcap_{j=1}^{\infty} E_j$ to be a positive set with strictly positive measure:

So suppose $\bigcap_{j=1}^{\infty} E_j$ is not positive. Then there exists $F \subseteq \bigcap_{j=1}^{\infty} E_j$ with strictly negative measure, i.e. $\mu(F) = -\varepsilon < 0$. Then

$$\mu(\bigcap_{j=1}^{\infty} E_j \setminus F) = \mu(\bigcap_{j=1}^{\infty} E_j) - \mu(F) = \mu(\bigcap_{j=1}^{\infty} E_j) + \varepsilon.$$

We know that $\mu(E_{j+1}) - \mu(E_j) \to 0$ as $j \to \infty$.
So in particular, $\mu(E_{j+1}) - \mu(E_j) < \frac{\varepsilon}{4}$ as $j \to \infty$.

$F \subseteq E_j$ for each j and $\mu(E_j \setminus F) = \mu(E_j) + \varepsilon$ for each j. So using

$$\sup_{E' \subseteq E_j} \mu(E') \geq \mu(E_j \setminus F) = \mu(E_j) + \varepsilon,$$

since we chose each E_j so that

$$\mu(E_{j+1}) \geq \frac{1}{2}\left(\mu(E_j) + \sup_{E' \subseteq E_j} \mu(E')\right),$$

we get

$$\mu(E_{j+1}) \geq \mu(E_j) + \frac{\varepsilon}{2}$$

for each j. But this contradicts e.g. $\mu(E_{j+1}) - \mu(E_j) < \frac{\varepsilon}{4}$ as $j \to \infty$.

So $\cap_{j=1}^{\infty} E_j$ is a positive set, meaning $\left(\cap_{j=1}^{\infty} E_j\right) \cup P$ is a positive set. Based on how we constructed the E_j's, $\mu\left(\cap_{j=1}^{\infty} E_j\right) > 0$, but then $\mu(\left(\cap_{j=1}^{\infty} E_j\right) \cup P) > \mu(P)$, contradicting the fact that P attains the sup. So N must have been negative.

For proving uniqueness, if $X = P \cup N = P' \cup N'$, then we can write $P \Delta P' = (P \setminus P') \cup (P' \setminus P)$.
$P \setminus P' \subseteq P \cap N'$ which is null, and $P' \setminus P \subseteq P' \cap N$, which is null.
Similarly $N \Delta N' = (N \setminus N') \cup (N' \setminus N)$.
$N \setminus N' \subseteq P' \cap N$ which is null, and $N' \setminus N \subseteq N' \cap P$, which is null.

$\square$

2.2 Lebesgue-Radon-Nikodym Theorem

Definition 6. *Given measures ν, μ (signed, positive, etc.) we say ν and μ are **mutually singular** when there exists a measurable set E so that E is a null set for ν and E^c is a null set for μ.*

Definition 7. *Given a signed measure ν and positive measure μ, we say ν is **absolutely continuous** with respect to μ when every set E so that $\mu(E) = 0$ has $\nu(E) = 0$ as well (Notation: $\nu \ll \mu$).*

Theorem 5 (Jordan Decomposition)**.** *Given a signed measure μ there exist unique positive measures μ^+ and μ^- so that $\mu = \mu^+ - \mu^-$ and $\mu^+ \perp \mu^-$.*

Proof. $\mu^+(E) = \mu(E \cap P)$ and $\mu^-(E) = -\mu(E \cap N)$. $\square$

Definition 8. *We say f is extended μ-integrable if when we let $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, then*

$$\int_X f^+ \, d\mu < \infty \quad \textit{or} \quad \int_X f^- \, d\mu < \infty$$

Example 2. *If μ is any positive measure and f is extended μ-integrable, then $E \mapsto \int_E f d\mu$ is a signed measure and $f d\mu \ll \mu$.*
Because: if E has μ-measure zero, then

$$\int_E f d\mu = \int_E (\textit{positive (or negative) function} + L^1 \textit{function}) d\mu$$

$$= \int \chi_E (\textit{positive (or negative) function} + L^1 \textit{ function}) d\mu = 0$$

Lemma 1. *If ν is a finite signed measure and μ is positive, then $\nu \ll \mu$ if and only if for all $\varepsilon > 0$, there exists $\delta > 0$ so that whenever $\mu(E) < \delta$, $|\nu(E)| < \varepsilon$.*

Proof. ($\Longleftarrow$): Suppose E is a measurable set so that $\mu(E) = 0$. For any $\varepsilon > 0$, let δ be given by the condition. $\mu(E) = 0 < \delta \implies |\nu(E)| < \varepsilon$. So if $\mu(E) = 0$ then $|\nu(E)| < \varepsilon$ for any $\varepsilon \implies \nu(E) = 0$.

($\implies$): Given that $\nu \ll \mu$, let us suppose the ε-δ condition fails. That is, suppose there exists $\varepsilon > 0$ so that for each $\delta > 0$ we can find E so that $\mu(E) < \delta$ but $|\nu(E)| \geq \varepsilon$. We want to show that ν cannot be absolutely continuous with respect to μ.

Recall $|\nu| := \nu^+ + \nu^-$. Since $\nu = \nu^+ - \nu^-$, if $|\nu(E)| \geq \varepsilon$, then $|\nu|(E) \geq \varepsilon$.

For each n, $\exists E_n$ measurable so that $\mu(E_n) < \frac{1}{n^2}$ but $|\nu|(E_n) \geq \varepsilon$.

Consider $E := \limsup_{n\to\infty} E_n = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n$.

$\mu(E) = 0$, because:

$$\mu(\bigcup_{n \geq N} E_n) \leq \sum_{n=N}^{\infty} \frac{1}{n^2} \leq \frac{1}{N-1} < \infty$$

so since $E \subseteq \bigcup_{n\geq N} E_n$, $\mu(E) \leq \frac{1}{N-1}$ for all N, hence $\mu(E) = 0$.

But $|\nu|(E) \geq \varepsilon > 0$, because:

Each E_n has $|\nu|(E_n) \geq \varepsilon$, so $|\nu|(\bigcup_{n\geq N} E_n) \geq \varepsilon$.

Since $|\nu|$ is a finite positive measure, $|\nu|(E) = \lim_{N\to\infty} |\nu|(\bigcup_{n\geq N} E_n)$ ($\{E_n\}$ is a decreasing family). Hence $|\nu|(E) \geq \varepsilon$.

So $\nu^+(E) + \nu^-(E) \geq \varepsilon \implies \nu^+(E)$ or $\nu^-(E) \geq \frac{\varepsilon}{2}$. This means E has a subset E' (either $E \cap P$ or $E \cap N$) so that $|\nu(E')| \geq \frac{\varepsilon}{2} > 0$ but $\mu(E') = 0$. So ν cannot be absolutely continuous with respect to μ. $\rightarrow\leftarrow$ $\square$

Corollary 4. *For any $f \in L^1(\mu)$, every $\varepsilon > 0$ admits $\delta > 0$ so that whenever $\mu(E) < \delta$, $|\int_E f d\mu| < \varepsilon$.*

Lemma 2 (LRN Key Lemma)**.** *Suppose ν, μ are finite positive measures on $(X, \mathcal{M})$. Then either $\nu \perp \mu$ or there exist $\varepsilon > 0$ and a measurable set E so that $\mu(E) > 0$ and $\nu \geq \epsilon\mu$ on E.*

Proof. Suppose no such ε, E exist.

Pick a sequence $\epsilon_n \searrow 0$. For each n, consider the signed measure $\phi_n = \nu - \epsilon_n \mu$. By Hahn Decomposition, there exists a positive set P_n and a negative set N_n for this signed measure ϕ_n. If P_n had $\mu(P_n) > 0$, we could take $E = P_n$ and would contradict our assumption, so we must have $\mu(P_n) = 0$.

Without loss of generality assume P_n are nested and increasing (Hahn Decomposition doesn't automatically give nested sets but we could define $P_n' = \bigcup_{k=1}^n P_k$, and since a finite union of μ-null sets is still a μ-null set, the logic holds.)

Let $X = P_n \cup N_n$ where $N_n = P_n^c$.

Then $\mu(\bigcup_n P_n) = \lim_{n\to\infty} \mu(P_n) = 0$ (μ is positive and the sets increase).

We claim: $\nu(\bigcap_n N_n) = 0$.

If E is any set contained in N_n, then $\nu - \varepsilon_n \mu$ is negative there
$\implies \nu(E) - \varepsilon_n \mu(E) \leq 0$.

So $\nu(\bigcap_n N_n) \leq \varepsilon_n \mu(\bigcap_n N_n)$.

We can let $\varepsilon_n \to 0$ (note $\mu(\bigcap_n N_n)$ is finite because μ is finite).

So $\nu(\bigcap_n N_n) = 0$, meaning $\mu \perp \nu$. $\square$

Theorem 6 (Lebesgue-Radon-Nikodym)**.** *If μ is a σ-finite (positive) measure and ν is a σ-finite signed measure, then*

$$\nu = f d\mu + d\lambda,$$

where λ is a σ-finite signed measure such that $\lambda \perp \mu$,
$f d\mu(E) := \int_E f d\mu$, and f is extended μ-integrable. And this decomposition is unique.

Proof. **Case 1:** Assume both ν, μ are finite positive.
Let $\mathcal{F} = \{f \geq 0 : f$ measurable and $\int_E f d\mu \leq \nu(E)$ for all E measurable$\}$.
Note that $\mathcal{F}$ is never empty because $0 \in \mathcal{F}$.

Let $S = \sup\{\int_X f d\mu : f \in \mathcal{F}\}$.
This sup must be finite because $\int_X f d\mu \leq \nu(X) < \infty$, because ν is a finite measure.

There exists a sequence $\{f_n\}_{n=1}^\infty \subseteq \mathcal{F}$ so that $\int_X f_n d\mu \to S$. We would like to show that the sup is attained, but as written, we do not know hard convergence properties of $\{f_n\}$.

We can fix this sequence to get pointwise monotone convergence.

Observe: if $f_1, f_2 \in \mathcal{F}$, then $\max(f_1, f_2) \in \mathcal{F}$.
Proof of observation: Let $E_1 = \{x : f_1 > f_2\}$ and $E_2 = E_1^c$. Then

$$\int_E \max(f_1, f_2) d\mu = \int_{E\cap E_1} f_1 d\mu + \int_{E\cap E_2} f_2 d\mu \leq \nu(E\cap E_1) + \nu(E\cap E_2) = \nu(E).$$

Since E was arbitrary, $\max(f_1, f_2) \in \mathcal{F}$.

Let $g_n = \max(f_1, \dots, f_n)$. Then $g_n \in \mathcal{F}$, $\{g_n\}$ is monotone pointwise non-decreasing, and $\int g_n d\mu \geq \int f_n d\mu$. Conclude:

$$\int \lim_{n\to\infty} g_n d\mu = S \text{ and } \lim_{n\to\infty} g_n \in \mathcal{F}.$$

Reason: Monotone Convergence $\implies$

$$\int_E \lim_{n\to\infty} g_n d\mu = \lim_{n\to\infty} \int_E g_n d\mu \leq \nu(E).$$

So there exists some $f^* \in \mathcal{F}$ so that

$$f^* = \lim_{n\to\infty} g_n$$

and

$$\int f^* d\mu = \sup\left\{\int f d\mu \mid f \in \mathcal{F}\right\}.$$

Notice: $d\nu - f^* d\mu$ is a finite positive measure, since $f^* \in \mathcal{F}$.

We claim that $d\nu - f^* d\mu$ is also mutually singular with respect to μ.

Plug in $d\lambda := d\nu - f^* d\mu$ and μ into our lemma: if $d\nu - f^* d\mu$ is not mutually singular with respect to μ, then $\exists \varepsilon > 0, E$ with $\mu(E) > 0$ so that $d\lambda \geq \varepsilon d\mu$ on E. Then $d\nu - f^* d\mu - \varepsilon \chi_E d\mu$ is still a positive measure, so $f^* + \varepsilon\chi_E \in \mathcal{F}$, a contradiction since then $\int (f^* + \varepsilon\chi_E) d\mu > \int f^* d\mu = S$. So $d\nu - f^* d\mu \perp \mu$.

Case 2: Suppose ν, μ are both σ-finite positive measures.

Then $\exists$ sequence of sets $Q_1 \subseteq Q_2 \subseteq \cdots$ such that $\bigcup_n Q_n = X$ and $\nu(Q_n) < \infty$ for all n.
There also exist sets $R_1 \subseteq R_2 \subseteq \cdots$ such that $\bigcup_n R_n = X$ and $\mu(R_n) < \infty$ for all n.

Then if $X_n = Q_n \cap R_n$, then $X_1 \subset X_2 \subset \cdots$, $\bigcup_n X_n = X$, and both of $\nu(X_n), \mu(X_n) < \infty$.

So let $H_n = X_n \setminus X_{n-1} (X_0 := \varnothing)$. Then $X = \bigsqcup H_n$ and $\nu(H_n), \mu(H_n) < \infty$.

On each H_n, $\exists$ a function f_n and a measure λ_n such that $\nu|_{H_n} = f_n d\mu|_{H_n} + d\lambda_n$ and $\lambda_n \perp \mu|_{H_n}$.

Let $f := \sum f_n \chi_{H_n}$ and $\lambda := \sum_{n=1}^\infty \lambda_n|_{H_n}$, i.e. $\lambda(E) = \sum_{n=1}^\infty \lambda_n(E \cap H_n)$.

Claim: This works. The fact that $d\nu = f d\mu + d\lambda$ is immediate from monotone convergence + equality on all sets H_n.

For showing $\lambda \perp \mu$:

For all $n, \exists E_n \subseteq H_n$ so that $\mu(E_n) = 0$ and $\lambda_n(H_n \setminus E_n) = 0$.
So let $E = \bigcup_n E_n$ (countable union).
Then $\mu(E) = \sum_n \mu(E_n) = 0$.
And $\lambda(E^c) = \sum_{n=1}^{\infty} \lambda(E^c \cap H_n) = \sum_{n=1}^{\infty} \lambda\left(\bigcap_m (E_m^c) \cap H_n\right) \leq \lambda(E_n^c \cap H_n) = 0$.

Case 3: If ν is σ-finite and signed and μ is σ-finite and positive:
Then

$$\nu = \nu^+ - \nu^-, \quad \nu^+ = f^+ d\mu + d\lambda^+, \quad \text{and } \nu^- = f^- d\mu + d\lambda^-$$

where λ^+ and λ^- are mutually singular with respect to μ.

Then $\nu = (f^+ - f^-)d\mu + d(\lambda^+ - \lambda^-)$. This is well-defined because at least one of ν^+, ν^- is a finite measure, and $(\lambda^+ - \lambda^-) \perp \mu$.

Proof of uniqueness:
Suppose $f d\mu + d\lambda = f' d\mu + d\lambda'$. We note that rearranging and subtracting is potentially problematic if the measures are not finite, so the cleanest strategy is to try something else.

Suppose E_1 = set of μ-measure zero such that $\lambda(E_1^c) = 0$.
Suppose E_2 = set of μ-measure zero such that $\lambda'(E_2^c) = 0$.
Let $E = E_1 \cup E_2$. Then $\mu(E) = 0$, $\lambda(E^c) = 0$, and $\lambda'(E^c) = 0$.

Restrict $f d\mu + d\lambda = f' d\mu + d\lambda'$ to E: we get $f d\mu|_E = 0$ and $f' d\mu|_E = 0$, so $d\lambda|_E = d\lambda'|_E$. And on $E^c, \lambda = \lambda' = 0$. Hence $\lambda = \lambda'$.

Now restrict to E^c: $d\lambda|_{E^c} = d\lambda|_{E^c} = 0 \implies f d\mu|_{E^c} = f' d\mu|_{E^c}$ as measures. Then for any set $F \subseteq X$, $F = (F \cap E) \sqcup (F \cap E^c)$ so

$$\int_F f d\mu = \int_{F \cap E} f d\mu + \int_{F \cap E^c} f d\mu = \int_{F \cap E} f d\mu = \int_{F \cap E} f' d\mu = \int_{F \cap E} f' d\mu + \int_{F \cap E^c} f' d\mu = \int_F f' d\mu,$$

meaning $f = f'$ μ-a.e. $\square$

3 Complex Measures

Definition 9. *A complex measure μ is a map $\mu : \mathcal{M} \to \mathbb{C}$ where $Re\, \mu$ and $Im\, \mu$ are finite signed measures, satisfying*

1. $\mu(\varnothing) = 0$
2. *If $\{E_j\}_{j=1}^{\infty}$ are disjoint and measurable, then $\sum_{j=1}^{\infty} \mu(E_j) = \mu(\bigcup_{j=1}^{\infty} E_j)$ with absolute convergence of series.*

Theorem 7 (Lebesgue-Radon-Nikodym for Complex Measures)**.** *If ν is a complex measure and μ is a σ-finite positive measure, then $\nu = f\, d\mu + d\lambda$ where $\lambda \perp \mu$, $d\lambda$ is a (finite) complex measure, and $f \in L^1(\mu)$.*

Proof. Take $Re\, \mu = f_1 d\mu + d\lambda_1$ and $Im\, \mu = f_2 d\mu + d\lambda_2$. Then $f_1, f_2 \in L^1$ because $Re\, \mu$ and $Im\, \mu$ are finite. λ_1 and λ_2 are also finite, for the same reason.

Take $\nu = Re\, \mu + iIm\, \mu = (f_1 + if_2)d\mu + d(\lambda_1 + i\lambda_2)$. □

Theorem 8 (Polar Decomposition)**.** *Given a complex measure μ there exists a measurable function ω so that $|\omega| = 1$ everywhere, and a positive measure $|\mu|$ so that $\mu = \omega|\mu|$. The function ω is unique up to a.e. equality.*

Proposition 4. *For any complex measure μ, there exist positive measures $\mu_1, \mu_2, \mu_3, \mu_4$ such that*

$$\mu = i\mu_1 + i^2\mu_2 + i^3\mu_3 + i^4\mu_4.$$

Proof. $Re\, \mu = \mu_4 - \mu_2$ and $Im\, \mu = \mu_1 - \mu_3$ where

$$\mu_4 = (Re\, \mu)^+, \quad \mu_2 = (Re\, \mu)^-, \quad \mu_1 = (Im\, \mu)^+, \quad \mu_3 = (Im\, \mu)^-$$

□

Proof. (Of Polar Decomposition) Now let $\mu_5 := \mu_1 + \mu_2 + \mu_3 + \mu_4$. This is a positive measure.

If $\mu_5(E) = 0$, then $\mu_5(E') = 0$ for all $E' \subseteq E \implies$
$\mu_1(E') = \cdots = \mu_4(E') = 0 \implies \mu(E') = 0$. So $\mu << \mu_5$.
Then apply LRN: there exists $f \in L^1(\mu_5)$ such that $\mu = fd\mu_5 + d\lambda$ and because $\mu << \mu_5$, we know $\lambda = 0$. So $\mu = fd\mu_5$.

Write $\mu = \frac{f}{|f|}|f|d\mu_5$.

Define $|\mu| = |f|d\mu_5$. This is a positive measure.

Define $\omega(x) := \begin{cases} \frac{f}{|f|}, & f(x) \neq 0 \\ 0, & f(x) = 0 \end{cases}$
Then $|\omega| = \frac{|f|}{|f|} = 1$ and

$$\mu = \omega|\mu|.$$

Proof of uniqueness:
Suppose μ is a complex measure and $d\mu = \omega_1 d\nu_1 = \omega_2 d\nu_2$ where ω_1, ω_2 are both magnitude 1 everywhere and ν_1, ν_2 are both positive.

Multiply the identity by $\overline{\omega_1}$:

$$\overline{\omega_1}d\mu = d\nu_1 = \overline{\omega_1}\omega_2 d\nu_2$$

so $\overline{\omega_1}\omega_2 d\nu_2$ is positive and so is $d\nu_2$.

Consider the complex-valued function of magnitude 1 $\overline{\omega_1}\omega_2$.

We note that $Im\,(\overline{\omega_1}\omega_2 d\nu_2) = Im\,(\overline{\omega_1}\omega_2)d\nu_2 = 0$ because the measure is positive. By Hahn Decomposition Theorem, if P and N are the positive and negative parts for $Im\,(\overline{\omega_1}\omega_2)d\nu_2$ then

$$Im\,(\overline{\omega_1}\omega_2)d\nu_2(P) = Im\,(\overline{\omega_1}\omega_2)d\nu_2(N) = 0 \implies Im\,(\overline{\omega_1}\omega_2) = 0 \quad \nu_2 \text{ a.e.}$$

because if not, of the sets where $Im\,(\overline{\omega_1}\omega_2) > 0$ or $Im\,(\overline{\omega_1}\omega_2) < 0$, at least one would have nonzero measure.

Thus $\overline{\omega_1}\omega_2 \in \mathbb{R}$ a.e. $\implies \overline{\omega_1}\omega_2 = \pm 1$ a.e.

And if $\{\overline{\omega_1}\omega_2 = -1\}$ had positive ν_2 measure, $\overline{\omega_1}\omega_2 d\nu_2$ would not be positive, so $\overline{\omega_1}\omega_2 = 1$ since $\overline{\omega_1}\omega_2 d\nu_2$ is a positive measure.

Conclude:
$\overline{\omega_1}\omega_2 = 1 \implies \omega_1 = \omega_2$ (μ-a.e.) and $d\nu_1 = (1)d\nu_2$, meaning $\nu_1 = \nu_2$.
Thus $|\mu|$ is unique and ω such that $d\mu = \omega d|\mu|$ is unique modulo $|\mu|$-a.e. $\square$

Definition 10. *The **total variation** of a complex measure ν is the positive measure $|\nu|$ determined by the property that if $d\nu = f d\mu$ where μ is a positive measure, then $d|\nu| = |f|d\mu$.*

This $|\nu|$ is independent of the choice of f and μ and is our ω from the polar decomposition.

The function f in $d\mu = f d|\mu|$ is often denoted as $\frac{d\mu}{d|\mu|}$.

Some consequences: If ν is a complex measure,

- $|\nu(E)| \leq |\nu|(E)$. Proof: $|\nu(E)| = \left|\int_E \omega d|\nu|\right| \leq \int_E |\omega| d|\nu| = |\nu|(E)$.
- $\nu << |\nu|$ and $\frac{d\nu}{d|\nu|} = \omega$.
- If ν_1, ν_2 are both complex measures, then $\nu_1 + \nu_2$ is also and $|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$. Proof:
$$\left|\int_E \omega(d\nu_1 + d\nu_2)\right| = \left|\int_E \omega(\omega_1 d|\nu_1| + \omega_2 |d\nu_2|)\right| \leq \left|\int_E \omega\omega_1 d|\nu_1|\right| + \left|\int_E \omega\omega_2 d|\nu_2|\right|$$
$$\leq \int_E |\omega\omega_1| d|\nu_1| + \int_E |\omega\omega_2| d|\nu_2| \leq \int_E d\nu_1 + \int_E d\nu_2 = |\nu_1|(E) + |\nu_2|(E)$$
- $||\mu|| := |\mu|(X)$ gives a norm. (We will see later that it's complete)

4 Lebesgue Differentiation Theorem

Definition 11. *The **Hardy-Littlewood maximal function** for $f \in L^1(\mathbb{R}^d)$ is:*

$$Mf(x) = \sup_{r>0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y)|\, dy$$

$Mf(x)$ is measurable, since:
$\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ pointwise on $\mathbb{R}^n \setminus \{y : |x_0 - y| = r_0\}$ as $x \to x_0$ and $r \to r_0$. Hence

$$\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)} \quad \text{a.e}$$

and

$$|\chi_{B_r(x)}| \leq \chi_{B_{r_0+1}(x_0)} \text{ if } r < r_0 + \frac{1}{2} \text{ and } |x - x_0| < \frac{1}{2}.$$

By dominated convergence,

$$\int_{B_r(x)} |f(y)| dy$$

is continuous in r and x, hence so is

$$\frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y)| dy.$$

Then $(Mf)^{-1}((a, \infty)) = \bigcup_{r>0} (\int_{B_r(\cdot)} |f(y)| dy)^{-1}((a, \infty))$ is open for any $a \in \mathbb{R}$.

Proposition 5 (Weak (1,1) Inequality)**.** *For all $\lambda \in (0, \infty)$ there is a constant C so that*

$$|\{x : Mf(x) > \lambda\}| \leq \frac{C\|f\|_1}{\lambda}.$$

We can rewrite this as

$$\lambda\, |\{x : Mf(x) > \lambda\}| \leq C\|f\|_1$$

thus

$$\int g \leq C\|f\|_1, \text{ where } g = \lambda \chi_{\{x: Mf(x) > \lambda\}} \leq Mf(x).$$

Lemma 3 (Vitali Covering Lemma)**.** *Suppose $\{B_i\}_{i=1}^N$ is a collection of balls in a metric space. Then there exists a subcollection $\{B_{i_j}\}_{j=1}^k$ of balls which are disjoint and*

$$\bigcup_{j=1}^{k} 3B_{i_j} \supseteq \bigcup_{i=1}^{N} B_i.$$

Proof. Work through the balls from largest to smallest.

For each ball, keep it is it is disjoint from the earlier kept balls, toss it if it is not.

Say $\{B_{i_j}\}_{j=1}^k$ = the kept balls. Pairwise disjointness follows from construction.

Triple property: Suppose B is a ball not kept. Then $B \cap B' \neq \varnothing$ for some B' that was kept, and the radius of B' is $\geq$ the radius of B. So $3B' \supseteq B$ because there exists a point in $B \cap B'$ so that all points in B are distance < 2 times the radius of B' to that intersection point. $\square$

Theorem 9 (Maximal Theorem)**.** *If $f \in L^1(\mathbb{R}^d)$ and $\lambda > 0$ then*

$$\big|\{x \in \mathbb{R}^d \mid Mf(x) > \lambda\}\big| \leq \frac{3^d ||f||_1}{\lambda}.$$

Proof. Let $E_\lambda = \{x : Mf(x) > \lambda\}$. Then $x \in E_\lambda \iff \exists r > 0$ so that $\frac{1}{|B_r(x)|}\int_{B_r(x)} |f| > \lambda$.

Let $K \subseteq E_\lambda$ be any compact set in E_λ. K is covered by balls so that

$$\frac{1}{|B|}\int_B |f| > \lambda, \text{ so } |B| < \frac{1}{\lambda}\int_B |f|.$$

Let $B_1, \ldots, B_N$ be a finite subcover. Then

$$|K| \leq |\bigcup_{i=1}^N B_i| \leq |\bigcup_{j=1}^k 3B_{i_j}| \leq \sum_{j=1}^k |3B_{i_j}| \leq \sum_{j=1}^k 3^d |B_{i_j}| \leq 3^d \sum_{j=1}^k |B_{i_j}|$$

$$\leq 3^d \sum_{j=1}^k \frac{1}{\lambda}\int_{B_{i_j}} |f| \leq 3^d \frac{1}{\lambda}\int \left(\sum_{j=1}^k \chi_{B_{i_j}}\right) |f| \leq \frac{3^d}{\lambda}\int |f| = \frac{3^d ||f||_1}{\lambda}.$$

Then $|K| \leq \frac{3^d ||f||_1}{\lambda}$ and

$$|E_\lambda| = \sup_{K \subseteq E_\lambda, K \text{ compact}} |K| \implies |E_\lambda| \leq \frac{3^d ||f||_1}{\lambda}.$$

$\square$

Definition 12. *A measurable function $f : \mathbb{R}^n \to \mathbb{C}$ is called **locally integrable** (with respect to Lebesgue measure) if $\int_K |f(x)|\,dx < \infty$ for every bounded measurable set $K \subset \mathbb{R}^n$.*

Theorem 10 (Lebesgue Differentiation)**.** *For $f \in L^1_{loc}(\mathbb{R}^d)$, for a.e. $x \in \mathbb{R}^d$:*

$$\lim_{r \to 0^+} \frac{1}{|B_r(x)|}\int_{B_r(x)} |f(y) - f(x)| dy = 0 \tag{1}$$

In particular, for a.e. $x \in \mathbb{R}^d$,

$$\lim_{r \to 0^+} \frac{1}{|B_r(x)|}\int_{B_r(x)} f(y) dy = f(x) \tag{2}$$

Points satisfying (2) are called **Lebesgue points.** Hence, the theorem says a.e. x is a Lebesgue point.

Proof. Without loss of generality we may assume $f \in L^1$ because the conclusions are local anyways.

Suppose $f \in L^1(\mathbb{R}^d)$ and g is continuous with compact support so that g approximates f in the $L^1(\mathbb{R}^d)$ norm.

Then by the triangle inequality and $f = f - g + g$,

$$\lim_{r\to 0^+} \sup \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| dy \leq$$

$$\lim_{r\to 0^+} \sup \frac{1}{|B_r(x)|} \int_{B_r(x)} |(f-g)(y)-(f-g)(x)| dy + \lim_{r\to 0^+} \sup \frac{1}{|B_r(x)|} \int_{B_r(x)} |g(y)-g(x)| dy.$$

Because g is continuous, this second term becomes 0. In truth we can actually show equality (adding and subtracting of g does not change the value of the $\limsup$).

So

$$\lim_{r\to 0^+} \sup \left[\frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| dy\right] \leq$$

$$\lim_{r\to 0^+} \sup \left[\frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - g(y)| dy + \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(x) - g(x)| dy\right]$$

$$\leq \left[\lim_{r\to 0^+} \sup \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - g(y)| dy + |f(x) - g(x)|\right]$$

$$\leq \sup_{r>0} \left[\frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - g(y)| dy + |f(x) - g(x)|\right]$$

$$= M(f-g)(x) + |f-g|(x).$$

Let $E_\varepsilon = \left\{x \in \mathbb{R}^d \mid \limsup_{r\to 0} \left[\frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| dy\right] > \varepsilon\right\}$.

Want to show: $|E_\varepsilon| = 0$ for all $\varepsilon > 0$.

$E_\varepsilon \subseteq \{x \mid M(f-g)(x) > \frac{\varepsilon}{2}\} \cup \{x \mid |f(x) - g(x)| > \frac{\varepsilon}{2}\}$.

We showed last time that $\left|\{x : M(f-g)(x) > \frac{\varepsilon}{2}\}\right| \leq \frac{3^d ||f-g||_1}{\frac{\varepsilon}{2}}$.
And if we recall Chebyshev's inequality, which says $\lambda |\{x : |f(x)| > \lambda\}| \leq ||f||_1$, then we know that $\left|\{x : |f(x) - g(x)| > \frac{\varepsilon}{2}\}\right| \leq \frac{||f-g||_1}{\frac{\varepsilon}{2}}$.

Thus
$$|E_\varepsilon| \leq \frac{2\cdot 3^d ||f-g||_1}{\varepsilon} + \frac{2||f-g||_1}{\varepsilon} = \frac{2(3^d+1)}{\varepsilon}||f-g||_1.$$

We can fix any $\eta > 0$. Then choose g such that $||f-g||_1 \leq \frac{\eta\cdot\varepsilon}{2(3^d+1)}$. Then $|E_\varepsilon| < \eta$.

Since η is arbitrary, $|E_\varepsilon| = 0$.
$$\left\{x : \left[\limsup_{r\to 0} \frac{1}{|B_r(x)|}\int_{B_r(x)} |f(y)-f(x)|dy\right] \neq 0\right\} = \bigcup_{n=1}^{\infty} E_{2^{-n}}$$
so the measure of the set on the left is zero.

To finish,
$$\left|\frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy - f(x)\right| = \left|\frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy - \frac{1}{|B_r(x)|}\int_{B_r(x)} f(x)dy\right| =$$
$$\frac{1}{|B_r(x)|}\left|\int_{B_r(x)}(f(y)-f(x))dy\right| \leq \frac{1}{|B_r(x)|}\int_{B_r(x)} |f(y)-f(x)|dy$$
We want to show (*) $\left|\frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy - f(x)\right|$ is $< \varepsilon$ for all sufficiently small r.
If the limit of (*) as $r \to 0$ does not exist or is not zero, then there exists $\varepsilon > 0$ so that infinitely many $r > 0$ have (*) $> \varepsilon$.
But then $\limsup_{r\to 0^+} \frac{1}{|B_r(x)|}\int_{B_r(x)} |f(y)-f(x)|dy > \varepsilon$, a contradiction. $\square$

Suppose Ω is a measurable set in $\mathbb{R}^d$. Then for a.e. $x \in \mathbb{R}^d$ we have
$$\chi_\Omega(x) = \lim_{r\to 0^+} \frac{1}{|B_r(x)|}\int_{B_r(x)} \chi_\Omega(y)dy = \lim_{r\to 0^+} \frac{|B_r(x)\cap\Omega|}{|B_r(x)|}$$
(which equals the proportion of the ball that is in Ω).

Connections to Radon-Nikodym:
If ν is a complex measure on $\mathbb{R}^d$, $\nu = f dm + d\lambda$ where m= Lebesgue measure and $\lambda \perp m$.

Question: Is $\lim_{r\to 0^+} \frac{\nu(B_r(x))}{m(B_r(x))} = f(x)$ a.e.?
Answer: True when ν is a **regular** measure, i.e. (1) $|\nu|$ is finite on every compact set and (2) $|\nu|(E) = \inf\{|\nu|(U) : U \supseteq E, U \text{ open}\}$.

Remarks about proof:

- Suffices to show $\lim_{r\to 0^+} \frac{\lambda(B_r(x))}{m(B_r(x))} = 0$ a.e. x when $\lambda \perp m$ and λ is regular
- Proof nearly reduces to same one we just did
- A modification of the Vitali argument gives what we want

5 Banach Spaces

5.1 Normed Vector Spaces, Cauchy Sequences, and Banach Spaces

We work on vector spaces X over $\mathbb{R}$ or $\mathbb{C}$.

Definition 13.
- *A **seminorm** $||\cdot||$: is a map $X \to [0,\infty)$ such that*
    1. *$||x+y|| \leq ||x|| + ||y||$ for all $x, y \in X$*
    2. *$||\lambda x|| = |\lambda| ||x||$ for $\lambda \in \mathbb{R}$ or $\mathbb{C}$ and $x \in X$*
- *A **norm** is a seminorm such that $||x|| = 0 \iff x = 0$*
- *A norm induces a metric topology on X via $d(x,y) := ||x-y||$*
- *Given norms $||\cdot||_1, ||\cdot||_2$, we say they are **equivalent** when there exists $C_1, C_2 > 0$ such that for all $x \in X, C_1||x||_1 \leq ||x||_2 \leq C_2||x||_1$*
- *A **Banach space** is a normed vector space which is complete in its induced metric space topology*

Basic Facts about Normed Vector Spaces:

1. If X is a normed vector space and $\sum_{n=1}^{\infty} x_n$ converges to x in X, then $||\sum_{n=1}^{\infty} x_n|| \leq \sum_{n=1}^{\infty} ||x_n||$.

    Proof: $\forall \varepsilon > 0$ $\exists N$ so that $n_1 \geq N \implies ||x - \sum_{n=1}^{n_1} x_n|| < \varepsilon$. Then
    $$||x|| \leq ||x - \sum_{n=1}^{n_1} x_n|| + ||\sum_{n=1}^{n_1} x_n|| < \varepsilon + \sum_{n=1}^{n_1} ||x_n||$$
    hence $||x|| < \varepsilon + \sum_{n=1}^{\infty} ||x_n||$. And since ε is arbitrary, $||x|| \leq \sum_{n=1}^{\infty} ||x_n||$.

2. If $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence in a normed vector space, then for any sequence $\{\phi_k\}_{k=1}^{\infty}$ of positive numbers, $\exists$ subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ so that $||x_{n_k} - x_{n_{k+1}}|| < \phi_k$ for each k

    Proof: Define n_k inductively:
    Let n_1 be such that whenever $\ell, m > n_1$, $||x_m - x_\ell|| < \varphi_1$.
    Assume $n_1, \ldots, n_{k-1}$ are well-defined (assuming $k > 1$).
    Let n_k be the smallest value of n such that $n > n_{k-1}$ and $n > N$ for N such that $||x_m - x_l|| < \varphi_k$ for all $m, l > N$.
    So $n_k > N$ (threshold where all remaining terms differ by $\leq \varphi_k$), meaning $||x_{n_k} - x_{n_{k+1}}|| < \varphi_k$ because $n_k > N$ and $n_{k+1} > n_k > N$ by induction.

3. A Cauchy sequence in a normed vector space converges to $x \iff$ it has a convergent subsequence

   Proof: ($\Rightarrow$:) If a sequence converges to x then all subsequences do as well.
   ($\Leftarrow$:)Suppose $x_{n_k} \to x$ i.e. for all $\varepsilon > 0$ there exists K such that $k > K \implies ||x_{n_k} - x|| < \varepsilon$.
   Fix $\varepsilon > 0$. Let K be such that $k > K \implies ||x_{n_k} - x|| < \varepsilon/2$.
   Let N be such that if $m, l > N$, then $||x_m - x_l|| < \varepsilon/2$.
   Fix some value of k such that $k > K$ and $n_k > N$. This is always possible since $n_k \to \infty$ as $k \to \infty$. Then for all $n > N$,

$$||x_n - x|| \leq ||x_n - x_{n_k}|| + ||x_{n_k} - x|| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

4. Every subsequence of a Cauchy sequence is Cauchy.

   Proof: $\{x_n\}$ Cauchy $\implies$ if $\{x_{n_k}\}$ is a subsequence, $n_1 < n_2 < \cdots \implies n_1 \geq 1, n_2 \geq 2, \ldots$.
   So in general, $n_k \geq k$. Then given $\varepsilon > 0$, let N be the threshold such that $m, l > N \implies ||x_l - x_m|| < \varepsilon$. Then if $k_1, k_2 > N$, then $n_{k_1}, n_{k_2} > N$, so $||x_{n_{k_1}} - x_{n_{k_2}}|| < \varepsilon$.

5. There is at most one $x \in X$ (normed vector space) so that the Cauchy sequence $\{x_n\}_{n=1}^{\infty}$ converges to x.

   Proof: If $\lim_{n\to\infty} x_n = a$ and $\lim_{n\to\infty} x_n = b$, then for all $\varepsilon > 0$, there exists N_1 such that $n > N_1 \implies ||x_n - a|| < \varepsilon$ and there exists N_2 such that $n > N_2 \implies ||x_n - b|| < \varepsilon$.
   Let $\varepsilon > 0$ and let $N > \max\{N_1, N_2\}$. Then $n > N \implies ||a - b|| \leq ||x_n - a|| + ||x_n - b|| < 2\varepsilon \implies a = b$.

Lemma 4. *Suppose X is a normed vector space. TFAE:*

1. *X is a Banach space*

2. *There is a strictly positive sequence $\{\varphi_n\}_{n=1}^{\infty}$ such that all Cauchy sequences satisfying $||x_n - x_{n+1}|| < \varphi_n$ for all n converge*

3. *All absolutely convergent series in X converge (i.e. $\sum_{n=1}^{\infty} b_n$ where $\sum_{n=1}^{\infty} ||b_n|| < \infty$)*

Proof. ((1) $\Rightarrow$ (2)): Immediate, since every Cauchy sequence is assumed to converge.
((2) $\Rightarrow$ (1)): Consequence of fact (2) above is that any Cauchy sequence has a subsequence such that $\|x_{n_k} - x_{n_{k+1}}\| < \varphi_k$ for all k. The subsequence is Cauchy so by assumption, it converges. So the original Cauchy sequence has a convergent subsequence, hence the original Cauchy sequence converges.

((1) $\Rightarrow$ (3)): Suppose $\sum_{k=1}^{\infty} b_k$ is an absolutely convergent series in X. Let $X_n := \sum_{k=1}^{n} b_k$ be the sequence of partial sums. Wlog assume that $n > m$:

$$||X_n - X_m|| = ||\sum_{k=1}^{n} b_k - \sum_{k=1}^{m} b_k|| = ||\sum_{k=m+1}^{n} b_k|| \leq \sum_{k=m+1}^{n} ||b_k||.$$

Let $A_n = \sum_{k=1}^{n} ||b_k||$. By hypothesis, absolute convergence means A_n has a finite limit. So $\{A_n\}$ is Cauchy, thus for all $\varepsilon > 0$ there exists N such that $n, m > N \implies |A_n - A_m| < \varepsilon$.
Conclude:

$$||X_n - X_m|| \leq \sum_{k=m+1}^{n} ||b_k|| = |A_n - A_m| < \varepsilon.$$

So $||X_n - X_m|| < \varepsilon$, meaning $\{X_n\}$ is Cauchy. So the sequence of partial sums is Cauchy. Assuming X is Banach, this means the partial sums converge.
((3) $\Rightarrow$ (1)): Here we assume that all absolutely convergent series converge, i.e.

$$\sum_{n=1}^{\infty} ||b_n|| < \infty \implies \lim_{N\to\infty} \sum_{n=1}^{N} b_n \text{ exists.}$$

Let $\{x_n\}$ be any Cauchy sequence.
Let $\{x_{n_k}\}$ be a subsequence such that $||x_{n_k} - x_{n_{k+1}}|| < 1/k^2$.
Let $b_k := x_{n_k} - x_{n_{k+1}}$. Then $||b_k|| < 1/k^2$, so $\sum_{k=1}^{\infty} ||b_k||$ is absolutely convergent, hence has a finite limit. So

$$\lim_{K\to\infty} \sum_{k=1}^{K} b_k$$

converges, and

$$\sum_{k=1}^{K} b_k = \sum_{k=1}^{K} (x_{n_k} - x_{n_{k+1}}) = x_{n_1} - x_{n_{K+1}}.$$

So $\lim_{K\to\infty}(x_{n_1} - x_{n_{K+1}})$ exists. But then $\lim_{K\to\infty} x_{n_{K+1}}$ exists. So our subsequence converges, meaning our original Cauchy sequence converges. $\square$

5.2 Maps Between Banach Spaces

We will think about maps between Banach spaces. In infinite dimensions, continuity of maps between normed vector spaces distinguishes how they behave.

Suppose X, Y are normed vector spaces over the same field. Let $T : X \to Y$ be a linear map. Recall: T is continuous at $x \iff$ for all $\varepsilon > 0$ there exists $\delta > 0$ such that $||x - x_1|| < \delta \implies ||T(x) - T(x_1)|| < \varepsilon$.

- In general, continuity means for all $x \in X$, for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x_1 \in X$, $||x - x_1|| < \delta \implies ||T(x) - T(x_1)|| < \varepsilon$.

- Continuity is equivalent to continuity of T at 0, because:
  ($\Rightarrow$): Continuous everywhere clearly implies continuous at 0.
  ($\Leftarrow$) : If T is continuous at 0, then for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x_1 \in X$, $||x_1|| < \delta \implies ||T(x_1)|| < \varepsilon$. Given this ε and this δ, apply the consequence to $x - x_1$:

  $$||x - x_1|| < \delta \implies ||T(x - x_1)|| = ||T(x) - T(x_1)|| < \varepsilon.$$

  In fact, also, continuity at 0 implies uniform continuity everywhere.

- Continuity at the origin is equivalent to boundedness.

  Definition 14. *We say that $T : X \to Y$ is* ***bounded*** *when there exists some finite C such that $||T(x)|| \leq C||x||$ for all $x \in X$.*

  Proof:
  ($\Rightarrow$): Suppose T is continuous at 0. Fix $\varepsilon = 1$. Then there exists $\delta > 0$ so that $||x|| < \delta \implies ||T(x)|| < 1$.
  Fix $x \in X$. Consider $x' := \frac{\delta}{2}||x||^{-1}x$. (assuming $x \neq 0$)
  Then $||x'|| = \frac{\delta}{2}||x||^{-1}||x|| = \delta/2 < \delta$.
  So $||T(x')|| = \frac{\delta}{2}||x||^{-1}||T(x)|| < 1$. Thus $x \neq 0 \implies ||T(x)|| < \frac{2}{\delta}||x||$. And if $x = 0$, then we also have $||T(x)|| < \frac{2}{\delta}||x||$, since both sides equal 0.

  Conclude: $||T(x)||_Y < \frac{2}{\delta}||x||_X$ $(C = 2/\delta)$.
  ($\Leftarrow$) : Assume $||T(x)||_Y < C||x||_X$. Fix $\varepsilon > 0$. Let $\delta = \frac{\varepsilon}{2C}$ (assume $C \neq 0$). So if $||x|| < \delta$, then

  $$||T(x)|| \leq C||x|| = C \cdot \frac{\varepsilon}{2C} = \frac{\varepsilon}{2} < \varepsilon.$$

Definition 15. *Given a bounded linear map $T : X \to Y$, we define the* ***operator norm*** *$||T||_{X\to Y}$ to be the infemum of all positive constants C such that $||T(x)|| \leq C||x||$ for all $x \in X$.*

Note: $||T||$ is also a bound, i.e. the infemum is attained.
Let $\mathcal{C} = \{C \geq 0 \mid ||Tx|| \leq C||x|| \forall x \in X\}$.
Then $||T|| = \inf \mathcal{C}$. Claim: $||T|| \in \mathcal{C}$. Proof:
Given $x \in X$, $||T(x)|| \leq C||x||$ for any $C \in \mathcal{C}$, i.e. $\frac{||T(x)||}{||x||}$ is a lower bound for $\mathcal{C}$.
So

$$\frac{||T(x)||}{||x||} \leq \inf \mathcal{C} = ||T||.$$

Thus $||T(x)|| \leq ||T||||x||$ whenever $x \neq 0$. But also $0 = ||T(x)|| \leq ||T||||x|| = 0$ whenever $x = 0$.

Alternatively,

$$||T|| = \sup_{x \in X \setminus \{0\}} \left\{ \frac{||T(x)||}{||x||} \right\}.$$

Proof:
Let
$$A := \sup_{x \in X \setminus \{0\}} \left(\frac{||T(x)||}{||x||} \right).$$
Then we can say that when A is finite, $\frac{||T(x)||}{||x||} \leq A$ for all $x \neq 0$, by definition. So $||T(x)|| \leq A||x||$ for all $x \neq 0$ (and this holds vacuously when $x = 0$). So A is a bound $\implies ||T|| \leq A$.

In the reverse direction, we know that $||T(x)|| \leq ||T||||x||$ for all x (so in particular this is true for all nonzero x). Then $\frac{||T(x)||}{||x||} \leq ||T||$ for all $x \neq 0 \implies$
$$A = \sup_{x \in X \setminus \{0\}} \left(\frac{||T(x)||}{||x||} \right) \leq ||T||.$$
This proves the claim.

Alternative equivalences:
$$||T|| = \sup_{x \in X, ||x||=1} ||T(x)|| = \sup_{x \in X, ||x|| \leq 1} ||T(x)||.$$

6 Hahn-Banach Theorem

Let X be a Banach space (vector space over $F = \mathbb{R}$ or $\mathbb{C}$), complete with respect to its norm.

A **functional** is a linear map $X \to F$.

In infinite dimensions we wish to study *bounded* linear functionals,
i.e. $\ell : X \to F$ which is linear and has some finite C for which $|\ell(x)| \leq C||x||$ for all $x \in X$.

The collection of all bounded linear functionals on X is called the dual space of X, denoted X^*.

In infinite dimensions, X and X^* can be very different.

Now: building linear functionals with given properties.
Typical scenario: you build part of a linear functional, usually on some finite-dimensional subspace of X, and you would like to extend it to all of X.

6.1 Hahn-Banach Statement and Proof

Theorem 11 (Hahn–Banach)**.** *Suppose X is a real vector space and $p : X \to \mathbb{R}$ is a sublinear functional on X, i.e.*

1. *$p(\lambda x) = \lambda p(x)$ for $\lambda \geq 0$ and $x \in X$*

2. *$p(x + y) \leq p(x) + p(y)$ (think about $p(x) = C\|x\|$)*

Say M is a subspace of X. Let $f : M \to \mathbb{R}$ be a linear functional satisfying $f(x) \leq p(x)\ \forall x \in M$.
Then $\exists$ a linear functional $F : X \to \mathbb{R}$ so that $F(x) \leq p(x)$ and $F(x) = f(x)$ for all $x \in M$.

Lemma 5 (Zorn's Lemma)**.** *Suppose S is a nonempty partially ordered set such that every chain (totally ordered subset) has an upper bound in S. Then S must contain a maximal element.*

Proof. (Of Hahn-Banach)
Let $S = \{\text{all partial extensions of } f\}$, i.e. pairs (D, f_D) where D is a vector space containing M and $f_D : D \to \mathbb{R}$ such that

1. $f_D|_M = f$

2. $f_D(x) \leq p(x)\ \forall x \in D$

We say $(D, f_D) \leq (D', f_{D'})$ when $D \subseteq D'$ and $f_{D'}|_D = f_D$.

Note that S is nonempty because it contains (M, f).

Let $\mathcal{C} \subseteq S$ be a chain (i.e. any two elements of $\mathcal{C}$ are related).
Construct an upper bound as follows:
Let
$$V = \bigcup_{(D, f_D) \in \mathcal{C}} D$$
V is a vector space: if $x, y \in V$, then $x \in D_1$, $y \in D_2$. Since it's a chain, (D_1, f_{D_1}) is related to (D_2, f_{D_2}). Either $D_1 \subseteq D_2$ or $D_2 \subseteq D_1$.
Wlog $D_1 \subseteq D_2$. Then $x, y \in D_2 \implies$ since D_2 is a vector space $x + y \in D_2 \subseteq V$.

Define $f_V(x) = f_D(x)$ for any (D, f_D) such that $x \in D$. This is well-defined:
If $x \in D_1$, $x \in D_2$, and (D_1, f_{D_1}) and (D_2, f_{D_2}) are related (wlog $D_1 \subseteq D_2$, $f_{D_2}|_{D_1} = f_{D_1}$). In particular, $x \in D_1 \cap D_2$, so $f_{D_2}(x) = f_{D_1}(x)$.

Check:

1. f_V is a linear functional

2. $f_V|_M = f$ ✓

3. $f_V(x) \leq p(x)$ for all $x \in V$ ✓

4. $(V, f_V) \geq (D, f_D)$ for all $(D, f_D) \in \mathcal{C}$

For (1) and (4): $x, y \in V \implies x \in D_1,\ y \in D_2$. Wlog $x, y \in D_2$ by being a chain.
(1)

$$f_V(x+y) = f_{D_2}(x+y) = f_{D_2}(x) + f_{D_2}(y) = f_V(x) + f_V(y)$$

(4) $D \subseteq V$ for all $(D, f_D) \in \mathcal{C}$ and $f_V|_D = f_D$.

By Zorn's Lemma, $\exists$ a maximal element (Y, f_Y) in S.

1. $f_Y|_M = f$
2. $f_Y(x) \leq p(x)\ \forall\, x \in Y$.
3. $\nexists\, (Z, f_Z) \in S$ so that $(Y, f_Y) \lneq (Z, f_Z)$

Need to show: if $Y \neq X$, then (Y, f_Y) is always extendable.
Suppose $Y \neq X$. Let $x' \in X \setminus Y$. Let $Z = \operatorname{span}\{x', Y\}$ (so $Y \subsetneq Z$).
Need $f_Z(x) = f_Y(x)$ for all $x \in Y$.

General element of Z has the form $z = cx' + y$ where $c \in \mathbb{R}$, $y \in Y$.

Need to define $f_Z(cx' + y) = cf_Z(x') + f_Z(y) = cf_Z(x') + f_Y(y)$.
It suffices to decide what $f_Z(x')$ equals. All we need is:

$$f_Z(cx'+y) \leq p(cx'+y) \quad \forall\, c \in \mathbb{R},\ y \in Y$$

$$\iff \quad cf_Z(x') + f_Y(y) \leq p(cx'+y) \quad \forall\, c \in \mathbb{R},\ y \in Y$$

$$\iff \quad cf_Z(x') \leq p(cx'+y) - f_Y(y) \quad \forall\, c \in \mathbb{R},\ y \in Y$$

Divide by $|c|$ on both sides (assume $c \neq 0$ because the inequality is true if not):

$$\frac{c}{|c|} f_Z(x') \leq p\left(\frac{c}{|c|}x' + \frac{y}{|c|}\right) - f_Z\left(\frac{y}{|c|}\right) \qquad \forall\, c \neq 0,\ y \in Y$$

So we need:

$$\pm f_Z(x') \leq p(\pm x' + y) - f_Z(y) \quad \forall\, y \in Y$$

Rewriting the negative one by multiplying by -1:

$$-p(-x'+y) + f_Z(y) \leq f_Z(x') \leq p(x'+y) - f_Z(y) \quad \text{for any } y \in Y$$

An acceptable value of $f_Z(x')$ can be chosen $\iff$ for all $y, y' \in Y$:

$$-p(-x'+y') + f_Z(y') \leq p(x'+y) - f_Z(y)$$

Q: Is this true? We're done if we can show this is true.

Observe: Move f_Z to one side and p to the other.
Q: $f_Y(y) + f_Y(y') \overset{?}{\leq} p(x'+y) + p(-x'+y')\ \forall\, y, y'$?

Note

$$\begin{aligned} f_Y(y) + f_Y(y') = f_Y(y+y') &\le p(y+y') \quad \text{by construction} \\ &\le p((x'+y) + (-x'+y')) \\ &\le p(x'+y) + p(-x'+y') \quad \text{by sublinearity.} \end{aligned}$$

Because (Y, f_Y) is maximal, we must have $Y = X$ (i.e. we've extended to everything). ☐

Note. *If $p(x) = C\|x\|$, then $F(x) \le C\|x\|$ and*

$$-F(x) = F(-x) \le C\| - x\| = C\|x\|$$

Conclusion: $|F(x)| \le C\|x\|$. So in particular, if f is a bounded linear functional, then so is F.

Theorem 12 (Hahn-Banach, Complex Version)**.** *If X is a complex vector space and $p : X \to \mathbb{R}$ is a seminorm, then for a linear functional $f : M \to \mathbb{C}$, where M is a subspace of X and $|f(x)| \le p(x)$ for all $x \in M$, $\exists$ a linear functional $F : X \to \mathbb{C}$ such that $F|_M = f$ and $|F(x)| \le p(x)$ for all $x \in X$.*

Proof. Complex linearity $\Rightarrow$ real linearity.
Given f, define $u(x) := \mathrm{Re}(f(x))$ on M. Check that this is real linear.
Extend to $U(x)$, a real linear functional on X (extend subject to $U(x) \le p(x)$ for all $x \in X$).
Now let $F(x) := U(x) - iU(ix)$.

Claim: this gives us what we want.

1. $F(x)$ is complex linear.
2. Check: $F|_M(x) = f(x)$.
3. Check: $|F(x)| \le p(x)\ \forall\, x \in X$.

Check (2):
Say $x \in M$.

$$\begin{aligned} F(x) = U(x) - iU(ix) &= u(x) - iu(ix) \\ &= \mathrm{Re}(f(x)) - i\,\mathrm{Re}(f(ix)) \\ &= \mathrm{Re}(f(x)) - i\,\mathrm{Re}(if(x)) \\ &= \mathrm{Re}(f(x)) - i(-\mathrm{Im} f(x)) \\ &= \mathrm{Re}(f(x)) + i\,\mathrm{Im} f(x) = f(x) \end{aligned}$$

Check (3):
Given x, let $\alpha \in \mathbb{C}$ be such that $|\alpha| = 1$ and $\alpha F(x)$ is real and nonnegative. Then

$$|F(x)| = |\alpha F(x)| = |F(\alpha x)| = F(\alpha x) = U(\alpha x) - iU(i\alpha x)$$

Since $F(\alpha x)$ is real and U is real, $U(i\alpha x) = 0$. So:

$$F(\alpha x) = U(\alpha x) \leq p(\alpha x) = p(x)$$

Since $|F(x)| = F(\alpha x), |F(x)| \leq p(x)\ \forall\, x \in X$. $\square$

6.2 Applications of Hahn–Banach

Theorem 13. *Suppose X is a normed vector space.*

(a) If M is a closed subspace of X and $x \notin M$, then $\exists$ a linear functional $f \in X^*$ so that $f(x) \neq 0$ and $f = 0$ on M.
Moreover, if $\delta = \inf\{\|x - y\| : y \in M\}$, one can say $f(x) = \delta$ and $\|f\| = 1$ (i.e. $|f(x)| \leq \|x\|\ \forall\, x \in X$ and 1 is the smallest bound possible).

(b) If $x \neq 0$, $\exists\, f \in X^*$ s.t. $\|f\| = 1$ and $f(x) = \|x\|$.

(c) X embeds isometrically into X^{**}.

Proof. (a) $\delta = \inf\{\|x - y\| : y \in M\}$.
If $\delta = 0$, $\exists$ a sequence $\{y_n\}_{n=1}^{\infty} \subset M$ such that $\lim_{n\to\infty} \|x - y_n\| = 0$, i.e. $x = \lim_{n\to\infty} y_n$. Since M is closed by hypothesis, $x \in M$.
If $x \notin M$, $\delta > 0$.

We want to find $f : X \to \mathbb{C}$ such that $f(x) = \delta$, $f = 0$ on M, and $|f(x)| \leq \|x\|$ $\forall\, x \in X$.

It suffices to define f on $M' = \operatorname{span}\{x, M\}$ such that $f(\alpha x + m) = \alpha\delta\ \forall\, \alpha \in \mathbb{C}$, $m \in M$.

Just check: $|\alpha\delta| \leq \|\alpha x + m\|$ for any $\alpha \in \mathbb{C}$, $m \in M$.
$\alpha = 0$: true.
$\alpha \neq 0$: divide by $|\alpha|$.
Need $\delta \leq \|x + \alpha^{-1}m\|$ for all $m \in M$, $\alpha \neq 0$.
($\alpha^{-1}m$ is some vector in M; call it $-m'$.)
Need to show $\delta \leq \|x - m'\|$ for any $m' \in M$.
But this is true by definition of δ. So this shows $||f|| \leq 1$.

To see $||f|| \geq 1$,

$$\frac{|f(x - m)|}{\|x - m\|} = \frac{\delta}{\|x - m\|}$$

and since

$$\delta = \inf_{m \in M} \|x - m\|, \exists \{m_n\} \subset M \text{ s.t. } \|x - m_n\| \to \delta$$

so

$$\|f\| = \sup_{x' \neq 0} \frac{|f(x')|}{\|x'\|} \geq \lim_{n\to\infty} \frac{\delta}{\|x - m_n\|} = \frac{\delta}{\delta} = 1.$$

(b): Suppose $x \neq 0$.
Define f so that $f(x) = \|x\|$.
Observe $|f(x)| \leq 1 \cdot \|x\|$ when x is fixed.
Let $M = \operatorname{span}\{x\}$.
$f(\alpha x) = \alpha\|x\|$ for $\alpha \in \mathbb{C}$.
$f(x) = \|x\|$ and $|f(y)| \leq \|y\|$ for all $y \in M$.

By Hahn–Banach, $\exists\, f : X \to \mathbb{C}$ such that $f(x) = \|x\|$ and $\|f\| \leq 1$ (i.e. $|f(x)| \leq \|f\|_{X^*}\|x\|_X$, i.e. $|f(x)| \leq \|x\|$).

(c): Given $x \in X$, let $\hat{x} \in X^{**}$:

$$\hat{x}(\ell) = \ell(x) \quad \text{for all } \ell \in X^*$$

$$\widehat{\alpha x + y}(\ell) = \ell(\alpha x + y) = \alpha\ell(x) + \ell(y) = \alpha\hat{x}(\ell) + \hat{y}(\ell) = (\alpha\hat{x} + \hat{y})(\ell)$$

Why is $\|\hat{x}\|_{X^{**}} = \|x\|_X$?
Define

$$\|\hat{x}\|_{X^{**}} = \sup_{\|\ell\|_{X^*} \leq 1} |\hat{x}(\ell)| = \sup_{\|\ell\|_{X^*} \leq 1} |\ell(x)|$$

Always:

$$|\ell(x)| \leq \|\ell\|_{X^*} \cdot \|x\|_X = \|x\|$$

(since $\|\ell\|_{X^*} = 1$), so $\|\hat{x}\|_{X^{**}} \leq \|x\|_X$.

If $x = 0$, clear that $||x|| \leq ||\hat{x}||$, and if not, by (b), $\exists\, \ell$ such that $\|\ell\| = 1$ and $\ell(x) = \|x\|$, so the sup is actually attained:

$$\|\hat{x}\|_{X^{**}} = \|x\|_X$$

□

7 Baire Category Theorem

7.1 Baire Category Theorem Statement and Proof

Theorem 14 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space. Then X is not a countable union of closed, nowhere dense sets.*

Definitions:

- E **nowhere dense** = closure of E has complement which is dense = E does not contain any balls.
- E **meager** = countable union of nowhere dense sets.
- E **generic** = E^c is meager.

Proof. Let $\{E_j\}_{j=1}^{\infty}$ be closed nowhere dense sets.
Need to show: $\exists\, p \in X \setminus \bigcup_{j=1}^{\infty} E_j$.
Idea: build a sequence which is Cauchy and whose limit could not possibly belong to any of the E_j's.
Step 1: Choose a ball B_0 in X.
E_1 nowhere dense $\implies B_0 \not\subseteq E_1 \implies \exists\, x_1 \in B_0 \setminus E_1$.

$B_1 =$ ball centered at x_1 such that:

1. radius of $B_1 < \frac{1}{2}$
2. $2B_1 \subseteq B_0 \setminus E_1$

In general, we get $B_0 \supset B_1 \supset \cdots \supset B_j$ where:

1. radius of $B_k < 2^{-k}$
2. $2B_k \subseteq B_{k-1} \setminus E_k$ for each k

Step 2: Repeat the process above. Get infinite sequence of balls $\{B_k\}_{k=0}^{\infty}$ and centers $\{x_k\}_{k=0}^{\infty}$.
Claim:

1. $\{x_k\}$ is Cauchy (complete $\implies$ has a limit x_*)
2. For each k, $x_* \notin E_k$
3. $x_* \in B_0$

Proof of claim:
(1) For each k, B_k has radius $< 2^{-k}$ and $B_{k+1} \subseteq B_k \subseteq \cdots$ and so on $\implies$
for x_n, x_m with $m > n$: x_n is the center of B_n and x_m is the center of $B_m \subseteq B_n \implies$
$d(x_n, x_m) < 2^{-n}$. i.e., given any $\varepsilon > 0$, if N satisfies $2^{-N} < \varepsilon$, then $n, m > N \implies d(x_n, x_m) < \varepsilon$. So $\{x_n\}$ is Cauchy.

(3) We know that the limit $x_* = \lim_{n\to\infty} x_n$ exists.
$B_0 \supseteq 2B_1$, and in general, $B_k \supseteq 2B_{k+1}$.

Suppose that $y \in B_k^c$, $x \in B_{k+1}$.
Then $d(y, x) \geq$ radius of $B_{k+1} > 0$.

We know that $x_k \in B_k$ and in fact $x_k \in B_j$ for $j < k$.
So $d(y, x_k) \geq$ radius of $B_1 > 0$ when $y \in B_0^c$.

So if we had $x_* \in B_0^c$, it would mean $d(x_*, x_k) >$ radius of $B_1 > 0 \rightarrow\leftarrow$
So we must have $x_* \in B_0$ (in fact $x_* \in \bigcap_{k=0}^{\infty} B_k$).

(2) Need to show: for each k, $x_* \notin E_k$.

Know: $y \in E_k$, $x \in B_k \implies d(y,x) >$ radius of $B_k > 0$.
Fix any k. Let $j \geq k$. Then if $y \in E_k$ and $x_j \in B_j$, since $B_j \subseteq B_k$, $d(y, x_j) >$ radius of $B_k > 0$.

Letting $j \to \infty$:
Whenever $y \in E_j$ and $j \geq k$, $d(y, x_*) \geq$ radius of $B_k > 0 \implies$
$y \neq x_* \implies x_* \notin E_k$.

So we found a Cauchy sequence in X that converges to a point not in any E_k. Since X is complete, this means $X \not\subset \bigcup_{k=1}^{\infty} E_k$. $\square$

Note. *We proved a stronger result: generic sets are dense in X.*

7.2 Applications of Baire Category Theorem

7.2.1 Open Mapping Theorem

Theorem 15 (Open Mapping Theorem)**.** *Let X, Y be Banach spaces and $T : X \to Y$ a continuous and linear map. If T is surjective, then T is open, i.e. the image of an open set is open.*

Proof. Goal: $\|x\|_X \leq C\|Tx\|_Y$.

Step 1: Show $\exists\, \delta > 0$ such that $\overline{T(B_s)}$ contains a ball of radius $\delta > 0$ for some s.
Suppose not. Then $\forall\, s$, $\overline{T(B_s)}$ is nowhere dense.

$$\bigcup_{s \in \mathbb{N}} \overline{T(B_s)} = Y$$

because every $x \in X$ belongs to B_s for some s, so $\bigcup_{s \in \mathbb{N}} T(B_s) \supseteq T(X) = Y$.

So by the Baire Category Theorem, not every $\overline{T(B_s)}$ is nowhere dense. i.e., $\exists\, \delta > 0$, $y_0 \in Y$ so that $\overline{T(B_s)} \supseteq B_\delta(y_0)$.

Step 2: Find a ball in Y which is centered at the origin.

Given $y \in B_\delta(y_0)$, $y = \lim_{n\to\infty} T(x_n)$ for some $\{x_n\}_{n=1}^{\infty} \subseteq B_s(0)$.

$y_0 = \lim_{n\to\infty} T(x_n')$ for some $\{x_n'\}_{n=1}^{\infty} \subseteq B_s(0)$.

$y - y_0 = \lim_{n\to\infty} T(x_n - x_n')$. Know: $\|x_n - x_n'\|_X < 2s$.

$y - y_0$ is an arbitrary point of $B_\delta(0) \subseteq Y$.
So every $z \in B_\delta(0) \subseteq Y$ is a limit $z = \lim_{n\to\infty} T(x_n'')$, $\{x_n''\} \subseteq B_{2s}(0) \subset X$.
So $B_\delta(0) \subseteq \overline{T(B_{2s})} \subseteq Y$.

Step 3: Want to say that every $z \in B_\delta(0)$ has $z = Tx$ for some $x \in B_{4s}(0)$.

By step 2, for all $z \in Y$ with $\|z\|_Y < \delta$, $\exists$ a sequence $\{x''_n\}_{n=1}^\infty$ such that $\|x''_n\| < 2s$ and $z = \lim_{n\to\infty} T(x''_n)$.

$z \in B_\delta(0) \implies \exists\, x_1 \in B_{2s}(0)$ such that $\|z - Tx_1\|_Y < \frac{\delta}{2}$.

Let $z_1 = z - Tx_1$. Then $z_1 \in B_{\frac{\delta}{2}}(0)$. So $2z_1 \in B_\delta(0)$. $\exists\, y_2 \in B_{2s}(0)$ such that
$\|2z_1 - Ty_2\|_Y < \frac{\delta}{2}$ ($\implies$ $\|z_1 - T\frac{y_2}{2}\|_Y < \delta/4$, where $\frac{y_2}{2} =: x_2$).

In general, we get $x_1, \dots, x_k$ such that

1. $\|x_k\| \le 2s \cdot 2^{-k+1}$ and
2. $$\left\| z - \sum_{j=1}^{k} T(x_j) \right\| < \delta \cdot 2^{-k}$$

Let $z_{k+1} = \left(z - \sum_{j=1}^k T(x_j)\right) 2^k$, so $\|z_{k+1}\| < \delta$.
Then $\exists\, y_{k+1} \in B_{2s}$ such that $\|z_{k+1} - Ty_{k+1}\| < \frac{\delta}{2}$.
Scale out by 2^k on both sides:

$$\left\| z - \sum_{j=1}^{k} T(x_j) - T\Big(\frac{y_{k+1}}{2^k}\Big) \right\| < \delta \cdot 2^{-k-1}$$

where $\frac{y_{k+1}}{2^k} =: x_{k+1}$. We get $\{x_k\}_{k=1}^\infty$ satisfying (1) and (2) above.

Let $x = \sum_{j=1}^\infty x_j$. This converges since $\|x_j\| \to 0$.
By continuity of T, $Tx = \sum_{j=1}^\infty Tx_j$ and $z = Tx$.
We know about x that:

$$\|x\| \le \sum_{j=1}^{\infty} \|x_j\| = \sum_{j=1}^{\infty} 2s \cdot 2^{-j+1} = s(2 + 1 + \frac{1}{2} + \cdots) = 4s$$

$\implies x \in B_{4s} \implies T(B_{4s}(0)) \supseteq B_\delta(0)$.
From here, we see T(open) is open. $\square$

Corollary 5. *Let X, Y be Banach spaces. If $T : X \to Y$ is linear, bounded and invertible, then T^{-1} is also bounded.*

7.2.2 Closed Graph Theorem

Theorem 16 (Closed Graph Theorem)**.** *Given Banach spaces X, Y, a linear $T : X \to Y$ is continuous $\iff$*

$$\Gamma_T := \{(x, y) \in X \times Y : x \in X,\, y = Tx\} \subseteq X \times Y$$

is a closed subset of $X \times Y$ (i.e. if $x_n \to x$ and $Tx_n \to y$, then $Tx = y$).

Proof. Think of $X \times Y$ as its own Banach space. Let

$$\|(x,y)\| := \|x\|_X + \|y\|_Y, \quad ((x,y)+(x',y') = (x+x', y+y'))$$

Check: this is a Banach space.
$\{(x_n,y_n)\}_{n=1}^{\infty}$ Cauchy $\implies \forall \varepsilon > 0$, $\exists N$ such that $n,m > N \implies \|(x_n,y_n) - (x_m,y_m)\| < \varepsilon$
$\implies \|x_n - x_m\|_X < \varepsilon$, $\|y_n - y_m\| < \varepsilon$
$\implies \{x_n\}$ Cauchy, $\{y_n\}$ Cauchy
$\implies x_n \to x_*$, $y_n \to y_*$.
Observe: $\|x_n - x_*\|_X + \|y_n - y_*\|_Y = \|(x_n,y_n) - (x_*,y_*)\| \to 0$.

Define projections:

$$\Pi_1 : X \times Y \to X, \quad (x,y) \mapsto x$$

$$\Pi_2 : X \times Y \to Y, \quad (x,y) \mapsto y$$

$$\|\Pi_1(x,y)\| = \|x\|_X \leq \|x\|_X + \|y\|_Y = \|(x,y)\|$$

So Π_1 is bounded with constant 1. Same for Π_2.

$\Gamma_T \subseteq X \times Y$ is a closed subspace by assumption. A closed subspace of a Banach space is a Banach space. Γ_T is a Banach space by inheriting norm.

$$\Pi_1 : \Gamma_T \to X, \qquad \Pi_2 : \Gamma_T \to Y$$

Still get Π_1, Π_2 bounded (restricting domain won't increase norm).
Π_1 is surjective because Tx is always defined.
Π_1 is injective also:

$$\Pi_1(x,y) = \Pi_1(x',y') \Rightarrow x = x' \Rightarrow y = Tx = Tx' = y'$$

So Π_1 has inverse Π_1^{-1} and it's continuous (by the Open Mapping Theorem / corollary).
So $\Pi_2 \circ \Pi_1^{-1}$ is continuous.

For $x \in X$: $\Pi_1^{-1}(x) = (x,y)$ where $(x,y) \in \Gamma_T$, so $= (x, Tx)$.
$\Pi_1^{-1}(x) = (x, Tx)$ is continuous.
$\implies \Pi_2 \circ \Pi_1^{-1}(x) = Tx$. So $T = \Pi_2 \circ \Pi_1^{-1}$ is continuous. $\square$

8 Alternate topologies

Begin with a Banach space $(X, ||\cdot||)$. We can look at some non-norm topologies that are sometimes useful.

Definition 16. *If X is a real or complex vector space, we call $||\cdot|| : X \to \mathbb{R}_{\geq 0}$ a **seminorm** when*

- $||cf|| = |c|||f||$

- $||f+g|| \leq ||f|| + ||g||$

(So $||f|| = 0$ might be possible for nonzero f)

New idea: Start with a real or complex vector space X and a family $\{||\cdot||_\alpha\}_{\alpha\in A}$ of seminorms.
Assume $||f||_\alpha = 0$ for all $\alpha \implies f = 0$.

Example 3. $X = L^2(\mathbb{R})$
Seminorms: $||f||_g := \left|\int_{\mathbb{R}} f\overline{g}\right|$ *for* $g \in L^2(\mathbb{R})$.
If $||f||_g = 0$ *for all* $g \in L^2(\mathbb{R})$*, then* $f = 0$ *(because e.g. take* $g = f$*)*

We can define a topology for a given family of seminorms.
Basis: Given $\alpha \in A$, $x \in X$, $\varepsilon > 0$, define

$$U_{\alpha,x,\varepsilon} := \{y \in X \mid ||y - x||_\alpha < \varepsilon\}$$

Let T= topology generated by the basis $\{U_{\alpha,x,\varepsilon}\}_{\alpha\in A, x\in X, \varepsilon>0}$ (that is, open sets in this topology are arbitrary unions of finite intersections of sets of type $U_{\alpha,x,\varepsilon}$)

Openness in this topology means: for every x in an open set O, there exist $\alpha_1, \ldots, \alpha_n$ and $\varepsilon_1, \ldots, \varepsilon_n$ and $x_1, \ldots, x_n$ so that if $||y - x_j||_{\alpha_j} < \varepsilon_j$ for each j, then $y \in O$ (and x belongs to that same set)
i.e. $||x - x_j||_{\alpha_j} < \varepsilon_j$ for each j and if $||y - x_j||_{\alpha_j} < \varepsilon_j$ for each j, then $y \in O$.

$||x - x_j||_{\alpha_j} =: \delta_j \implies \delta_j < \varepsilon_j$.
Claim: if $||y - x||_{\alpha_j} < \varepsilon_j - \delta_j$ for each j, then

$$||y - x_j||_{\alpha_j} \leq ||y - x||_{\alpha_j} + ||x - x_j||_{\alpha_j} < \varepsilon_j \text{ for each } j.$$

Facts:

1. (Just proved) $O \subseteq X$ is open $\iff$ for all $x \in O$ there exist $\alpha_1, \ldots, \alpha_n \in A$, $\varepsilon_1, \ldots, \varepsilon_n > 0$ such that $||y - x||_{\alpha_j} < \varepsilon_j$ forces $y \in O$.

2. If $\{x_n\}_{n=1}^\infty$ is a sequence in X, then $x_n \to x$ in this topology $\iff$ $||x_n - x||_\alpha \to 0$ for all $\alpha \in A$ as $n \to \infty$ (i.e. $\lim_{n\to\infty} ||x_n - x||_\alpha = 0$ for all $\alpha \in A$)

Proof of (2):
Given $\alpha \in A$, $\varepsilon > 0$,

$$U_{\alpha,\varepsilon} := \{y : ||y - x||_\alpha < \varepsilon\}$$

is an open set.
If $x_n \to x$ as $n \to \infty$, topologically there exists N such that $x_n \in U_{\alpha,\varepsilon}$ for all $n > N$ i.e. $||x_n - x||_\alpha < \varepsilon$ for all $n > N$ (i.e. $\lim_{n\to\infty} ||x - x_n|| = 0$)
For the reverse direction, we assume $\lim_{n\to\infty} ||x - x_n||_\alpha = 0$ for all α.
Let U be any open set containing x. Then there are $\alpha_1, \ldots, \alpha_k$ and $\varepsilon_1, \ldots, \varepsilon_k$ such that $x \in U_{\alpha_1,\varepsilon_1} \cap \cdots \cap U_{\alpha_k,\varepsilon_k} \subseteq U \implies$ know $||x - x_n||_{\alpha_j} \to 0 \implies$ for $n > N_j$, $||x - x_n||_{\alpha_j} < \varepsilon_j$ for each j.

Then if $n > \max\{N_1, \dots, N_k\} =: N_*$, $||x - x_n||_{\alpha_j} < \varepsilon_j$ for each $j \implies x_n \in U_{\alpha_1,\varepsilon_1} \cap \cdots \cap U_{\alpha_k,\varepsilon_k} \implies x_n \in U$.

Weak topology:
If X is a Banach space, define the seminorms $||f|| := |\ell(f)|$ for $\ell \in X^*$. Then take the family of seminorms $\{|| \cdot ||_\ell\}_{\ell \in X^*}$.
Observe: by Hahn Banach, if $||x||_\ell = 0$ for all $\ell \in X^*$, $x = 0$.
Build a topology from these seminorms: e.g. we say $x_n \to x$ **weakly** as $n \to \infty$ when

$$\lim_{n\to\infty} \ell(x_n) = \ell(x)$$

for all $\ell \in X^*$.
Fact: $x_n \to x$ as $n \to \infty \implies x_n \to x$ weakly as $n \to \infty$.
Proof of fact: This is a consequence of continuity of linear functionals $\ell \in X^*$. Given $\ell \in X^*$, $\ell(x_n) \to \ell(x) \implies \lim_{n\to\infty} \ell(x_n) = \ell(x)$ for all $\ell \in X^*$, thus $x_n \to x$ weakly.

Example 4. *For now, let's use for free the fact that $(L^2(\mathbb{R}))^* = L^2(\mathbb{R})$ (we will prove this later).*
$f_n \to f$ weakly in $L^2(\mathbb{R})$ as $n \to \infty \iff \int f_n\overline{g} \to \int f\overline{g}$ for all $g \in L^2(\mathbb{R})$.

Example 5. $f_n := \chi_{[n,n+1]}$.
If $n \neq m$,

$$||f_n - f_m|| = \int |\chi_{[n,n+1]} - \chi_{[m,m+1]}|^2 dx = \int \chi_{[n,n+1]} + \chi_{[m,m+1]} = 2$$

So this does not converge in norm.

But $f_n \to 0$ weakly as $n \to \infty$, i.e.

$$\lim_{n\to\infty} \int \chi_{[n,n+1]} g = 0$$

for all $g \in L^2(\mathbb{R})$. Cauchy-Schwarz $\implies$

$$\left|\int \chi_{[n,n+1]} g\right| = \left|\int \chi_{[n,n+1]}\chi_{[n,n+1]} g\right| \le ||\chi_{[n,n+1]}||_2 ||\chi_{[n,n+1]} g||_2 = \left(\int \chi_{[n,n+1]} |g|^2\right)^{\frac{1}{2}}$$

Now $|g|^2 \in L^1$ by definition so $\chi_{[n,n+1]}|g|^2 \le |g|^2$ for all n and $\lim_{n\to\infty} \chi_{[n,n+1]}(x)|g(x)|^2 = 0$ for all $x \implies$ by Dominated Convergence Theorem,

$$\lim_{n\to\infty} \int \chi_{[n,n+1]} g = 0.$$

Example 6. *Let X = Banach space whose dual is $L^1(\mathbb{R})$.*
If $\chi_{[n,n+1]} \in X$, the same thing happens: $\chi_{[n,n+1]} \to 0$ weakly as $n \to \infty$ because we need to compute $\int \chi_{[n,n+1]} g dx$ for $g \in L^1(\mathbb{R})$ fixed. Again Dominated Convergence Theorem $\implies$ the integral goes to 0.

For maps between Banach spaces, there are even more topologies:

- Norm operator topology: $T_n \to T$ if $||T_n - T||_{X\to Y} \to 0$.
- Strong operator topology: $T_n \to T$ strongly if $T_n x \to Tx$ in norm Y for each $x \in X$
- Weak operator topology: $T_n \to T$ weakly if $T_n x \to Tx$ weakly for each $x \in X$

Note. *The unit ball of L^2 is not compact in the weak topology because $\{\chi_{[n,n+1]}\}$ live in this unit ball but have no convergent subsequence.*

Definition 17. *If $X = Y^*$ where Y is a Banach space, we define the weak* topology on X by taking seminorms $y \in Y$ i.e. $||f||_y = |f(y)|$*

If $X = X^{**}$, weak topology = weak* topology.
But if $X \neq X^{**}$, weak topology $\neq$ weak* topology.

Theorem 17 (Banach-Alaoglu)**.** *If X is a normed vector space, the closed unit ball in X^* is compact in the weak* topology.*

Note. *$X \hookrightarrow X^{**}$ always.*
X is a closed subspace.
*Norm $\implies$ Weak $\implies$ Weak**

Proof. (of Banach-Alaoglu)
Idea: embed closed unit ball into a larger space. Show the larger space is compact. Show the closed unit ball is a closed subset of the larger space.

Step 1: Embed B = closed unit ball into some compact space. Define

$$P = \prod_{x\in X} \overline{B_{||x||}(0)}$$

(When $x = 0$, $\overline{B_{||x||}(0)} := \{0\}$).
By the Tychonoff Product Theorem, P is compact in the product topology.
The product topology is generated by a basis of open sets of the form $\prod_{x\in X} U_x$ where U_x is open in $\overline{B_{||x||}(0)}$ for all x and $U_y = \overline{B_{||y||}(0)}$ for all but finitely many $y \in X$.

Embed: Define

$$\varphi : X^* \to P, \quad \varphi(f)|_x = f(x)$$

- $||f|| \leq 1 \implies |f(x)| \leq ||f||||x|| \leq ||x||$ so φ lands in P
- $\varphi(f) = \varphi(g) \iff f(x) = g(x)$ for all $x \in X \implies f = g$, so φ is injective

Check: φ respects topology, i.e. is a homeomorphism of B (with weak* topology) and $\varphi(B)$
Basis elements of the weak* topology look like

$$\bigcap_{i=1}^{n}\{g \in X^* \mid |g(x_i) - g_i(x_i)| < \varepsilon_i\}$$

= sets of functionals that on a finite number of points $x_1, \ldots, x_n$ are close to some fixed functional $g_1, \ldots, g_n$ (respectively).

Noting that $\varphi(g)|_{x_i} = g(x_i)$, these open sets map under φ to

$$\prod_{x \in X} U_x$$

where at the specific points $x_1, \ldots, x_n$, the set U_x is a small neighborhood (an open disk), and at all other points x, the set U_x is just the full range $B_{\|x\|}(0)$

That is,

$$U_x = \begin{cases} \{z \in K : |z - g_i(x)| < \varepsilon_i\} & \text{if } x \in \{x_1, \ldots, x_n\} \\ B_{||x||}(0) & \text{otherwise} \end{cases}$$

Proof of this: use the fact that $\varphi(g) = g(x_i)$ in coordinate direction x_i.

So φ preserves open sets. Then φ^{-1} is continuous, so
Step 2: Suffices to show $\varphi(B)$ is compact. $\varphi(B) \subseteq P$ and P is compact so want to show $\varphi(B)$ is closed i.e. $P \setminus \varphi(B)$ is open.

Claim:

$$\varphi(B) = \{\ell \in P \mid \ell|_{\lambda x+y} = \lambda \ell|_x + \ell|_y \text{ for all } \lambda \in \mathbb{C}, x, y \in X\}$$

Check: for any $g \in B$, $\varphi(g)|_{\lambda x+y} = g(\lambda x+y) = \lambda g(x)+g(y) = \lambda\varphi(g)|_x+\varphi(g)|_y$. So $\varphi(B)$ lies in the above set.
Conversely, if ℓ is in the set, then $\ell \in P$ and $\ell|_{\lambda x+y} = \lambda\ell|_x + \ell|_y$.
$g(x) := \ell|_x$. By construction, $g(\lambda x + y) = \lambda g(x) + g(y) \implies g$ is linear.
Also $g(x) \in \overline{B_{||x||}(0)} \implies |g(x)| \le ||x||$.
So g is linear and bounded of norm at most 1, meaning $g \in B$.

$P \setminus \varphi(B)$ = everything which is not linear. If $\ell \in P \setminus \varphi(B)$, there exists some $\lambda \in \mathbb{C}, x, y \in X$ such that $\ell|_{\lambda x+y} \neq \lambda\ell|_x + \ell|_y$.

Want to show: there exists an open set containing ℓ on which this same linearity failure happens.
Notice: $\lambda \neq 0$.
Define $\varepsilon := \ell|_{\lambda x+y} - \lambda\ell|_x - \ell|_y$. So $\varepsilon \neq 0$.

My open set:

$$U := \left\{ p \in P \mid |p|_{\lambda x+y} - \ell|_{\lambda x+y}| < \frac{|\varepsilon|}{3}, |p|_x - \ell|_x| < \frac{|\varepsilon|}{3|\lambda|}, \text{ and } p|_y - \ell|_y| < \frac{|\varepsilon|}{3} \right\}$$

Claim: for all $p \in U$, $p|_{\lambda x+y} \neq \lambda p|_x + p|_y$. Because:

$$|(p|_{\lambda x+y} - \lambda p|_x - p|_y)| \geq |(\ell|_{\lambda x+y} - \lambda \ell|_x - \ell|_y)| - \Big(|(p|_{\lambda x+y} - \ell|_{\lambda x+y})| + \lambda|(p|_x - \ell|_x)| + |(p|_y - \ell|_y)| \Big)$$

$$> |\epsilon| - \left(\frac{|\epsilon|}{3} + \frac{|\epsilon|}{3} + \frac{|\epsilon|}{3} \right) = 0$$

□

9 Hilbert Spaces

A **pre-Hilbert space** $\mathcal{H}$ (otherwise known as inner product space) is a vector space over $\mathbb{R}$ or $\mathbb{C}$ equipped with an inner product $\langle , \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ (or $\mathbb{R}$) so that

1. For each y, $x \mapsto \langle x, y \rangle$ is linear as a function of x
2. For all $x, y \in \mathcal{H}$, $\langle x, y \rangle = \overline{\langle y, x \rangle}$
   Note $y \mapsto \langle x, y \rangle$ is conjugate linear.
3. $\langle x, x \rangle \in \mathbb{R}_{\geq 0}$ and $\langle x, x \rangle = 0 \iff x = 0$.

Define $||x|| := \sqrt{\langle x, x \rangle}$.

Theorem 18 (Cauchy-Schwarz)**.**

$$|\langle x, y \rangle| \leq ||x|| ||y||$$

for all $x, y \in \mathcal{H}$.

Proof. First: For $\theta \in \mathbb{R}$,

$$\langle e^{i\theta} x, e^{i\theta} x \rangle = e^{i\theta} \langle x, e^{i\theta} x \rangle = e^{i\theta} e^{-i\theta} \langle x, x \rangle$$

So let θ be chosen such that $e^{i\theta} \langle x, y \rangle$ is real and non-negative.
Let's prove the inequality for $\langle x', y \rangle$ where $x' := e^{i\theta} x$.
Observe: $||x|| = ||x'||$ and $|\langle x, y \rangle| = |\langle x', y \rangle|$. So if the statement holds true for x', y then it holds true for x, y.
Takeaway: it suffices to assume $\langle x, y \rangle$ is real and nonnegative.
Now define for $t \in \mathbb{R}$,

$$f(t) = ||x+ty||^2 = \langle x+ty, x+ty \rangle = \langle x, x \rangle + t\langle y, x \rangle + t\langle x, y \rangle + t^2 \langle y, y \rangle = ||x||^2 + 2t\langle x, y \rangle + t^2 ||y||^2.$$

This is a quadratic, and by assumption $f(t) \geq 0$. Note: if $||y|| = 0, f(t) = ||x||^2 + 2t\langle x, y \rangle \geq 0 \implies \langle x, y \rangle = 0$ (otherwise the line is not parallel to the axis and crosses the line of vanishing).

So assume $||y|| \neq 0$.

$$t_0 = \frac{-\langle x, y\rangle}{||y||^2}$$

is a particular value of t that we will consider. Compute:

$$f(t_0) = ||x||^2 - 2\frac{\langle x, y\rangle^2}{||y||^2} + \frac{\langle x, y\rangle^2}{||y||^4}||y||^2 \geq 0.$$

But

$$f(t_0) = ||x||^2 - 2\frac{\langle x, y\rangle^2}{||y||^2} + \frac{\langle x, y\rangle^2}{||y||^4}||y||^2 = ||x||^2 - \frac{\langle x, y\rangle^2}{||y||^2} \implies$$
$$||x||^2||y||^2 \geq \langle x, y\rangle^2 \implies ||x||||y|| \geq |\langle x, y\rangle|.$$

□

Corollary 6. *$x \mapsto ||x||$ is a norm.*

Proof. Hard part: show $||x + y|| \leq ||x|| + ||y||$.

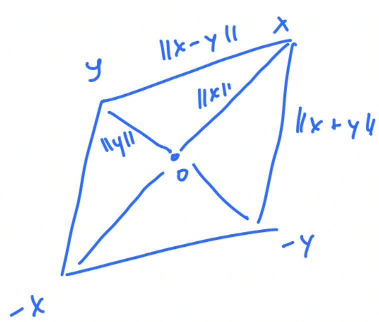
$$||x+y||^2 = \langle x+y, x+y\rangle = ||x||^2+\langle x, y\rangle+\langle y, x\rangle+||y||^2 = ||x||^2+2\mathrm{Re}\langle x, y\rangle+||y||^2 \leq$$
$$||x||^2 + 2|\langle x, y\rangle| + ||y||^2 \leq ||x||^2 + 2||x||||y|| + ||y||^2 = (||x|| + ||y||)^2.$$

So $||x + y|| \leq ||x|| + ||y||$. □

Definition 18. *A **Hilbert space** is a pre-Hilbert space which is complete in its norm topology.*

Facts:

1. $x, y \mapsto \langle x, y\rangle$ is continuous on $\mathcal{H} \times \mathcal{H}$
   $|\langle x, y\rangle - \langle x', y'\rangle| \leq |\langle x - x', y\rangle| + |\langle x', y - y'\rangle| \leq ||x - x'||||y|| + ||x'||||y - y'||$
2. Parallelogram law:
   $||x + y||^2 + ||x - y||^2 = 2(||x||^2 + ||y||^2)$



Corollary 7. *If K is a closed convex subset of a Hilbert space $\mathcal{H}$ and if $z \in \mathcal{H}$ then there exists a unique $z' \in K$ such that*

$$||z - z'|| = \inf_{z'' \in K} ||z - z''||$$

Proof. Let $\{z_n\}_{n=1}^{\infty} \subseteq K$ satisfy

$$\lim_{n \to \infty} ||z - z_n|| = \inf_{z'' \in K} ||z - z''||.$$

Claim: This sequence must be Cauchy.
Proof: Use the parallelogram law on $x = z - z_n, y = z - z_m$.

$$2(||z-z_m||^2+||z-z_n||^2) = ||z_n-z_m||^2+||2z-z_n-z_m||^2 = ||z_n-z_m||^2+4\left|\left|z - \frac{z_n + z_m}{2}\right|\right|^2$$

and $\frac{z_n+z_m}{2} \in K$ by convexity, so

$$4\left|\left|z - \frac{z_n + z_m}{2}\right|\right|^2 \geq 4\left(\inf_{z'' \in K} ||z - z''||\right)^2$$

Now let

$$\delta := \inf_{z'' \in K} ||z - z''||.$$

We know: $||z - z_m||^2 \to \delta^2$ and $||z - z_n||^2 \to \delta^2$. Then since the right hand side is at least $||z_n - z_m||^2 + 4\delta^2$,

$$||z_n - z_m||^2 \leq 2||z - z_n||^2 + 2||z - z_m||^2 - 4\delta^2$$

and thus there exists N such that whenever $n, m \geq N$, $||z - z_n||^2 < \delta^2 + \frac{\varepsilon}{2}$, $||z - z_m||^2 < \delta^2 + \frac{\varepsilon}{2}$, so $||z_n - z_m||^2 < \varepsilon$. So the sequence is Cauchy.

Then $z_n \to z'$ for some $z' \in K$, meaning $||z - z'|| = \delta$.

For uniqueness, if z'' attains the minimum also, then interleave your z_n sequence with the z'''s. Now the sequence is still Cauchy and has a subsequence with your limit and a subsequence with limit $z'' \implies$ they are equal, i.e. the minimizer is unique. $\square$

Note that this is not true in general: some Banach spaces admit no closest point and some admit multiple closest points.

From this,
If K is a closed convex subspace of $\mathcal{H}$, $x \in \mathcal{H}$, and $y \in K$ is the unique closest point in K to x, then
Claim: For this y, $x - y \perp K$, i.e. $\langle x - y, k \rangle = 0$ for all $k \in K$.

Proof. Given $k \in K$, define

$$f(t) := ||x - y - tk||^2 = ||x - y||^2 - 2\text{Re}\langle x - y, tk\rangle + t^2||k||^2,$$

assuming $t \in \mathbb{R}$. Observe:

$$\frac{d}{dt} f(t)|_{t=0} = -2\text{Re}\langle x - y, k\rangle.$$

Since $y + tk \in K$ for all t, $||(x - (y + tk))||^2 \geq ||x - y||^2$.
But: if $\frac{d}{dt} f(t)|_{t=0} \neq 0$ then $f(t)$ does not have a minimum at $t = 0$, contradicting y being the minimizer. So $\text{Re}\langle x - y, k\rangle = 0$ for all $k \in K$.

If $\mathcal{H}$ is complex, then $ik \in K$ so we can use the same argument to show $\text{Re}\langle x - y, ik\rangle = 0$. But $\text{Re}\langle x - y, ik\rangle = \text{Im}\langle x - y, k\rangle$. So we have shown that $\langle x - y, k\rangle = 0$.

So when $x \in \mathcal{H}$ and y is the unique element of K closest to x, we can write $x = y + (x - y)$ where $y \in K$ and $x - y \in K^\perp$.

Quick facts about $K^\perp$:

- $K^\perp = \{z \in \mathcal{H} : \langle z, k\rangle = 0 \text{ for all } k \in K\}$
- $K^\perp$ is subspace, since: $\langle z_1 + z_2, k\rangle = \langle z_1, k\rangle + \langle z_2, k\rangle = 0$.
- $K^\perp$ is in particular a closed subspace: If $z_n \to z$ and $\langle z_n, k\rangle = 0$ for all $k \in K$, $\langle z, k\rangle = \lim_{n\to\infty}\langle z_n, k\rangle = 0$.
- $K^\perp \cap K = \{0\}$.

So given any $x \in \mathcal{H}$, $x = y + z$ where $y \in K, z \in K^\perp$.
And this decomposition is unique: if $x = y' + z'$ where $y' \in K$ and $z' \in K^\perp$, $y + z = y' + z' \implies y' - y = z' - z \implies$ since the LHS is in K and the RHS is in $K^\perp$, $y' - y, z - z' \in K \cap K^\perp = \{0\}$. □

Summary of what we showed here:

Theorem 19. *If K is a closed subspace of a Hilbert Space $\mathcal{H}$, then $\mathcal{H} = K \oplus K^\perp$ (that is, each $x \in \mathcal{H}$ can be expressed uniquely as $x = y + z$ where $y \in K$ and $z \in K^\perp$), and y and z are the unique elements of K and $K^\perp$ whose distance to x is minimal.*

9.1 Riesz Representation Theorem

Consequently, the map $x \mapsto$ nearest point y of K must be linear:
Let $P_x = y, Q_x = x$, i.e. $x = P_x + Q_x$ where $P_x \in K$ and $Q_x \in K^\perp$.

$$x + x' = P_x + Q_x + P_{x'} + Q_{x'} = (P_x + P_{x'}) + (Q_x + Q_{x'})$$

where $P_x + P_{x'} \in K$ and $Q_x + Q_{x'} \in K'$. So by uniqueness, $P_{(x+x')} = P_x + P_{x'}$ and $Q_{(x+x')} = Q_x + Q_{x'}$. So P, Q must be linear.

Moreover, $\langle P_x, Q_x \rangle = 0 \implies ||x||^2 = ||P_x + Q_x||^2 = ||P_x||^2 + ||Q_x||^2 \implies ||P_x|| \leq ||x||$ and $||Q_x|| \leq ||x|| \implies P, Q$ are bounded linear operators (norm ≥ 1). They are called **orthogonal projections**.

Theorem 20 (Riesz Representation Theorem)**.** *Suppose $\ell \in \mathcal{H}^*$. There exists $y \in \mathcal{H}$ such that $\ell(x) = \langle x, y \rangle$ for all $x \in \mathcal{H}$ and the map $\ell \mapsto y$ is a conjugate-linear isometry.*

Proof. Let $\ell \in \mathcal{H}^*$ be a bounded linear functional. Thus $\ker \ell$ is a closed subspace (closed because ℓ is continuous). If $\ell = 0$, take $y = 0$ and we're done. Otherwise, if $\ell \neq 0$ then $\ker \ell \neq \mathcal{H}$ and so there exists $y_0 \in \mathcal{H}$ such that $\ell(y_0) \neq 0$. In particular, $y_0 \in (\ker \ell)^{\perp}$.

If y_1 also has $\ell(y_1) \neq 0$ then choose c so that $c\ell(y_0) = \ell(y_1)$:

$$\ell(y_1 - cy_0) = 0 \implies y_1 - cy_0 \in \ker \ell.$$

So if $y_1 \notin \ker \ell$, $y_1 = cy_0 + k$ for $k \in \ker \ell$.

Then if you pick any $y_1 \in (\ker \ell)^{\perp}$ then either:

- $y_1 \notin \ker \ell \implies \ell(y_1) \neq 0 \implies y_1 = cy_0 + k$ for some $k \in \ker \ell$.
  Here we'll have $k = 0$ since $y_0, y_1 \in (\ker \ell)^{\perp} \implies y_1 - cy_0 = k \in (\ker \ell)^{\perp}$ but also $k \in \ker \ell$, and $\ker \ell \cap (\ker \ell)^{\perp} = 0$.
- Or $y_1 = 0$.

This implies that any element in $(\ker \ell)^{\perp}$ is in the span of y_0, so $(\ker \ell)^{\perp}$ is 1-dimensional.

Now pick $0 \neq y \in (\ker \ell)^{\perp}$. Then $y \notin \ker \ell \implies \ell(y) \neq 0$.
Notice that since $y \neq 0$, $\langle y, y \rangle \neq 0$.

Now we want to find $c \in \mathbb{C}$ such that $\ell(x) = \langle x, cy \rangle$ for all $x \in \mathcal{H}$. To have this we would need $\ell(y) = \langle y, cy \rangle = \overline{c}||y||^2$.

So take

$$c := \frac{\overline{\ell(y)}}{||y||^2}, \quad \text{and define } Y := \frac{\overline{\ell(y)}}{||y||^2} y.$$

Observe: $\ell(y) = \langle y, Y \rangle$ because we picked Y to make this be true.
For any $x \in \mathcal{H}$, $x = k + c'y$ where $k \in \ker \ell$.

$$\ell(x) = \ell(k) + \ell(c'y) = 0 + c'\langle y, Y \rangle = \langle 0 + c'y, Y \rangle = \langle k + c'y, Y \rangle = \langle x, Y \rangle$$

because $\langle k, Y\rangle = 0$ due to $Y \in (\ker \ell)^{\perp}$.

Now observe:

$$||\ell|| = \sup_{x\in\mathcal{H}:||x||=1} |\ell(x)| = \sup_{x\in\mathcal{H}:||x||=1} |\langle x, Y\rangle \leq \sup_{x\in\mathcal{H}:||x||=1} ||x||||Y|| = ||Y||$$

by Cauchy-Schwarz. Moreover, choosing $x = \frac{Y}{||Y||}$ (assuming $Y \neq 0$),

$$\left\langle \frac{Y}{||Y||}, Y \right\rangle = \frac{||Y||^2}{||Y||} = ||Y||$$

so $\sup_{x\in\mathcal{H}:||x||=1} |\langle x, Y|\rangle \geq ||Y||$.

Thus $||\ell|| = ||Y||$. □

This establishes a lot of information, including the fact that $\mathcal{H}^{**} = \mathcal{H}$.

9.2 Bases

Given a collection of vectors $\{e_\alpha\}_{\alpha\in A}$ we say it is an **orthonormal set** when

$$\langle e_\alpha, e_\beta\rangle = \begin{cases} 1, & \alpha = \beta \\ 0, & \alpha \neq \beta \end{cases}$$

$\operatorname{span}\{e_\alpha\}_{\alpha\in A}$ = finite linear combinations of elements $\{e_\alpha\}_{\alpha\in A}$.
An orthonormal set is called an **orthonormal basis** when the span of the set is dense in $\mathcal{H}$.

The distinction is between countable and uncountable bases.

Example 7. *(Uncountable)*
$\ell^2(\mathbb{R}) = \{f : \mathbb{R} \to \mathbb{C} \mid \sum_{x\in\mathbb{R}} |f(x)|^2 < \infty\}$
(Note that $\sum_{x\in\mathbb{R}} |f(x)|^2 < \infty$ *requires* f *to be countably supported)*

$\langle f, g\rangle = \sum_{x\in\mathbb{R}} f(x)\overline{g(x)}$.

Let

$$e_x(y) = \begin{cases} 1, & x = y \\ 0, & x \neq y \end{cases}$$

$$||e_x||^2 = \sum_{y\in\mathbb{R}} |e_x(y)|^2 = 1$$

$$\langle e_x, e_{x'}\rangle = \sum_{y\in\mathbb{R}} e_x(y)\overline{e_{x'}(y)} = 0$$

because $e_x(y)\overline{e_{x'}(y)} = 0$ *for every* $y \in \mathbb{R}$.

For $f \in \ell^2(\mathbb{R})$, $f = \sum_{x\in\mathbb{R}} f_x e_x$.

Observe: if $\mathcal{H}$ has a countable orthonormal basis, then $\mathcal{H}$ must be separable. $\sum_{i=1}^{N} a_i e_i$ are dense in $\mathcal{H}$, and each such vector is a limit of linear combinations where a_i has a rational real and imaginary part; this establishes separability.

Now suppose you know $\mathcal{H}$ is separable. Then there exists $\{y_n\}_{n=1}^{\infty}$ a sequence of vectors that are dense in $\mathcal{H}$.
For each $d \in \mathbb{N}, \exists$ some i_d minimal such that $\dim \operatorname{span}\{y_{i_1}, \ldots, y_{i_d}\} = d$.
Take $y_{i_1}, \ldots, y_{i_d}$ and do Graham-Schmidt on these. We get $e_{i_1}, e_{i_2}, \ldots$ which is orthonormal with the same span.

9.3 Bessel's inequality and Parseval's identity

Bessel's inequality:
Suppose $\mathcal{H}$ is a Hilbert space with orthonormal set $\{e_i\}_{i=1}^{\infty}$. Then for any $x \in \mathcal{H}$,

$$||x||^2 \geq \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$$

Parseval's identity:
Suppose $\mathcal{H}$ is a separable Hilbert space with orthonormal basis $\{e_i\}_{i=1}^{\infty}$. Then

$$||x||^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$$

The proof uses:

$$||x - \sum_{i=1}^{N} c_i e_i||^2 = ||x - \sum_{i=1}^{N} \langle x_i, e_i \rangle e_i||^2 + \sum_{i=1}^{N} |c_i - \langle x_i, e_i \rangle|^2$$

and

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i.$$

The conditions for a Parseval frame are weaker than those for an orthonormal basis.

10 L^p Spaces

10.1 Defining L^p Spaces

Let $(X, \mathcal{M}, \mu)$ be a measure space. For $1 \leq p < \infty$ define L^p = measurable functions f modulo equivalence a.e. such that $\int |f|^p d\mu < \infty$.
Define $||f||_p := \left(\int |f|^p d\mu\right)^{\frac{1}{p}}$.

Given $p, p' \in [1, \infty]$, we say that they are **conjugate exponents** when

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

(By convention $\frac{1}{\infty} := 0$.)
Note: when μ is σ-finite, $(L^1(\mu))^* = L^\infty(\mu)$ but usually, $L^\infty(\mu)$ does not make sense.

Proposition 6 (Young's Inequality)**.** *Suppose p, p' are conjugate, both in $(1, \infty)$. For any $a, b \geq 0$,*

$$ab \leq \frac{1}{p}a^p + \frac{1}{p'}b^{p'}$$

and equality holds $\iff a^p = b^{p'}$.

Proof. $f(t) = e^t$ is strictly convex; i.e. for $\theta \in [0, 1]$,

$$f(\theta t_1 + (1-\theta)t_2) \leq \theta f(t_1) + (1-\theta)f(t_2)$$

with equality $\iff t_1 = t_2$. Let $t_1 = \ln(a^p)$ and $t_2 = \ln(b^{p'})$. Let $\theta = \frac{1}{p}$.

$$\exp(\frac{1}{p}\ln(a^p) + \frac{1}{p'}\ln(b^{p'})) \leq \frac{1}{p}\exp(\ln(a^p)) + \frac{1}{p'}\exp(\ln(b^{p'})) \implies ab \leq \frac{1}{p}a^p + \frac{1}{p'}b^{p'}.$$

□

Proposition 7 (Holder's Inequality)**.** *If $f \in L^p(\mu), g \in L^{p'}(\mu)$ and p, p' are conjugate exponents in $[1, \infty]$, then $fg \in L^1(\mu)$ and*

$$\left|\int fg d\mu\right| \leq ||f||_p ||g||_{p'}$$

with equality $\iff \alpha|f|^p = \beta|g|^{p'}$ a.e. for some nonzero constants α, β.

Proof. Assume $1 < p, p' < \infty$. By Young's inequality,

$$|fg| \leq \frac{1}{p}|f|^p + \frac{1}{p'}|g|^{p'}$$

The function on the right is integrable by assumption, so $|fg| \in L^1(\mu)$.

$|fg| \in L^1(\mu) \implies fg \in L^1(\mu)$ and $|\int fg| \leq \int |fg|$.

$$\int |fg| d\mu \leq \frac{1}{p}||f||_p^p + \frac{1}{p'}||g||_{p'}^{p'}$$

If $||f||_p = 0$ or $||g||_{p'} = 0$ then f (resp. g) is 0 μ a.e., in which case $fg = 0$ a.e $\implies \int |fg| = 0$.

Then assume $||f||_p$, $||g||_{p'} \neq 0$.
So we can normalize so that $||f||_p = 1$ and $||g||_{p'} = 1$. In this case, the RHS of the inequality agrees with the promised RHS, so $|\int fg| \leq ||f||_p ||g||_{p'}$ when $||f||_p = ||g||_{p'} = 1$.

So when $f, g \neq 0$,

$$\left|\int \frac{f}{||f||_p} \frac{g}{||g||_{p'}} d\mu\right| \leq 1.$$

Scale out the constants to get the result.

Now if $||f||_p = 1 = ||g||_{p'}$ and we have equality, i.e. $\left|\int fg d\mu\right| = ||f||_p ||g||_{p'} = 1$, then

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^{p'}}{p'}$$

but both fractions have the same integral, so

$$|f(x)g(x)| = \frac{|f(x)|^p}{p} + \frac{|g(x)|^{p'}}{p'}$$

for μ-a.e. x.

We know that Young's inequality is sharp when $|f(x)|^p = |g(x)|^{p'}$, so we have equality $\iff$

$$|f(x)|^p = |g(x)|^{p'} \text{ in magnitude a.e.}$$

i.e. f and g are powers of each other in magnitude.

In general, we have equality $\iff$ $|f|^p, |g|^{p'}$ are linearly dependent.

The case for $p = 1, p' = \infty$ is left as homework. $\square$

Proposition 8 (Minkowski's Inequality). *For all $f, g \in L^p(\mu)$ ($p \in [1, \infty]$)*

$$||f + g||_p \leq ||f||_p + ||g||_p.$$

In particular, $L^p(\mu)$ is a vector space.

Proof. Assume $1 \leq p < \infty$. Observe:

$$|f(x) + g(x)|^p \leq (|f(x)| + |g(x)|)^p$$

for all x. Now,

$$|f(x)| + |g(x)| \leq 2 \max\{|f(x)|, |g(x)|\} \implies$$

$$(|f(x)| + |g(x)|)^p \leq 2^p \max\{|f(x)|^p, |g(x)|^p\} \leq 2^p \left(|f(x)|^p + |g(x)|^p\right).$$

Integrate:

$$\int |f + g|^p d\mu \leq \int (|f| + |g|)^p d\mu \leq 2^p (||f||_p^p + ||g||_p^p) < \infty.$$

So $f+g \in L^p(\mu)$.

Now to prove our desired inequality, run a similar argument but enhance it:

$$(|f|+|g|)^p = |f|(|f|+|g|)^{p-1} + |g|(|f|+|g|)^{p-1}$$

Integrate:

$$\int (|f|+|g|)^p d\mu = \int |f|(|f|+|g|)^{p-1} d\mu + \int |g|(|f|+|g|)^{p-1} d\mu \implies$$

by Holder's inequality,

$$\int (|f|+|g|)^p d\mu \leq ||f||_p ||(|f|+|g|)^{p-1}||_{p'} + ||g||_p ||(|f|+|g|)^{p-1}||_{p'} \leq (||f||_p + ||g||_p) ||(|f|+|g|)^{p-1}||_{p'}.$$

If $p=1$ we're done just using $\int |f+g|^p d\mu \leq \int (|f|+|g|)^p d\mu$. Assume $p \in (1,\infty)$:

$$\frac{1}{p} + \frac{1}{p'} = 1 \implies p' = \frac{p}{p-1}.$$

Then

$$||(|f|+|g|)^{p-1}||_{p'} = \left(\int (|f|+|g|)^{(p-1)\cdot\frac{p}{p-1}} d\mu\right)^{\frac{p-1}{p}} = \left(\int (|f|+|g|)^p d\mu\right)^{\frac{p-1}{p}} = ||(|f|+|g|)||_p^{p-1}.$$

Conclude: $||(|f|+|g|)||_p^p \leq (||f||_p + ||g||_p) \cdot ||(|f|+|g|)||_p^{p-1}$.
Since we know that $||(|f|+|g|)||_p < \infty$, we can divide both sides of above by $||(|f|+|g|)||_p^{p-1}$ (unless $||(|f|+|g|)||_p = 0$, in which case both sides are 0). So we get

$$||f+g||_p \leq ||(|f|+|g|)||_p \leq ||f||_p + ||g||_p.$$

Note that in L^∞, Minkowski is almost a pointwise statement. The proof is different but not harder. We leave it as an exercise. □

Side comment:
Chebyshev's Inequality in L^p:
Let $f \in L^p(\mu)$, $1 \leq p < \infty$. Then for all $\lambda > 0$,

$$\mu(\{x \in X : |f(x)| > \lambda\}) \leq \frac{||f||_p^p}{\lambda^p}.$$

Proof. $\lambda \chi_{\{x:|f(x)|>\lambda\}} \leq |f(x)|$ for a.e. $x \in X$.
Raise both sides by a power of p and integrate:

$$\lambda^p \mu(\{x \in X : |f(x)| > \lambda\}) \leq ||f||_p^p.$$

□

So we now have the following properties of $||\cdot||_p$ for $1 \leq p < \infty$:

- $||cf||_p = |c|||f||_p$ for all $c \in \mathbb{C}, f \in L^p(\mu)$
- $||f+g||_p \leq ||f||_p + ||g||_p$ for all $f, g \in L^p(\mu)$
- $||f||_p \geq 0$ with equality $\iff f = 0$

Note that this last property necessitates using equivalence classes.
Now to show that $L^p(\mu)$ is a Banach space, we need to show that Cauchy sequences always converge in norm.

Note that for L^∞ this is easier: the L^∞ norm is just the sup norm but with a null set of exceptions allowed. So once we throw out the bad points, what is left is a sup norm, which is Cauchy in the uniform norm.

Proof of completeness for $1 \leq p < \infty$:
Let $\{f_n\}_{n=1}^\infty$ be a Cauchy sequence in $L^p(\mu)$. Want to show: there exists some $f \in L^p(\mu)$ such that $f_n \to f$ in norm.

(Warning: there exist Cauchy sequences of L^p functions which are pointwise convergent nowhere)

We will show that every Cauchy sequence has a subsequence which converges pointwise a.e.

We can always pick a subsequence $\{f_{n_k}\}$ such that $||f_{n_k} - f_{n_{k+1}}||_p < 2^{-k}$.

If we take such a subsequence and relabel, for ease of notation assume that $\{f_n\}_{n=1}^\infty$ has $||f_n - f_{n+1}||_p < 2^{-n}$ for each n.

Let

$$F_N := |f_1| + \sum_{n=1}^{N} |f_n - f_{n+1}|$$

where f_n is a chosen representative for each n. Then $F_N \in L^p(\mu)$ because

$$||F_N||_p \leq ||f_1||_p + \sum_{n=1}^{N} ||f_n - f_{n+1}||_p \leq ||f_1||_p + \sum_{n=1}^{N} 2^{-n} \leq ||f_1||_p + 1$$

So $F_N(x)$ is monotone increasing pointwise, and by Monotone Convergence

$$F(x) := \lim_{N\to\infty} F_N(x) \implies \int |F|^p d\mu = \int \lim_{N\to\infty} |F_N|^p = \lim_{N\to\infty} \int |F_N|^p < \infty.$$

So $F \in L^p$ and so $F(x) < \infty$ for μ-a.e. x.
Then for μ a.e. x,

$$|f_1(x)| + \sum_{n=1}^{\infty} |f_n(x) - f_{n+1}(x)| < \infty$$

$\implies f_1(x) + \sum_{n=1}^{\infty}(-f_n(x) + f_{n+1}(x))$ is an absolutely convergent series. But

$$f_1(x) + \sum_{n=1}^{N}(-f_n(x) + f_{n+1}(x)) = f_{N+1}(x)$$

so $\lim_{N\to\infty} f_N(x)$ exists for μ-a.e. $x \in X$. Let $f(x) := \lim_{N\to\infty} f_N(x)$.
f is measurable.
By construction $|f_N(x)| \leq F(x)$, so $|f_N(x)|^p \leq F(x)^p \in L^1$.
And $|f_N(x)|^p \to |f(x)|^p$ for μ-a.e. x. So by dominated convergence,

$$\int |f|^p d\mu = \lim_{N\to\infty} \int |f_N|^p d\mu < \infty \implies f \in L^p(\mu).$$

So our Cauchy sequence converges a.e. pointwise to f and $f \in L^p$.

Now need to show: $f_n \to f$ in the L^p norm.

$$|f_n(x) - f(x)| = \lim_{m\to\infty} |f_n(x) - f_m(x)| \text{ for } \mu\text{-a.e.} x$$

and

$$|f_n(x) - f_m(x)| \leq \sum_{k=n}^{m} |f_k(x) - f_{k+1}(x)|.$$

So

$$|f_n(x) - f(x)| \leq \sum_{k=n}^{\infty} |f_k(x) - f_{k+1}(x)| \leq F(x).$$

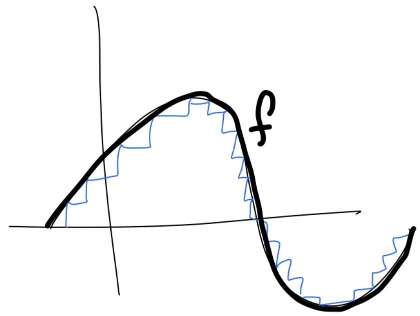
Also $\sum_{k=n}^{\infty} |f_k(x) - f_{k+1}(x)| \to 0$ for a.e. x as $n \to \infty$, by being the tail of a convergent series. Now $|f_n(x) - f(x)|^p \leq F(x)^p \in L^1(\mu)$ and $|f_n(x) - f(x)|^p \to 0$ for a.e. x. So dominated convergence $\implies$

$$\int |f_n(x) - f(x)|^p d\mu \to 0 \text{ as } n \to \infty \implies ||f_n - f||_p \to 0 \text{ as } n \to \infty.$$

So we have a Cauchy sequence with a convergent subsequence, hence the Cauchy sequence converges.

Basic facts about L^p spaces:

1. When $1 \leq p \leq \infty$, simple functions are dense in L^p, i.e. finite linear combinations of indicator functions of measurable sets *
   * If $1 \leq p < \infty$, we may assume these sets have finite measure.
   Because:
   $f = f_{\text{real}} + f_{\text{imag}}$
   $f \sim \sum_{j=0}^{\infty} j\varepsilon \chi_{\{j\varepsilon \leq f(x) \leq (j+1)\varepsilon\}}$. Do this on the positive part and on the negative part, and to get a simple function, truncate the sum.
   Alternatively, Folland's proof: There exists a sequence of simple functions



$\{f_n\}$ such that $|f_n| \leq |f|$ everywhere and $f_n(x) \to f(x)$ a.e.
For these functions, $\int |f_n - f|^p dx \to 0$ as $n \to \infty$ because

$$|f_n - f|^p \leq (|f_n| + |f|)^p \leq (2|f|)^p < \infty$$

and $\int (2|f|)^p < \infty$, so we can use dominated convergence.

2. These are lots of other dense classes of functions: in particular, on $\mathbb{R}^d$, C^∞ functions of compact support are dense in L^p when $1 \leq p < \infty$. The proof is very similar to the L^1 case.

3. Continuous functions are not dense in $L^\infty(\mathbb{R}^d)$. Recall that $||f||_\infty = \operatorname{ess\,sup}_{x\in X} |f(x)| = \inf\{a \geq 0 : \mu(x : |f(x)| > a) = 0\}$.
Note that for continuous functions, ess sup = sup.

10.2 Interpolation of L^p spaces

Think of the parameter p as something you measure.
Given f we can ask: $f \in L^p(\mu)$ for which p?
Roughly speaking, p measures integrability. If f is a function that is not bounded, then $|f|^p$ is in a sense worse than $|f|$ because when $|f|$ is large, for $p > 1$ $|f|^p$ is larger.
So if $\int |f|^p < \infty$, this gives us information about the set where f is large.
Recall:

$$\mu(\{x : |f(x)| > \lambda\}) \leq \frac{||f||_p^p}{\lambda^p}.$$

If you know that $||f||_1 < \infty$ and $||f||_3 < \infty$ then the inequality you get from $p = 3$ is sharper than the inequality with $p = 1$, when λ is really large.

Question: Given f, what is the set $F \subseteq [1, \infty]$ such that $f \in L^p \iff p \in F$?
Such sets F are always convex, i.e. the possibilities are:

- $F = \varnothing$

- F is an interval which may or may not contain its endpoints

We will build up to this result by talking about relationships between L^p spaces.

Theorem 21. *Suppose* $1 \leq p < q < r \leq \infty$. *Then*

1. $L^p \cap L^r \subseteq L^q$
2. $L^q \subseteq L^p + L^r$
3. $$\frac{1}{q} = \frac{1-\theta}{p} + \frac{\theta}{r} \implies ||f||_q \leq ||f||_p^{1-\theta} ||f||_r^{\theta}$$

   This result is called log-convexity because it says: if $F(t) = ||f||_{\frac{1}{t}}$ *then* $\ln F$ *is a convex function on* $[0,1]$

Proof. 1. Say $f \in L^p \cap L^r$. Assume $p < q < r$.
Claim: For all $t \geq 0$, $t^q \leq C(t^p + t^q)$ for all t ($C < \infty$ and does not depend on t). Proof:

$$\frac{t^q}{t^p + t^r}$$

is continuous on $(0, \infty)$ and

$$\lim_{t \to 0^+} \frac{t^q}{t^p + t^r} = 0$$

because when $t \in (0,1)$, $t^p + t^r \geq t^p$ so $\frac{t^q}{t^p+t^r} < t^{q-p}$.

Since $q - p > 0$, $\lim_{t \to 0} t^{q-p} = 0$.

On the other hand

$$\lim_{t \to \infty} \frac{t^q}{t^p + t^r} < \lim_{t \to \infty} \frac{t^q}{t^r} = \lim_{t \to \infty} t^{q-r}$$

and since $q - r < 0$, $\lim_{t \to \infty} t^{q-r} = 0$. Thus $\frac{t^q}{t^p+t^r}$ is bounded on $(0, \infty) \implies t^q \leq C(t^p + t^r)$ for all $t \in (0, \infty)$.

So pointwise,

$$|f(x)|^q \leq C(|f(x)|^p + |f(x)|^r) \text{ for all } x \in X.$$

Integrate:

$$\int |f|^q d\mu \leq C \left(\int |f|^p d\mu + \int |f|^r d\mu \right) < \infty.$$

So we're done if $r \neq \infty$. And if $r = \infty$, we have $f \in L^p$ for some finite p and $f \in L^\infty$ and $p < q < \infty$. Then

$$|f|^q = |f|^{q-p+p} = |f|^{q-p}|f|^p \leq ||f||_\infty^{q-p}|f|^p \text{ a.e.}$$

so since $f \in L^p$, if we integrate both sides of this we see that $f \in L^q$.

2. Let $p < q < r$. Pick $\alpha \in (0, \infty)$. Let

$$f^\alpha(x) := \begin{cases} f(x), & |f(x)| \geq \alpha \\ 0 & \text{otherwise} \end{cases}$$

and

$$f_\alpha(x) := \begin{cases} f(x), & |f(x)| < \alpha \\ 0 & \text{otherwise} \end{cases}$$

It is clear that $f = f^\alpha + f_\alpha$. Since $p \neq \infty$,

$$||f^\alpha||_p^p = \int |f^\alpha|^p = \int |f|^p \chi_{\{|f(x)| \geq \alpha\}} = \int |f|^q |f|^{p-q} \chi_{\{|f(x)| \geq \alpha\}}.$$

$p - q < 0$ so when $|f(x)| \geq \alpha$, $|f|^{p-q} = |f|^{-(q-p)} \leq \alpha^{-(q-p)}$. Then we get

$$\int |f|^q |f|^{-(q-p)} \chi_{\{|f(x)| \geq \alpha\}} \leq \int |f|^q \alpha^{-(q-p)} \chi_{\{|f(x)| \geq \alpha\}} = \alpha^{-(q-p)} ||f^\alpha||_q^q.$$

Note: $||f^\alpha||_q \leq ||f||_q$. Then

$$||f^\alpha||_p \leq \alpha^{-(\frac{q}{p}-1)} ||f||_q^{\frac{q}{p}} < \infty.$$

For f_α, we need to do cases.
$f_\alpha = f \cdot \chi_{\{|f| < \alpha\}}$.
$f_\alpha \in L^\infty$ and $||f_\alpha||_\infty \leq \alpha$, so if $r = \infty$ we're done.
If $r < \infty$, then

$$\int |f_\alpha|^r d\mu = \int |f_\alpha|^q |f_\alpha|^{r-q} d\mu = \int |f_\alpha|^q |f|^{r-q} \chi_{\{|f| < \alpha\}}$$

and since $r - q > 0 \implies |f(x)|^{r-q} \leq \alpha^{r-q}$ on $\{|f| < \alpha\}$,

$$\int |f_\alpha|^q |f|^{r-q} \chi_{\{|f| < \alpha\}} \leq \int |f_\alpha|^q \alpha^{r-q} = \alpha^{r-q} ||f_\alpha||_q^q < \infty.$$

Then here we get

$$||f_\alpha||_r \leq \alpha^{1-\frac{q}{r}} ||f_\alpha||_q^{\frac{q}{r}} \leq \alpha^{1-\frac{q}{r}} ||f||_q^{\frac{q}{r}} < \infty.$$

Optimally, pick α somewhere in the middle. If you pick $\alpha := ||f||_q$ then you get $||f^\alpha||_p \leq ||f||_q$ and $||f_\alpha||_p \leq ||f||_q$.

3. We leave the third result as a homework exercise.

$\square$

On $\mathbb{R}^n$ for example, there are no containment relationships:

- $L^p \not\subset L^q$ when $p \neq q$

- $L^p \cap L^r \subsetneq L^q$
- $L^q \subsetneq L^p + L^r$

Two situations in which we expect containment:

1. When $\mu(X) < \infty$
2. When μ is a counting measure

Fact 1: If μ is a finite measure then $L^p(\mu) \subseteq L^q(\mu)$ when $p > q$, i.e. all spaces are nested, L^1 is the largest space and L^∞ is the smallest. Proof:
Want to show $||f||_p < \infty \implies ||f||_q < \infty$ when $q < p$. Let $t = \frac{p}{q}$ and apply Holder's inequality:

$$||f||_q^q = \int |f|^q = \int |f|^q \cdot 1 \leq ||f^q||_t ||1||_{t'}.$$

Now $t > 1 \implies t' \neq \infty \implies$

$$||1||_{t'} = \left(\int |1|^{t'} d\mu \right)^{\frac{1}{t'}} = \mu(X)^{\frac{1}{t'}} < \infty$$

and

$$||f^q||_t = \left(\int |f|^{qt} \right)^{\frac{1}{t}} = \left(\int |f|^p \right)^{\frac{q}{p}} = ||f||_p^q < \infty.$$

Fact 2: Suppose μ is counting measure on X. Then $\ell^p \subseteq \ell^q$ when $p < q$ i.e. ℓ^1 is the smallest space and ℓ^∞ is the largest. Proof:
Want to show: $\ell^p \subseteq \ell^q$. So $||f||_p^p < \infty \implies ||f||_q^q < \infty$ where now $p < q$. If $q \neq \infty$, consider

$$\sum_{x \in X} |f(x)|^q = \sum_{x \in X} |f(x)|^p |f(x)|^{q-p}.$$

Observe that

$$|f(x)|^p \leq \left(\sum_{y \in X} |f(y)|^p \right) \implies |f(x)| \leq \left(\sum_{y \in X} |f(y)|^p \right)^{\frac{1}{p}} \implies |f(x)|^{q-p} \leq ||f||_p^{q-p}.$$

Thus

$$\sum_{x \in X} |f(x)|^q = \sum_{x \in X} |f(x)|^p |f(x)|^{q-p} \leq \sum_{x \in X} |f(x)|^p ||f||_p^{q-p} = ||f||_p^{q-p} ||f||_p^p = ||f||_p^q.$$

So $||f||_q \leq ||f||_p$.

For $q = \infty$, observe that $|f(x)| \leq ||f||_p$ for all x so $||f||_\infty \leq ||f||_p$.

10.3 Riesz-Thorin Interpolation

The idea of the Riesz-Thorin Interpolation Theorem:
Say $T : L^{p_0}(\mu) \to L^{q_0}(\nu)$ and $T : L^{p_1}(\mu) \to L^{q_1}(\nu)$. Then it is always the case that there is a whole family of interpolating inequalities:
If $\theta \in [0, 1]$ and

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{q_0} \text{ and } \frac{1}{q_\theta} = \frac{1-\theta}{p_1} + \frac{\theta}{q_1}$$

then $T : L^{p_\theta} \to L^{q_\theta}$.

Moreover,

$$||T||_{p_\theta \to q_\theta} \leq ||T||_{p_0 \to q_0}^{1-\theta} ||T||_{p_1 \to q_1}^{\theta}.$$

Some technical issues:
Oftentimes we want to show some operator T is bounded from $L^p(\mu)$ to $L^q(\nu)$ but we have trouble defining T on these spaces.

We say that T is **densely defined** on some Banach space X and bounded into Y if there exists a dense subset $S \subset X$ such that $T(x)$ is defined for $x \in S$, $T(x) \in Y$, and there exists some constant C such that $||T(x)||_Y \leq C||x||_X$ for all $x \in S$, where C is independent of x.

Theorem 22. *If X, Y are Banach spaces and T is a linear map defined on a dense subspace S of X and maps boundedly into Y, then there exists a unique bounded extension $T : X \to Y$ which agrees with T on S. The bound for the extension is no larger than the bound for the densely defined T.*

Proof. Let $S \subseteq X$ be a dense subspace. Let $x \in X \setminus S$. Then there exists a sequence $\{x_n\}$ so that $\{x_n\} \subseteq S$ and $x_n \to x$ in the topology of X.

So $\{x_n\}$ is Cauchy, so since T is bounded,

$$||Tx_n - Tx_m||_Y = ||T(x_n - x_m)||_Y \leq C||x_m - x_n||_X \to 0 \text{ for large } n, m.$$

So $\{Tx_n\}$ is Cauchy in Y so a limit exists.
Define $Tx := \lim_{n\to\infty} Tx_n$.

Observe: if $\{y_n\} \subseteq S$ and $y_n \to x$ as $n \to \infty$, then $||x_n - y_n|| \to 0$ as $n \to \infty$, so $||Tx_n - Ty_n|| \leq ||T|| ||x_n - y_n|| \to 0$ as $n \to \infty$.

Thus

$$\lim_{n\to\infty} Tx_n = \lim_{n\to\infty} Ty_n$$

so the extension does not depend on the sequence we chose. Then this implies that T is linear because if $x = \lim_{n\to\infty} x_n$ and $y = \lim_{n\to\infty} y_n$ then

$$\lambda x + y = \lim_{n\to\infty} (\lambda x_n + y_n) = \lambda \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$$

so

$$T(\lambda x+y)=\lim_{n\to\infty}T(\lambda x_n+y_n)=\lambda\lim_{n\to\infty}T(x_n)+\lim_{n\to\infty}T(y_n)=\lambda T(x)+T(y).$$

Also, T is bounded:
If $x_n\to x$ then $||x_n||\to||x||$ (by continuity of norm) so $||Tx_n||\to||Tx||$.

$||Tx_n||\le C||x_n||$ so, in the limit, $||Tx||\le C||x||$.

So T is linear and bounded. To show uniqueness, suppose we are given a bounded linear map $U:X\to Y$ so that U agrees with T on the dense subspace. Suppose $x\in X$, $x=\lim_{n\to\infty}x_n$, and $\{x_n\}\subseteq S$. Suppose $Tx_n=Ux_n$ for all n. Then letting $n\to\infty$, both T and U are bounded so continuous. Then

$$Tx=\lim_{n\to\infty}Tx_n=\lim_{n\to\infty}Ux_n=Ux.$$

□

Next let's think about a way to compute norms that is "linearized." This process is called dualization.

Given a measure space $(X,\mathcal{M},\mu)$ let $S^p(\mu)=\{\text{simple functions }\}\cap L^p(\mu)$.

This is a dense "non-closed" subspace of $L^p(\mu)$.

If $p\neq\infty$, $S^p(\mu)=$ finite linear combinations of indicator functions of finite measure.

Proposition 9. *Suppose $p,q\in[1,\infty]$. Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be measure spaces. If $q=\infty$, suppose ν is semifinite (i.e. every set with positive measure has a finite positive measure subset). Suppose $T:S^p(\mu)\to L^q(\nu)$ is linear. And suppose there exists $C<\infty$ such that*

$$\left|\int(Tf)g\,d\nu\right|\le C||f||_p||g||_{q'}$$

for all $f\in S^p(\mu),g\in S^{q'}(\nu)$ (where q,q' are dual). Then T extends uniquely to a bounded map $T:L^p(\mu)\to L^q(\nu)$ with norm at most C.

Proof. It suffices to show that

$$||Tf||_{L^q(\nu)}\le C||f||_{L^p(\mu)}$$

for all $f\in S^p(\mu)$. Once we show this, we can apply the previous theorem.
Issue: We know that $Tf\in L^q(\nu)$ for all $f\in S^p(\mu)$ but we do not know anything about the norm of Tf.
Idea: Fix $f\in S^p(\mu)$. Take a sequence of simple functions tending to $h:=|Tf|^{q-1}\frac{\overline{Tf}}{|Tf|}$ (assuming $q\neq\infty$).

We know that $|h| = |Tf|^{q-1}$. So there exists some sequence $\{g_n\}$ of simple functions that that $g_n \to h$ pointwise everywhere and $|g_n|$ increases to $|h|$ everywhere.

If $q \neq \infty$ and $q \neq 1$, $|g_n| \leq |h| \implies |g_n| \leq |Tf|^{q-1} \implies |g_n|^{q'} \leq |Tf|^q \in L^1$.
So $\int |g_n|^{q'} < \infty \implies g_n \in S^{q'}(\nu)$.
If $q = 1, |g_n| \leq 1 \implies g_n \in S^\infty(\nu)$. Then by assumption,

$$\left| \int Tfg_n d\nu \right| \leq C||f||_p||g_n||_{q'}.$$

Apply Dominated Convergence:
For the LHS,

$$\lim_{n\to\infty} \int (Tf)g_n d\nu = \int Tf|Tf|^{q-1} \cdot \frac{\overline{Tf}}{|Tf|} = \int |Tf|^q = ||Tf||_q^q.$$

For the RHS,

$$\left(\int |g_n|^{q'} \right)^{\frac{1}{q'}} \to \left(\int (|Tf|^{q-1})^{q'} \right)^{\frac{1}{q'}} = \left(\int |Tf|^q \right)^{\frac{1}{q'}}.$$

$1/q' = \frac{q-1}{q}$ so

$$\left(\int |Tf|^q \right)^{\frac{1}{q'}} = ||Tf||_q^{q-1}.$$

so

$$\lim_{n\to\infty} C||f||_p||g_n||_{q'} = C||f||_p||Tf||_q^{q-1}.$$

Then

$$||Tf||_q^q \leq C||f||_p||Tf||_q^{q-1}.$$

We know that $||Tf||_q$ is finite. If it is 0, the inequality we want is clear. If not, divide by $||Tf||_q^{q-1}$ and get the desired inequality $||Tf||_q \leq C||f||_p$.

Remaining case: $q = \infty$. Suppose we know that

$$Tf \in L^\infty(\nu) \text{ and } \int (Tf)gd\nu \leq C||f||_p||g||_1$$

for any f, g simple. We want to show: if Tf has large L^∞ norm, it is not possible for $\int (Tf)gd\nu$ to be small for all g.

Assume $||Tf||_\infty = A$. To prove the inequality, we need to show $A \leq C\|f\|_p$.
If $E_1 := \{x : |Tf(x)| > A - \varepsilon\}$, $\nu(E_1) > 0$. Since ν is semifinite, E_1 has a subset E_0 whose measure is positive and finite.

Let $\{h_n\}$ be a sequence of simple functions tending to $\frac{\overline{Tf}}{|Tf|}$ that are bounded in

magnitude by 1.
Let $g_n := h_n\chi_{E_0}$. Then $g_n \in L^1$ and is simple and tends to a function $g = \frac{\overline{Tf}}{|Tf|}\chi_{E_0}$ which has $|g| = \chi_{E_0}$ ν-a.e. We can apply Dominated Convergence, since $|(Tf)h_n\chi_{E_0}| \leq |Tf|\chi_{E_0} \in L^1(\nu)$, and we get

$$\lim_{n\to\infty}\int(Tf)g_n d\nu = \int(Tf)gd\nu$$

but

$$\int(Tf)gd\nu = \int Tf\frac{\overline{Tf}}{|Tf|}\chi_{E_0}d\nu = \int_{E_0}|Tf|d\nu \geq (A-\varepsilon)\nu(E_0).$$

So

$$(A-\varepsilon)\nu(E_0) \leq \lim_{n\to\infty}\left|\int(Tf)g_n d\nu\right| \leq C||f||_p \lim_{n\to\infty}||g_n||_1 = C||f||_p\nu(E_0).$$

Then since $\nu(E_0) \in (0,\infty)$, we can divide by it. We get

$$||Tf||_\infty - \varepsilon = A - \varepsilon \leq C||f||_p.$$

Since ε is arbitrary, $||Tf||_\infty \leq C||f||_p$. □

Lemma 6 (Three Lines Lemma, AKA Phragment Lindeloff Theorem)**.** *Suppose $P = \{z \in \mathbb{C} : 0 < Re(z) < 1\}$. Let F be a function defined on $\overline{P}$. Suppose F is holomorphic on P, continuous on $\overline{P}$, and bounded on $\overline{P}$. Then if:*

- $|F(0+it)| \leq M_0$ *for all* $t \in \mathbb{R}$ *and*
- $|F(1+it)| \leq M_1$ *for all* $t \in \mathbb{R}$

then $|F(\theta+it)| \leq M_0^{1-\theta}M_1^\theta$ *for all* $\theta \in (0,1), t \in \mathbb{R}$.

Proof. $\varphi(z) := M_0^{-(1-z)}M_1^{-z}$ is holomorphic everywhere (i.e. entire).
We can rewrite:

$$M_0^{-(1-z)} = e^{-(1-z)\ln M_0} = e^{-(1-\mathrm{Re}z)\ln M_0}\cdot e^{i(\mathrm{Im}z\ln M_0)}$$

and

$$M_1^{-z} = e^{-z\ln M_1} = e^{-(\mathrm{Re}z)\ln M_1}\cdot e^{-i(\mathrm{Im}z\ln M_1)}$$

so

$$\varphi(z) = M_0^{-(1-z)}M_1^{-z} = e^{-(1-\mathrm{Re}z)\ln M_0-(\mathrm{Re}z)\ln M_1}\cdot e^{i\cdot\mathrm{stuff}}.$$

Now note that $\varphi(z)$ is bounded on $\overline{P}$ since $0 \leq \mathrm{Re}z \leq 1$, and $e^{i\cdot\mathrm{stuff}}$ has magnitude 1. So $F(z)\varphi(z)$ is holomorphic and bounded on P and continuous on $\overline{P}$. And

$$|\varphi(0+it)| = e^{-(1-0)\ln M_0-0\ln M_1} = e^{-\ln M_0} = 1/M_0.$$
$$|\varphi(1+it)| = e^{-(1-1)\ln M_0-1\ln M_1} = e^{-\ln M_1} = 1/M_1.$$

So $|F(z)\varphi(z)| \leq 1$ on ∂P so also $F(z)\varphi(z)$ is bounded on $\overline{P}$.
So it suffices to prove that $|F(z)\varphi(z)| \leq 1$ everywhere, because

$$F(z) = (F(z)\varphi(z))\frac{1}{\varphi(z)} \text{ and } \left|\frac{1}{\varphi(z)}\right| \leq M_0^{1-\mathrm{Re} z} M_1^{\mathrm{Re} z}.$$

Assume without loss of generality that F is bounded and $|F| \leq 1$ on ∂P.
Let $F_\varepsilon(z) := F(z)e^{\varepsilon(z^2-1)}$ for $\varepsilon > 0$.
Let $z = x + iy$. Then $\varepsilon(z^2 - 1) = \varepsilon(x^2 - y^2 - 1) + 2i\varepsilon xy$.
So if $x \in [0, 1]$,

$$\left|e^{\varepsilon(z^2-1)}\right| = e^{\varepsilon(x^2-y^2-1)} \leq e^{-\varepsilon y^2} \to 0 \text{ uniformly in } x \text{ as } y \to \infty.$$

Thus $F_\varepsilon(z)$ is holomorphic on P, continuous on $\overline{P}$, ≤ 1 in magnitude on ∂P, and $\to 0$ uniformly in x as $y \to \pm\infty$.

Let $K = [0, 1] \times [-R, R]$. The Maximum Modulus Principle applies to K. (Maximum Modulus Principle states: if a function f is holomorphic on a bounded, connected domain and continuous on its closure, the maximum value of its modulus occurs on the boundary). For R sufficiently large,

$$\max_{z\in K} |F_\varepsilon(z)| = \max_{z\in\partial K} |F_\varepsilon(z)| \leq 1.$$

Take the union of the rectangles in the sequence

$$[0, 1] \times [-1, 1] \subset [0, 1] \times [-2, 2] \subset \cdots$$

The union is all of $\overline{P}$. So $|F_\varepsilon(z)| \leq 1$ on all of $\overline{P}$.
Thus $|F(z)e^{\varepsilon(z^2-1)}| \leq 1$ for all $z \in \overline{P}$, for all $\varepsilon > 0$.
Fix $z \in \overline{P}$ and let $\varepsilon \to 0$. Then $e^{\varepsilon(z^2-1)} \to 1$, meaning

$$|F(z)| \leq |e^{\varepsilon(z^2-1)}| \to 1 \text{ as } \varepsilon \to 0.$$

So $|F(z)| \leq 1$ everywhere. $\square$

Theorem 23 (Riesz-Thorin)**.** *Suppose $p_0, p_1, q_0, q_1 \in [1, \infty]$. Let T be a linear map defined on $S^{p_0}(\mu) \cap S^{p_1}(\mu)$. Suppose $||Tf||_{q_j} \leq M_j ||f||_{p_j}$ for $j = 0, 1$ for all $f \in S^{p_j}(\mu)$. If $q_0, q_1 = \infty$ assume ν is semifinite. Then if $\theta \in (0, 1)$, if*

$$\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1} \text{ and } \frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}$$

then T extends uniquely to a linear map $T : L^{p_\theta} \to L^{q_\theta}$ and

$$||Tf||_{q_\theta} \leq M_0^{1-\theta} M_1^\theta ||f||_{p_\theta}.$$

Proof. Suffices to show:

$$\left|\int Tfg\right| \leq M_0^{1-\theta}M_1^{\theta}||f||_{p_\theta}||g||_{q'_\theta}$$

for all $f \in S^{p_\theta}(\mu)$, $g \in S^{q'_\theta}(\nu)$.

f, g are both simple. Let's introduce a complex parameter z, so we will have f_z, g_z be simple functions depending on $z \in \mathbb{C}$. Want:

1. $z \mapsto \int (Tf_z)g_z d\nu$ holomorphic on $0 \leq \mathrm{Re}z \leq 1$.
2. $\int (Tf_z)g_z d\nu$ bounded on $\mathrm{Re}z = 0$ based entirely on the $L^{p_0} \to L^{q'_0}$ bounds.
3. $\int (Tf_z)g_z d\nu$ bounded on $\mathrm{Re}z = 1$ based entirely on the $L^{p_1} \to L^{q'_1}$ bounds.
4. Three lines lemma gives info about $L^{p_\theta} \to L^{q'_\theta}$ when $z = \theta$.

Assume

$$f = \sum_{j=1}^{N} |c_j|e^{i\varphi_j}\chi_{E_j}$$

where $\varphi_j \in \mathbb{R}$, $c_j \in \mathbb{R}, E_j$ pairwise disjoint.

Similarly

$$g = \sum_{k=1}^{N'} |d_k|e^{i\psi_k}\chi_{F_k}$$

where $\psi_k \in \mathbb{R}$, $d_k \in \mathbb{R}, F_k$ pairwise disjoint.

And assume f, g are normalized in $L^{p_\theta}, L^{q'_\theta}$ respectively to have norm 1. Given $r_0, r_1 \in [1, \infty]$, define

$$\alpha_r(z) := \frac{\frac{1-z}{r_0} + \frac{z}{r_1}}{\frac{1-\theta}{r_0} + \frac{\theta}{r_1}}$$

if $r_0, r_1 \neq \infty$. And $\alpha_r(z) := 1$ if $r_0, r_1 = \infty$.
Observations:

1. $\alpha_r(z)$ is holomorphic as a function of z.
2. When $\mathrm{Re}z = 0$, $\alpha_r(z) = \frac{r_\theta}{r_0} +$ imaginary stuff, where $\frac{1}{r_\theta} = \frac{1-\theta}{r_0} + \frac{\theta}{r_1}$.
3. When $\mathrm{Re}z = 1$, $\alpha_r(z) = \frac{r_\theta}{r_1} +$ imaginary stuff (r_θ is the same as in (2))
4. When $z = \theta$, $\alpha_r(z) = 1$.

Define

$$f_z := \sum_{j=1}^{N} |c_j|^{\alpha_p(z)}e^{i\varphi_j}\chi_{E_j}$$

$$g_z := \sum_{k=1}^{N'} |d_k|^{\alpha_{q'}(z)} e^{i\psi_k} \chi_{F_k}$$

Now $f_\theta = f$, $g_\theta = g$, and

$$|f_{0+it}| = |f|^{p_\theta/p_0}, \quad |f_{1+it}| = |f|^{p_\theta/p_1},$$

and similarly for g. Then when $z = 0+it$, f_z is normalized in $L^{p_0}(\mu)$ with norm 1. And when $z = 1+it$, f_z is normalized in $L^{p_1}(\mu)$ with norm 1. And similarly for g. Now

$$\int (Tf_z) g_z d\nu = \sum_{j=1}^{N} \sum_{k=1}^{N'} e^{i(\varphi_j+\psi_k)} \left(\int (T\chi_{E_j}) \chi_{F_k} d\nu \right) |c_j|^{\alpha_p(z)} |d_k|^{\alpha_{q'}(z)}.$$

Note that

- $e^{i(\varphi_j+\psi_k)} \left(\int (T\chi_{E_j})\chi_{F_k} d\nu\right)$ is constant with respect to z
- $|c_j|^{\alpha_p(z)} |d_k|^{\alpha_{q'}(z)}$ is holomorphic.

Let

$$F(z) := \int (Tf_z) g_z d\nu.$$

F is entire, and F is bounded on the strip $0 \leq \mathrm{Re} z \leq 1$.
Using Holder's,

$$|F(0+it)| = \left| \int (Tf_{0+it}) g_{0+it} d\nu \right| \leq ||Tf_{0+it}||_{q_0} ||g_{0+it}||_{q_0'} \leq M_0 ||f_z||_{p_0} ||g_z||_{q_0'} \leq M_0$$

for all $t \in \mathbb{R}$. And similarly,

$$|F(1+it)| \leq M_1 ||f_z||_{p_1} ||g_z||_{q_1'} \leq M_1$$

for all $t \in \mathbb{R}$.

So we can apply the Three Lines Lemma: $|F(\theta)| \leq M_0^{1-\theta} M_1^{\theta} \implies$

$$\left| \int (Tf) g d\nu \right| \leq M_0^{1-\theta} M_1^{\theta}$$

for any normalized $f \in S^{p_\theta}(\mu)$, $g \in S^{q_\theta'}(\nu)$, meaning

$$\left| \int (Tf) g d\nu \right| \leq M_0^{1-\theta} M_1^{\theta} ||f||_{p_\theta} ||g||_{q_\theta'}$$

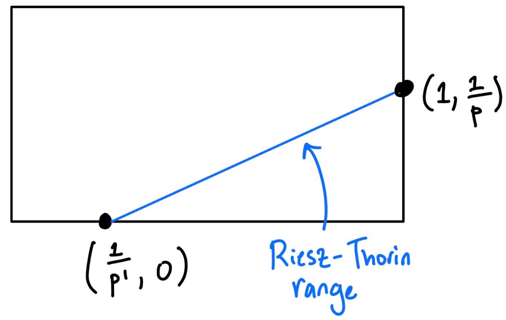
for all $f \in S^{p_\theta}(\mu)$, $g \in S^{q_\theta'}(\nu)$. So the proposition applies and we're done. $\square$

Fix $F \in L^p(\mathbb{R}^n)$. Let $Tg := f * g$. We know:

$$g \in L^1(\mathbb{R}^n) \implies ||Tg||_p \leq ||f||_p||g||_1$$

$$g \in L^{p'}(\mathbb{R}^n) \implies ||Tg||_\infty \leq ||f||_p||g||_{p'}.$$

$T : L^1 \to L^p$ corresponds to $(1, \frac{1}{p})$.
$T : L^{p'} \to L^\infty$ corresponds to $(\frac{1}{p}, 0)$.



Each time we apply Riesz-Thorin, we have to extend our restriction of T to some spaces $L^{p_\theta} \to L^{q_\theta}$, and when we do this, we have no say over how that happens.
Q: What if extending T in different $L^p \to L^q$ pairs gives a different extension?
Answer:

- $f \in L^p(\mu)$ is approximated by simple functions
- $f \in L^{p_1}(\mu) \cap L^{p_2}(\mu) \implies \exists$ a sequence of simple functions which converges to f in both norms at the same time $\implies Tf \in L^{q_0}$ has to equal $Tf \in L^{q_1}$.

10.4 Duality of L^p Spaces

Theorem 24. *Let $(X, \mathcal{M}, \mu)$ be a measure space. If $1 < p < \infty$ and p' is conjugate to p, then $(L^p(\mu))^* = L^{p'}(\mu)$ i.e. for every $\ell \in (L^p(\mu))^*, \exists g \in L^{p'}(\mu)$ so that $\ell(f) = \int gf d\mu$ (and $||\ell||_{(L^p)^*} = ||g||_{p'}$).*
If μ is σ-finite (sufficient but not necessary), then $(L^1(\mu))^* = L^\infty(\mu)$.

Idea for the proof:

1. Verify that if $g \in L^{p'}$, the map $\ell : L^p \to \mathbb{R}$, $f \mapsto \int gf d\mu$ is in $(L^p)^*$ and $||\ell||_{(L^p)^*} = ||g||_{p'}$.
2. Take linear functional ℓ, act on indicator functions of sets. This is a measure. Use Radon-Nikodym : the Radon-Nikodym derivative is the thing you will be integrating against.

Technicalities: need to restrict to finite set and glue together.

Proposition 10. *$L^{p'}(\mu)$ canonically embeds in $(L^p)^*$ with linear isometry, i.e. given $g \in L^{p'}(\mu)$, the map $\ell(f) := \int fg d\mu$ is in $(L^p(\mu))^*$ and $||\ell||_{(L^p)^*} = ||g||_{p'}$.*

Proof. By Holder's inequality, if $f \in L^p(\mu)$ and $g \in L^{p'}(\mu)$, $|\ell(f)| \leq ||f||_p||g||_{p'} \implies \ell \in (L^p(\mu))^*$ and $||\ell||_{(L^p)^*} \leq ||g||_{p'}$.

For the reverse inequality, we'll exhibit $f \in L^p$ so that $|\ell(f)| = ||f||_p||g||_{p'}$. Take $f := |g|^{p'-2} \cdot \overline{g}\chi_{\{|g|>0\}}$. We need to check: as long as $p' \neq \infty$,

$$\int fg = \int |g|^{p'-2} \cdot \overline{g} \cdot g\chi_{\{|g|>0\}} = \int |g|^{p'} < \infty.$$

So $||\ell||_{(L^p)^*} = ||g||_{p'}$.

Next, we manually check:

$$|f| = \left||g|^{p'-2} \cdot \overline{g}\right| = |g|^{p'-2} \cdot |g| = |g|^{p'-1} \implies$$

$$||f||_p = \left(\int |f|^p\right)^{\frac{1}{p}} = \left(\int (|g|^{p'-1})^p\right)^{\frac{1}{p}} = \left(\int |g|^{p'}\right)^{\frac{1}{p}} = ||g||_{p'}^{\frac{p'}{p}} = ||g||_{p'}^{p'-1}.$$

This is exactly what we need because

$$||\ell|| = \sup_{f\neq 0} \frac{|\ell(f)|}{||f||_p}$$

and for our given f,

$$\frac{|\ell(f)|}{||f||_p} = \frac{||g||_{p'}^{p'}}{||g||_{p'}^{p'-1}} = ||g||_{p'}$$

so conclude $||\ell||_{(L^p)^*} = ||g||_{p'}$.

When $p' = \infty$, $p = 1$, assume $g \in L^\infty$. We have $f \in L^1 \mapsto \int fgd\mu$ via ℓ.
Want to show: $||\ell||_{(L^p)^*} = ||g||_\infty$.
Already know: $||\ell||_{(L^p)^*} \leq ||g||_\infty$. Let $E \subseteq X$ be a set where $|g| > ||g||_\infty - \varepsilon$ for some $\varepsilon > 0$. σ-finite $\implies$ semifinite so E has a subset with finite positive measure. Choose $f := \chi_E \frac{\overline{g}}{|g|}$. So $fg = |g|\chi_E$. Then

$$\int fg = \int g\chi_E \frac{\overline{g}}{|g|} = \int \chi_E|g| \geq \mu(E) \cdot (||g||_\infty - \varepsilon) \implies$$

$$\mu(E)(||g||_\infty - \varepsilon) \leq \left|\int fgd\mu\right| = |\ell(f)| \leq ||\ell||_*||f||_1 = ||\ell||_*\mu(E).$$

Divide by $0 < \mu(E) < \infty$: get $||g||_\infty - \varepsilon \leq ||\ell||_*$. Then let $\varepsilon \to 0^+$. □

So either $L^{p'}$ is the dual space or a closed subset of it.

Next, we will show that when $1 \leq p < \infty$, σ-finiteness is essentially free and does not need to be assumed.

Proposition 11. *Suppose p, p' are dual exponents and $p > 1$. Let $\ell \in (L^p(\mu))^*$. For each measurable set E, let $\ell|_E$ be $f \mapsto \ell(f\chi_E)$. Then $\ell|_E \in (L^p(\mu))^*$ and if E_1, E_2 are disjoint,*

$$||\ell|_{E_1\cup E_2}||_{(L^p)^*} = \left(||\ell|_{E_1}||_{(L^p)^*}^{p'} + ||\ell|_{E_2}||_{(L^p)^*}^{p'}\right)^{1/p'}.$$

Proof.

$$||\ell|_E||_{(L^p)^*} = \sup_{f\in L^p(\mu),||f||\le 1} |\ell(f\chi_E)| \le \sup_{f\in L^p(\mu),||f||\le 1} ||\ell||_{(L^p)^*}||f\chi_E||_p \le$$

$$\sup_{f\in L^p(\mu),||f||\le 1} ||\ell||_{(L^p)^*}||f||_p = ||\ell||_{(L^p)^*}.$$

So

$$||\ell|_E||_{(L^p)^*} \le ||\ell||_{(L^p)^*},$$

meaning $\ell|_E \in (L^p)^*$ (since linear is clear).
Now

$$(\ell|_{E_1\cup E_2})(f) = \ell(f\chi_{E_1}) + \ell(f\chi_{E_2}).$$

We can normalize $f\chi_{E_1}$ by defining

$$f_1 := \frac{f\chi_{E_1}}{\|f\chi_{E_1}\|_p} \implies f\chi_{E_1} = f_1 \cdot \|f\chi_{E_1}\|_p$$

and similarly f_2 for $f\chi_{E_2}$. Then

$$\ell(f\chi_{E_1}) = \ell(f_1 \cdot \|f\chi_{E_1}\|_p) = \ell(f_1)\cdot \|f\chi_{E_1}\|_p$$

and

$$\ell(f\chi_{E_2}) = \ell(f_2 \cdot \|f\chi_{E_2}\|_p) = \ell(f_2)\cdot \|f\chi_{E_2}\|_p$$

and by triangle inequality,

$$|\ell|_{E_1\cup E_2}(f)| \le |\ell(f\chi_{E_1}) + \ell(f\chi_{E_2})| \le |\ell(f_1)|\|f\chi_{E_1}\|_p + |\ell(f_2)|\|f\chi_{E_2}\|_p.$$

and $|\ell(f_1)| \le ||\ell|_{E_1}||_{(L^p)^*}$ and $|\ell(f_2)| \le ||\ell|_{E_2}||_{(L^p)^*}$.

We can choose f_1 so that

$$|\ell(f_1)| - \varepsilon \ge ||\ell|_{E_1}||_{(L^p)^*}$$

and similarly f_2. So for a given f with $||f||_p \le 1$,

$$|\ell|_{E_1\cup E_2}(f)| = ab + cd$$

where $a \le ||\ell|_{E_1}||_{(L^p)^*}$ and $c \le ||\ell|_{E_2}||_{(L^p)^*}$. By choice of f, we may assume these are ε-close. Notice

$$b = ||f\chi_{E_1}||_p, \quad d = ||f\chi_{E_2}||_p \implies$$

$b^p + d^p \le ||f||_p^p \le 1$ (may assume equality if we want).

By the numerical version of Holders,

$$ab + cd \le (a^{p'} + c^{p'})^{\frac{1}{p'}}(b^p + d^p)^{\frac{1}{p}} \le \left(||\ell|_{E_1}||^{p'}_{(L^p)^*} + ||\ell|_{E_2}||^{p'}_{(L^p)^*}\right)^{\frac{1}{p'}} \cdot 1$$

so

$$|\ell|_{E_1\cup E_2}(f)| \le \left(||\ell|_{E_1}||^{p'}_{(L^p)^*} + ||\ell|_{E_2}||^{p'}_{(L^p)^*}\right)^{\frac{1}{p'}}.$$

But also the sup over $ab+cd$ is

$$\ge \left(||\ell|_{E_1}||^{p'}_{(L^p)^*} + ||\ell|_{E_2}||^{p'}_{(L^p)^*}\right)^{\frac{1}{p'}} \cdot 1$$

since we can choose things ε-close. So

$$||\ell|_{E_1\cup E_2}||^{p'} = ||\ell|_{E_1}||^{p'}_{(L^p)^*} + ||\ell|_{E_2}||^{p'}_{(L^p)^*}.$$

□

Summary of the above proof:
Take f on X, write f as $f = f\chi_{E_1} + f\chi_{E_2} + f\chi_{(E_1\cup E_2)^c}$.
Let f_1, f_2, f_3 be functions with $L^p(\mu)$ norm 1 so that $f_1 = 0$ on E_1^c, $f_2 = 0$ on E_2^c, and $f_3 = 0$ on $E_1 \cup E_2$.

Then $f = \alpha f_1 + \beta f_2 + \gamma f_3$ with $|\alpha|^p + |\beta|^p + |\gamma|^p = ||f||_p^p$.

$$||\ell|_{E_1\cup E_2}|| = \sup_{f_1,f_2,f_3,|\alpha|^p+|\beta|^p+|\gamma|^p=1} |\ell|_{E_1\cup E_2}(\alpha f_1 + \beta f_2 + \gamma f_3)| =$$

$$\sup_{f_1,f_2,f_3,|\alpha|^p+|\beta|^p+|\gamma|^p=1} |\alpha\ell|_{E_1}(f_1) + \beta\ell|_{E_2}(f_2)|$$

And then for fixed f_1, f_2, f_3, $\sup_{\alpha,\beta,\gamma}$ is the $\ell^{p'}$ norm so we get:

$$= \sup_{f_1,f_2,f_3 \text{ normalized, nonzero on right set}} \left(|\ell|_{E_1}(f_1)|^{p'} + |\ell|_{E_2}(f_2)|^{p'}\right)^{\frac{1}{p'}} = (||\ell|_{E_1}||^{p'} + ||\ell|_{E_2}||^{p'})^{\frac{1}{p'}}.$$

Proposition 12. *For $1 < p < \infty$, given $\ell \in (L^p(\mu))^*$, there exists a measurable set $E \subseteq X$ which is σ-finite such that $\ell = \ell|_E$.*

Proof. Let

$$\mathcal{M}^\sigma := \{E \in \mathcal{M} \mid E \text{ is a countable union of sets of finite measure}\}.$$

$\mathcal{M}^\sigma$ is closed under countable unions and is nonempty (contains $\varnothing$).
Consider $\sup_{E\in\mathcal{M}^\sigma} ||\ell|_E||$.

$\sup_{E\in\mathcal{M}^\sigma} ||\ell|_E||$ is well-defined and finite, since $||\ell|_E|| \le ||\ell||$.

Let $\{E_n\}_{n=1}^\infty$ be a sequence in $\mathcal{M}^\sigma$ such that $||\ell|_{E_n}|| \to \sup_{E\in\mathcal{M}^\sigma} ||\ell|_E||$.

$$E_\infty := \bigcup_{n=1}^\infty E_n \in \mathcal{M}^\sigma$$

must attain the sup, because:
$E_\infty = E_n \sqcup F_n$, $F_n = E_\infty \setminus E_n$,

$$||\ell|_{E_\infty}|| = (||\ell|_{E_n}||^{p'} + ||\ell|_{F_n}||^{p'})^{\frac{1}{p'}} \geq ||\ell|_{E_n}||$$

for all $n \implies$

$$||\ell|_{E_\infty}|| \geq \sup_n ||\ell|_{E_n}|| = \sup_{E\in\mathcal{M}^\sigma} ||\ell|_E||.$$

Since $E_\infty \in \mathcal{M}^\sigma$, $||\ell|_{E_\infty}|| = \sup_{E\in\mathcal{M}^\sigma} ||\ell.|_E||$.

Now suppose $\ell|_{E_\infty^c} \neq 0$, i.e. $\exists f \in L^p(\mu)$ so that $f = 0$ on E_∞ but $\ell(f) \neq 0$.
Let $E_{\text{bad}} = \{x : |f| > 0\} = \bigcup_{n=1}^\infty \{x : |f| > \frac{1}{n}\}$. Note that the sets in this union are finite, by Chebyshev.
Then $E_{\text{bad}} \cap E_\infty = \varnothing$ since $f = 0$ on E_∞, and

$$||\ell|_{E_\infty \cup E_{\text{bad}}}|| = (||\ell|_{E_\infty}||^{p'} + ||\ell|_{E_{\text{bad}}}||^{p'})^{\frac{1}{p'}} > ||\ell|_{E_\infty}||,$$

since $||\ell|_{E_{\text{bad}}}||^{p'}$ is nonzero by how we constructed E_{bad} (recall $\ell|_{E_\infty^c}(f) \neq 0 \implies \ell|_{E_{\text{bad}}} \neq 0$).

This contradicts maximality of $||\ell|_{E_\infty}||$, because $E_\infty \cup E_{\text{bad}} \in \mathcal{M}^\sigma$ too. So it must be that $\ell|_{E_\infty^c}$ is zero, meaning $\ell = \ell|_E$. □

Proof. (Of Duality Theorem)
So far we have shown

1. $L^{p'}$ canonically embeds in $(L^p)^*$ with linear isometry, so either $L^{p'}$ is the dual space or a closed subset of it.

2. We can always find a σ-finite set X_0 with $\ell|_{X_0} = \ell$, so we can eventually reduce to the σ-finite case, and that is as far as we need to reduce.

Now pick $p \in [1, \infty)$. Fix $E \in \mathcal{M}$ with finite measure. Pick $\ell \in (L^p(\mu))^*$. We need to show that ℓ is given by integration against something. Define

$$\nu(F) := \ell|_E(\chi_F)$$

for all $F \in \mathcal{M}$ (wlog may assume $F \subseteq E$). Want to show: ν is a complex measure.

Firstly, given a sequence of sets $\{F_j\}_{j=1}^\infty$ disjoint and measurable, want to show that $\nu(\bigcup_{j=1}^\infty F_j) = \sum_{j=1}^\infty \nu(F_j)$ with absolute convergence of the sum.

Define $F := \bigcup_{j=1}^\infty F_j$.
Then we want to show: $\chi_F \chi_E = \sum_{j=1}^\infty \chi_{F_j} \chi_E$ with convergence in L^p. Since ℓ is continuous on L^p, we would then have

$$\ell(\sum_{j=1}^\infty \chi_{F_j} \chi_E) = \sum_{j=1}^\infty \ell(\chi_F \chi_E),$$

which, by our definition of ν, would mean $\nu(\bigcup_{j=1}^{\infty} F_j) = \sum_{j=1}^{\infty} \nu(F_j)$.
Write:

$$\int \left| \chi_F \chi_E - \sum_{j=1}^{N} \chi_{F_j} \chi_E \right|^p d\mu = \int \chi_E \left| \chi_F - \sum_{j=1}^{N} \chi_{F_j} \right|^p d\mu = \int \chi_E \chi_{(F \setminus \bigcup_{j=1}^{N} F_j)} d\mu$$

and $\int \chi_E \chi_{(F \setminus \bigcup_{j=1}^{N} F_j)} d\mu \to 0$ as $N \to \infty$, by dominated convergence. Thus $\chi_F \chi_E = \sum_{j=1}^{\infty} \chi_{F_j} \chi_E$ with convergence in L^p. So

$$\nu(F) = \nu(\bigcup_{j=1}^{\infty} F_j) = \sum_{j=1}^{\infty} \nu(F_j).$$

Left to show: the series is absolutely convergent. Let $c_j \in \mathbb{C}$ such that $|c_j| = 1$ and $c_j \nu(E_j)$ is real and nonnegative.

$$\sum_{j=1}^{\infty} c_j \chi_E \chi_{F_j} \text{ also converges in } L^p,$$

since:

$$\int |\sum_{i=1}^{N} c_j \chi_E \chi_{F_j}|^p d\mu = \int \sum_{i=1}^{N} |c_j \chi_E \chi_{F_j}|^p d\mu = \sum_{i=1}^{N} \int |c_j \chi_E \chi_{F_j}|^p d\mu$$

and

$$\lim_{N\to\infty} \sum_{i=1}^{N} \int |c_j \chi_E \chi_{F_j}|^p d\mu = \int \chi_{E\cap F} d\mu = \mu(E \cap F).$$

So

$$\ell(\sum_{j=1}^{\infty} c_j \chi_E \chi_{F_j}) = \sum_{j=1}^{\infty} c_j \nu(F_j) = \sum_{j=1}^{\infty} |\nu(E_j)| < \infty.$$

So ν is a (finite) complex measure.

$\nu << \mu|_E$, because:
If $F \subseteq E$ such that $\mu(F) = 0$, then $\chi_F = 0$ in the L^p sense (since $\chi_F = 0$ a.e.) $\implies \nu(F) = \ell(0) = 0$. Since $\mu|_E$ is finite, LRN applies. Thus there exists g such that $\nu(E) = \int_F g d\mu$ and $g \in L^1(\mu)$.

So for all measurable sets F,

$$\ell(\chi_E \chi_F) = \int g \chi_E \chi_F d\mu,$$

meaning that for all simple functions $f \in L^p(\mu)$,

$$\ell(\chi_E f) = \int g \chi_E f d\mu.$$

Since ℓ is bounded,

$$\int g\chi_E f d\mu \leq ||\ell|| \cdot ||f||_p$$

for all f simple and in $L^p(\mu)$. We claim that this inequality forces g to be in $L^{p'}$. To prove it:

- Take a sequence of functions $\{f_n\}$ which are simple and converge appropriately to $|g|^{p'-2} \cdot \overline{g}$.
- By plugging these f_n into the inequality and using the Monotone Convergence Theorem, we force the integral $\int |g|^{p'} d\mu$ to be finite (in particular, $||g||_{p'} \leq ||\ell|_E||$).

Thus $g \in L^{p'}(\mu)$ and that $||g||_{p'} \leq ||\ell|_E||$. Conclude: when $\mu(E) < \infty$, $\ell|_E$ is integration against some $g \in L^{p'}(\mu)$, and $||\ell|_E|| = ||g||_{p'}$, where $g = 0$ on E^c.

Final step: generalize to non-finite case.
If $p = 1$, assume that X is σ-finite.
If $p > 1$, there exists $X_0 \subseteq X$ which is σ-finite so that $\ell|_{X_0^c} = 0$.
Write $X_0 = \sqcup_{j=1}^{\infty} E_j$ where the E_j's have finite measure. So

$$\ell = \ell|_{\sqcup_{j=1}^{\infty} E_j}.$$

We know: each E_j admits some $g_j \in L^{p'}(\mu)$ such that $\ell(\chi_{E_j} f) = \int g_j f d\mu$. Let

$$G_N := \sum_{j=1}^{N} g_j \in L^{p'}(\mu) \quad (L^{p'} \text{ is a vector space})$$

Notice that by our earlier proposition,

$$||G_N||_{p'} = ||\{g_j||_{p'}\}_{j=1}^N||_{\ell^{p'}(\mu)} = \left(\sum_{j=1}^{N} \|g_j\|_{p'}^{p'}\right)^{1/p'} = ||\ell|_{E_1 \cup \cdots \cup E_n}|| < \infty$$

independent of N.

Letting $G := \sum_{n=1}^{\infty} g_j = \lim_{N\to\infty} G_N$,

$$\int |G_N|^p d\mu = \int |G|^p \chi_{E_1 \cup \cdots \cup E_N}$$

and by Monotone Convergence,

$$\int |G|^p \chi_{E_1 \cup \cdots \cup E_N} \to \int |G|^p d\mu \text{ as } N \to \infty.$$

So $G \in L^{p'}(\mu)$. Then for all $f \in L^p(\mu)$,

$$\ell(f|_{E_1 \cup \cdots \cup E_n}) = \int G f|_{E_1 \cup \cdots \cup E_n}$$

and

$$\lim_{N\to\infty} f|_{E_1\cup\cdots\cup E_n} \to f\chi_{X_0}$$

in L^p, so by continuity of ℓ,

$$\ell(f) = \ell(\lim_{N\to\infty} f|_{E_1\cup\cdots\cup E_n}) = \lim_{N\to\infty} \ell(f|_{E_1\cup\cdots\cup E_n}) \implies$$

$$\ell(f) = \lim_{N\to\infty} \int_{E_1\cup\cdots\cup E_n} Gf\, d\mu = \int_{X_0} Gf d\mu.$$

Conclude

$$\int Gf\, d\mu = \ell(f).$$

□

Other facts about dual spaces:
If X = compact topological space,
$C(X)$ = continuous functions on X
$(C(X))^*$ = regular Radon measures on X
$\ell \in (C(X))^* \implies \ell(g) = \int g d\mu$ where μ is a regular Radon measure (= finite Borel measures with inner and outer regularity properties)

11 Schwartz Class

Uses of functional analysis:

- Fourier theory: what is the most general setting in which Fourier theory makes sense?
- Theory of tempered distributions: dual of tempered distributions, i.e. Schwartz functions

Definition 19. *We say f on $\mathbb{R}^d$ is a **Schwartz function** if $f \in C^\infty$ and $x^\alpha \partial^\beta f(x) \in L^\infty$ (for all α, β).*

Examples: C^∞ functions of compact support, Gaussians

Topology is generated by seminorms

$$||f||_{\alpha,\beta} = \sup_{x\in\mathbb{R}^d} |x^\alpha \partial^\beta f(x)|$$

α, β are multi-indices, i.e. $\alpha = (\alpha_1, \ldots, \alpha_d) \implies x^\alpha = x_1^{\alpha_1}\cdots x_d^{\alpha_d}$,
$\beta = (\beta_1, \ldots, \beta_d) \implies \partial^\beta f = \partial_{x_1}^{\beta_1}\cdots\partial_{x_d}^{\beta_d} f$.

If all seminorms are 0, then the $(\alpha, \beta) = (0, 0)$ seminorm is 0 so $f = 0$.

Recall:
$\varphi_n \to \varphi \iff ||\varphi - \varphi_n||_{\alpha,\beta} \to 0$ as $n \to \infty$ for all α, β.

$U \subseteq S$ is open $\iff$ for all $\varphi \in U$, there exist $\alpha_j, \beta_j, \varepsilon_j$ ($j = 1, \ldots, N$) so that whenever $||\psi - \varphi||_{\alpha_j, \beta_j} < \varepsilon_j$ for all $j = 1, \ldots, N$, $\psi \in U$.

This is a Hausdorff space (given two Schwartz functions there exists a seminorm that separates them).

Now given $\{|| \cdot ||_\eta\}_{\eta \in H}$ and $\{|| \cdot ||_\gamma\}_{\gamma \in G}$ two families of seminorms on the same vector space, we may ask: do they generate the same topology?

Proposition 13. *Suppose that for all $\eta \in H$, there exists $\alpha_1, \ldots, \alpha_N \in G$ and $C < \infty$ such that*

$$||f||_\eta \leq C \sum_{j=1}^{N} ||f||_{\gamma_j},$$

and also for all $\gamma \in G$, there exist $\eta_1, \ldots, \eta_N \in H$ and $D < \infty$ such that

$$||f||_\gamma \leq D \sum_{j=1}^{N} ||f||_{\eta_j}.$$

Then $\{|| \cdot ||_\eta\}_{\eta \in H}$, $\{|| \cdot ||\}_{\gamma \in G}$, generate the same topology.

Proof. Take $U \subseteq X$ which is open with respect to a family of seminorms. Show it's open in the others as well:

Suppose $U \subseteq X$ is open with respect to the topology generated by the H-seminorms. Then for all $\varphi \in U$ there exist $\eta_1, \ldots, \eta_k, \varepsilon_1, \ldots, \varepsilon_k$ such that if $||\psi - \varphi||_{\eta_i} < \varepsilon_i$ for all i, then $\psi \in U$. By assumption, for each $i = 1, \ldots, k$ there exists C_i so that

$$|| \cdot ||_{\eta_i} \leq C_i \sum_{j=1}^{N} || \cdot ||_{\gamma_j}.$$

Want to show: if $||\varphi - \psi||_{\gamma_j}$ is small enough, then $\psi \in U$, i.e. U is open in the topology generated by the G-seminorms. Choose δ so that $C_j N \delta < \varepsilon_j$ (in particular, take $\delta = \frac{1}{2} \min \frac{\varepsilon_j}{C_j N}$).

Then whenever $||\psi - \varphi||_{\gamma_j} < \delta$,

$$||\psi - \varphi||_{\eta_i} \leq C_i \cdot N \cdot \delta \leq C_i \cdot N \cdot \frac{1}{2} \frac{\varepsilon_i}{C_i N} = \frac{\varepsilon_i}{2} \implies \psi \in U.$$

Thus if U is open with respect to the topology generated by the H-seminorms, then U is also open with respect to the topology generated by the G-seminorms. The second half of the argument is exactly symmetric. $\square$

Example 8. *(Equivalent family for $\mathcal{S}$(= Schwartz class))*
$||\partial^\beta (x^\alpha \varphi)||_\infty$ *(α, β multi-indices)*
Would need to show:

$$||\partial^\beta (x^\alpha \varphi)||_\infty \leq C \sum_{\alpha', \beta' \text{ finite sum}} ||x^{\alpha'} \partial^{\beta'} \varphi||_\infty.$$

Also need to show:

$$||x^\alpha(\partial^\beta\varphi)||_\infty \leq C \sum_{\alpha',\beta' \text{ finite sum}} ||\partial^{\beta'}(x^{\alpha'}\varphi)||_\infty.$$

To show the first inequality, we could simply use product rule. To show the second inequality, we would need induction.

Proposition 14. *If $\{||\cdot||_a\}_{a\in A}$ and $\{||\cdot||'_b\}_{b\in B}$ are known to generate the same topology, then for all $a \in A$, there exists a finite a finite $N \in \mathbb{N}$, a finite subset $\{b_1, \dots, b_N\} \subseteq B$, and C_a such that*

$$||\varphi||_a \leq C_a \sum_{j=1}^N ||\varphi||'_{b_j},$$

and also for all $b \in B$, there exists a finite $N \in \mathbb{N}$, a finite subset $\{a_1, \dots, a_N\} \subseteq A$, and C_b such that

$$||\varphi||'_b \leq C_b \sum_{j=1}^N ||\varphi||_{a_j}$$

Proof. If the topologies are the same, then any seminorm $||\cdot||_a$ for $a \in A$ must be continuous with respect to the topology generated by $\{||\cdot||'_b\}$. in particular, we should have continuity at the origin. So for all $\varepsilon > 0$, there exists an open set U with respect to $\{||\cdot||'_b\}$ (containing 0) such that if $\psi \in U$, $||\psi||_a < \varepsilon$.

U open with respect to $\{||\cdot||'_b\}$ means that since $0 \in U$, there exist $b_1, \dots, b_N$, $\varepsilon_1, \dots, \varepsilon_N$ so that

$$\{\psi : ||\psi||'_{b_j} < \varepsilon_j \text{ for all } j = 1, \dots, N\} \subseteq U.$$

Thus if

$$\sum_{j=1}^N ||\psi||'_{b_j} < \min_j \varepsilon_j,$$

then

$$||\psi||'_{b_j} < \varepsilon_j \text{ for all } j \implies \psi \in U \implies ||\psi||_a < \varepsilon.$$

Claim: there exists a constant C such that $||\psi||_a \leq C\sum_{j=1}^N ||\psi||'_{b_j}$.
Proof of claim:
Let $N(\psi) = \sum_{i=1}^N ||\psi||'_{b_j}$. We know from above that if

$$N(\psi) < \min_j \varepsilon_j, \text{ then } ||\psi||_a < \varepsilon.$$

But also N and $||\cdot||_a$ are homogeneous $\implies$ given ψ, we can test the inequality on

$$\left(\frac{1}{2}\min_j \varepsilon_j\right)\frac{\psi}{N(\psi)} =: \tilde{\psi}.$$

Observe that

$$N(\tilde{\psi}) = \frac{1}{2}\min_j \varepsilon_j < \min_j \varepsilon_j,$$

so $||\tilde{\psi}||_a < \varepsilon$. But this means

$$\left|\left|\left(\frac{1}{2}\min_j \varepsilon_j\right)\frac{\psi}{N(\psi)}\right|\right|_a < \varepsilon \implies ||\psi||_a \le \frac{2\varepsilon N(\psi)}{(\min_j \varepsilon_j)},$$

i.e. $C = \frac{2\varepsilon}{\min_j \varepsilon_j}$ gives the promised inequality.
Symmetry gives the other direction. ☐

This tools gives us lots of ways to characterize $\mathcal{S}$.

Proposition 15. *β multi-index, $N \in \mathbb{N}_{\ge 0} \implies$*

$$\{||(1+|x|)^N \partial^\beta \psi||_{\beta,N}\}_{\beta,N}$$

is an equivalent family of seminorms generating the same topology on $\mathcal{S}$.

Proof. $(1+|x|)^N \le 2^N + (2|x|)^N$ because $1+|x| \le \max\{1+1, |x|+|x|\}$. So

$$(1+|x|)^N \le 2^N + (2|x|)^N \le 2^N + 2^N|x|^N = 2^N + 2^N\left(\sum_{j=1}^n x_j^2\right)^{\frac{N}{2}}$$

And then by convexity of $t \mapsto t^{N/2}$ for $N \ge 2$ and Jensen's Inequality applied to $x_1^2, \ldots, x_n^2$,

$$\left(\sum_{j=1}^n x_j^2\right)^{\frac{N}{2}} \le \sum_{j=1}^n (nx_j^2)^{\frac{N}{2}},$$

meaning

$$(1+|x|)^N \le 2^N\left(1 + \sum_{j=1}^n n^{\frac{N}{2}}|x_j|^N\right).$$

Then

$$||(1+|x|)^N \partial^\beta \psi||_\infty \le \left|\left|2^N\left(1+\sum_{j=1}^n n^{\frac{N}{2}}|x_j|^N\right)\partial^\beta\psi\right|\right|_\infty$$

and

$$\left|\left|2^N\left(1+\sum_{j=1}^n n^{\frac{N}{2}}|x_j|^N\right)\partial^\beta\psi\right|\right|_\infty \le 2^N\left(||\partial^\beta\psi||_\infty + n^{\frac{N}{2}}\sum_{j=1}^n |||x_j|^N\partial^\beta\psi||_\infty\right)$$

$$\le 2^N \max(1, n^{\frac{N}{2}})\left(||\partial^\beta\psi||_\infty + \sum_{j=1}^n ||x_j^N \partial^\beta\psi||_\infty\right).$$

For the reverse direction, we want to show $||x^\alpha \partial^\beta \psi||_\infty \leq C||(1+|x|)^N \partial^\beta \psi||_\infty$. We have

$$|x^\alpha| = |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n}$$

so $|x_j| \leq |x| \implies |x_j| \leq 1 + |x|$. Thus

$$|x|^\alpha = |x_1|^{\alpha_1} \cdots |x_n|^{\alpha_n} \leq (1+|x|)^{\alpha_1 + \cdots + \alpha_n} = (1+|x|)^{|\alpha|} \implies$$

$$||x^\alpha \partial^\beta \psi||_\infty \leq ||(1+|x|)^{|\alpha|} \partial^\beta \psi||_\infty.$$

$\square$

Note. *It's also okay to take seminorms $||x^\alpha \partial^\beta \psi||_{L^1(\mathbb{R}^n)}$.*
Easier direction to prove:
$||\psi||_{L^1} \leq C_n ||(1+|x|)^{n+1} \psi||_\infty$ *for some C_n.*

$$|\psi(x)| \leq \frac{1}{(1+|x|)^{n+1}} |(1+|x|)^{n+1} \psi(x)|$$

and $\frac{1}{(1+|x|)^{n+1}} \in L^1$ and $|(1+|x|)^{n+1}\psi(x)| \in L^\infty$, so apply Holder's.
The other direction is trickier.

So the equivalent families for $\mathcal{S}$ that we have found are:

- $\sup_{x \in \mathbb{R}^d} |x^\alpha \partial^\beta f(x)|$
- $\sup_{x \in \mathbb{R}^d} |(1+|x|)^\alpha \partial^\beta f(x)|$
- $||x^\alpha \partial^\beta f||_{L^1(\mathbb{R}^n)}$
- $||\partial^\beta (x^\alpha f)||_\infty$

Q: How do we know $\mathcal{S}$ is not simply a normed vector space? How do we know there does not exist a norm $||\cdot||$ on $\mathcal{S}$ which generates the same topology?
A: If there were, say $\{||\cdot||\}$ generates the topology. Then by our proposition, we would get

$$||\psi|| \leq C \sum_{j=1}^{N} ||\psi||_{\alpha_j, \beta_j}$$

and

$$||\psi||_{\alpha,\beta} \leq C_{\alpha,\beta} ||\psi|| \text{ for all } \alpha, \beta$$

so combining, for all α, β there exists $C_{\alpha,\beta}$ so that

$$||\psi||_{\alpha,\beta} \leq C_{\alpha,\beta} \sum_{j=1}^{N} ||\psi||_{\alpha_j, \beta_j}.$$

We can show that this family of inequalities could not possibly be true. Consider the following counterexample:

If $\psi_\lambda = e^{i\lambda x_1} e^{-\pi x^2}$, note that $\psi_\lambda \in \mathcal{S}$.
As $\lambda \to \infty$, ψ_λ becomes very large.
Differentiating,

$$\partial^\beta \psi_\lambda(x) = P_\beta(x, \lambda)\, e^{i\lambda x_1} e^{-\pi|x|^2},$$

where P_β is a polynomial in x and λ of degree $|\beta|$ in λ, thus

$$\|\psi_\lambda\|_{\alpha,\beta} = \sup_x |x^\alpha P_\beta(x,\lambda)|\, e^{-\pi|x|^2} \le C_{\alpha,\beta}(1+|\lambda|)^{|\beta|}.$$

By similar reasoning:

$$||(\partial x_1)^k \psi_\lambda||_\infty \ge C_k|\lambda|^k$$

as $\lambda \to \infty$ because

$$(\partial x_1)^k \psi_\lambda = (i\lambda)^k \psi_\lambda + O(\lambda^k).$$

Conclude: if $k > \max_j |\beta_j|$, then as $\lambda \to \infty$,

$$||(\partial x_1)^k \psi_\lambda||_\infty \text{ is not } \le C \sum_{j=1}^N ||\psi||_{\alpha_j,\beta_j}.$$

because LHS $\ge |\lambda|^k$ and RHS $\le 1 + |\lambda|^{\max \beta_j}$.

11.1 Maps on the Schwartz Class

Note. *For a linear operator $T : \mathcal{S} \to \mathcal{S}$, continuity is equivalent to continuity at the origin. T is continuous at 0 if for every target seminorm $\|\cdot\|_b$ and every $\epsilon > 0$, there exists an open neighborhood U in the domain such that $T(U) \subseteq \{\psi : \|\psi\|_b < \epsilon\}$.*

In the Schwartz topology, by definition, U open $\implies$ U must contain a set of the form

$$U \supseteq \{\varphi \in \mathcal{S} : \|\varphi\|_{a_j} < \delta \text{ for } j = 1, \ldots, N\}$$

In fact, $T : \mathcal{S} \to \mathcal{S}$ is continuous $\iff$ for every output seminorm $\|\cdot\|_b$, there exists $C > 0$ and $\{\|\cdot\|_{a_j}\}_{j=1}^N$ such that:

$$\|T\varphi\|_b \le C \sum_{j=1}^N \|\varphi\|_{a_j} \quad \forall \varphi \in \mathcal{S}$$

This works because:
If this inequality holds, we can satisfy the ϵ-condition for any output by choosing $\delta = \epsilon/C$.
Conversely, if T is continuous, $T^{-1}(\{\psi : \|\psi\|_b < 1\})$ is an open neighborhood of 0, hence it contains a set of the form $U = \{\varphi : \sum_{j=1}^N \|\varphi\|_{a_j} < \delta\}$. Thus, $\sum \|\varphi\|_{a_j} < \delta \implies \|T\varphi\|_b < 1$.

For any $\varphi \neq 0$, let $\lambda = \frac{\delta}{2\sum \|\varphi\|_{a_j}}$. Then $\sum \|\lambda\varphi\|_{a_j} = \delta/2 < \delta$, so $\|T(\lambda\varphi)\|_b < 1$. By linearity and homogeneity:

$$\lambda\|T\varphi\|_b < 1 \implies \|T\varphi\|_b < \frac{1}{\lambda} = \frac{2}{\delta}\sum_{j=1}^{N} \|\varphi\|_{a_j}$$

Choose $C = 2/\delta$.

Proposition 16. *Addition is continuous $\mathcal{S} \times \mathcal{S} \to \mathcal{S}$.*

Proof. $A(\varphi, \psi) := \varphi + \psi$.
We need the seminorms of $A(\varphi, \psi)$ to be controlled in terms of the seminorms in φ, ψ. We know that:

$$||A(\varphi, \psi)||_{\alpha,\beta} \leq ||\varphi||_{\alpha,\beta} + ||\psi||_{\alpha,\beta}$$

for all α, β. We will show that the preimage of an open set is an open set:

If $U \subseteq \mathcal{S}$ is open, pick $\varphi \in U$. Openness $\implies \exists \alpha_1, \beta_1, \varepsilon_1, \ldots, \alpha_N, \beta_N, \varepsilon_N$ so that if $||\psi - \tilde{\psi}||_{\alpha_j,\beta_j} < \varepsilon_j$ for all $j = 1, \ldots, N$, then $\tilde{\psi} \in U$.

Suppose $(\varphi_1, \varphi_2) \in \mathcal{S} \times \mathcal{S}$ has the property that $A(\varphi_1, \varphi_2) = \psi \in U$. Want to show: there exists an open set around $(\varphi_1, \varphi_2) \in \mathcal{S} \times \mathcal{S}$ which maps into U.

If we want $A(\varphi_1', \varphi_2') \in U$, suffices to show

$$||A(\varphi_1' - \varphi_1, \varphi_2' - \varphi_2)||_{\alpha_j,\beta_j} = ||A(\varphi_1', \varphi_2') - A(\varphi_1, \varphi_2)||_{\alpha_j,\beta_j} < \varepsilon_j$$

for all $j = 1, \ldots, N$. Our inequality from the start of the proof gives

$$||A(\varphi_1' - \varphi_1, \varphi_2' - \varphi_2)||_{\alpha_j,\beta_j} \leq ||\varphi_1' - \varphi_1||_{\alpha_j,\beta_j} + ||\varphi_2' - \varphi_2||_{\alpha_j,\beta_j}$$

so if we know

$$||\varphi_1' - \varphi_1||_{\alpha_j,\beta_j} < \frac{\varepsilon_j}{2} \text{ for all } j$$

and

$$||\varphi_2' - \varphi_2||_{\alpha_j,\beta_j} < \frac{\varepsilon_j}{2} \text{ for all } j$$

then

$$||A(\varphi_1' - \varphi_1, \varphi_2' - \varphi_2)||_{\alpha_j,\beta_j} < \varepsilon_j$$

for all $j = 1, \ldots, N \implies ||A(\varphi_1', \varphi_2')||_{\alpha_j,\beta_j} \in U$.

So the preimage of the open set U contains a product of open neighborhoods in $\mathcal{S} \times \mathcal{S}$, specifically the set

$$\{\varphi_1' : \|\varphi_1' - \varphi_1\|_{\alpha_j,\beta_j} < \frac{\varepsilon_j}{2} \text{ for all } j\} \times \{\varphi_2' : \|\varphi_2' - \varphi_2\|_{\alpha_j,\beta_j} < \frac{\varepsilon_j}{2} \text{ for all } j\}.$$

Since every point in the preimage is contained within such an open neighborhood, the preimage is open. □

Proposition 17. *$\tau_h\varphi(x) := \varphi(x-h) \implies \tau_h : \mathcal{S} \to \mathcal{S}$ continuously.*

Proof.

$$||\tau_h\varphi||_{\alpha,\beta} = ||x^\alpha\partial^\beta\varphi(x-h)||_{L^\infty(X)} = ||(x+h)\partial^\beta\varphi(x)||_{L^\infty(X)}.$$

$$(x+h)^\alpha = \sum_{\alpha'\leq\alpha} C_{\alpha,\alpha'}h^{\alpha-\alpha'} \implies |(x+h)^\alpha\partial^\beta\varphi(x)| \leq \sum_{\alpha'\leq\alpha} C_{\alpha,\alpha'}h^{\alpha-\alpha'}|x^{\alpha'}\partial^\beta\varphi(x)|$$

and $|x^{\alpha'}\partial^\beta\varphi(x)| \in L^\infty$ since the definition of $\mathcal{S} \implies$ this value is finite. Consequently

$$||\tau_h\varphi||_{\alpha,\beta} \leq C_{\alpha,\beta}\sum_{\alpha'\leq\alpha}||\varphi||_{\alpha',\beta}.$$

□

Proposition 18. *$\varphi \mapsto \partial^\beta\varphi$ is continuous on $\mathcal{S}$.*

Proof. $||x^\alpha\partial^{\beta'}(\partial^\beta\varphi)||_\infty = ||x^\alpha(\partial^{\beta+\beta'}\varphi)||_\infty$. □

Proposition 19. *Suppose $\psi \in C^\infty$ has at most polynomial growth, i.e. $|\partial^\beta\psi(x)| \leq C_\beta(1+|x|)^{N_\beta}$ for some N_β. Then $T\varphi := \varphi\psi \implies T$ is a continuous map $\mathcal{S} \to \mathcal{S}$.*

Proof.

$$x^\alpha\partial^\beta(\varphi\psi) = \sum_{\beta_1+\beta_2=\beta} C_{\alpha,\beta_1,\beta_2}x^\alpha\partial^{\beta_1}\varphi\partial^{\beta_2}\psi.$$

□

Summary:
Continuous maps of Schwartz functions:

- Addition
- Multiplication by C^∞ functions of polynomial growth
- Translation
- Differentiation

We say f is an **integrable function of rapid decrease** when

$$\int |f(x)||x^\alpha|dx < \infty$$

for all multi-indices α.

Proposition 20. *If f is an integrable function of rapid decrease, $\varphi \mapsto \varphi * f$ where $\varphi \in \mathcal{S}$ maps $\mathcal{S}$ continuously to itself.*

Proof. Idea: $x^\alpha \partial^\beta(\varphi * f) = x^\alpha(\partial^\beta \varphi * f)(x) = x^\alpha \int \partial^\beta \varphi(x-y) f(y) dy$.
Know: $x^\alpha \partial^\beta \varphi(x) \in L^1$ and $y^\alpha f(y) \in L^1$.
Since we can write $x = x - y + y$,

$$x^\alpha = (x-y+y)^\alpha = \prod_{j=1}^{n} \sum_{k=0}^{\alpha_j} \binom{\alpha_j}{k} (x_j - y_j)^{k_j} y_j^{\alpha_j - k_j}$$

where $\alpha! := \alpha_1! \cdots \alpha_n!$. And

$$\prod_{j=1}^{n} \frac{\alpha_j!}{k_j!(\alpha_j - k_j)!} = \frac{\alpha!}{k!(\alpha-k)!}$$

so we can write

$$\sum_{k_1=0}^{\alpha_1} \cdots \sum_{k_n=1}^{\alpha_n} \prod_{j=1}^{n} \binom{\alpha_j}{k_j} (x_j - y_j)^{k_j} y_j^{\alpha_j - k_j} = \sum_{k \leq \alpha} \binom{\alpha}{k} (x-y)^k y^{\alpha-k}$$

where $\binom{\alpha}{k} := \frac{\alpha!}{k!(\alpha-k)!}$. Thus:

$$x^\alpha \partial^\beta (\varphi * f) = \sum_{k \leq \alpha} \binom{\alpha}{k} \int (x-y)^k \partial^\beta \varphi(x-y) y^{\alpha-k} f(y) dy \implies$$

$$||\varphi * f||_{\alpha,\beta} = \sup_x |(x-y)^k \partial^\beta (\varphi * f)(x)| \leq \sum_{k \leq \alpha} \binom{\alpha}{k} ||\varphi||_{k,\beta} \int |y^{\alpha-k} f(y)| dy$$

where the integrand on the RHS is finite by assumption, since f integrable of rapid decrease $\implies \int |y^{\alpha-k} f(y)| dy < \infty$.
The output seminorm is bounded by a finite sum of input seminorms, so
$\varphi \mapsto \varphi * f$ is continuous. □

Corollary 8. *Convolutions of Schwartz functions are Schwartz functions.*

Note. *$\mathcal{S}$ is metrizable. Metric:*

$$d(\varphi, \psi) = \sum_{\alpha,\beta} \partial^{-|\alpha|} \partial^{-|\beta|} \min(||\varphi - \psi||_{\alpha,\beta}, 1).$$

This is one of many possible metrics.

Proposition 21. *C^∞ functions of compact support are dense in $\mathcal{S}$. In particular, every $\varphi \in \mathcal{S}$ is a limit in $\mathcal{S}$ of a sequence of C^∞ functions of compact support.*

Proof. Let $\eta \in C^\infty$ have compact support such that $\eta(0) = 1$. (We claim this always exists).
Given $\varphi \in \mathcal{S}$, let $\varphi_N(x) = \varphi(x)\eta(N^{-1}x)$. Observe:

- $\varphi_N \in C^\infty$

- φ_N is compactly supported
- $\varphi_N \in \mathcal{S}$

Note: if $f \in C_c^\infty(\mathbb{R}^n)$, $f \in \mathcal{S}$ because $|x^\alpha \partial^\beta \varphi(x)|$ is C^∞, compactly supported $\implies$ has a maximum at some point $\implies$ is bounded.
Need to show:
$||\varphi - \varphi_N||_{\alpha,\beta} \to 0$ as $N \to \infty$ for any α, β. We know

$$(\varphi - \varphi_N)(x) = \varphi(x)\left(1 - \eta(N^{-1}x)\right)$$

so

$$x^\alpha \partial^\beta(\varphi(x) - \varphi_N(x)) = \sum_{\beta_1+\beta_2=\beta} C_{\beta_1,\beta_2} x^\alpha \partial^{\beta_1}\varphi(x)\partial^{\beta_2}\left(1 - \eta(N^{-1}x)\right).$$

When $\beta_2 \neq 0$,

$$\partial^{\beta_2}(1 - \eta(N^{-1}x)) = \pm N^{-|\beta_2|}(\partial^{\beta_2}\eta)(N^{-1}x)$$

so $||\partial^{\beta_2}(1 - \eta(N^{-1}x))||_\infty \leq C_{\beta_2} \cdot N^{-|\beta_2|} \to 0$ as $N \to \infty$. Now

$$|x^\alpha \partial^\beta(\varphi(x) - \varphi_N(x))| = \sum_{\beta_1+\beta_2=\beta} C_{\beta_1,\beta_2} x^\alpha \partial^{\beta_1}\varphi(x)\partial^{\beta_2}\left(1 - \eta(N^{-1}x)\right) =$$

$$|(x^\alpha \partial^\beta \varphi(x))(1-\eta(N^{-1}x))| + \sum_{\beta_1+\beta_2=\beta, \beta_2\neq 0} C_{\beta_1,\beta_2} x^\alpha \partial^{\beta_1}\varphi(x)\partial^{\beta_2}\left(1 - \eta(N^{-1}x)\right) =$$

$$|(x^\alpha \partial^\beta \varphi(x))(1-\eta(N^{-1}x))| + \sum_{\beta_1+\beta_2=\beta, \beta_2\neq 0} C'_{\beta_1,\beta_2} ||\varphi||_{\alpha,\beta_1} N^{-|\beta_2|}.$$

By Mean Value Theorem, for some ξ,

$$|1 - \eta(N^{-1}x)| = |\eta(0) - \eta(N^{-1}x)| \leq |\nabla\eta(\xi)| \cdot |0 - N^{-1}x| \leq C_\xi \cdot N^{-1}|x|.$$

So

$$|(x^\alpha \partial^\beta \varphi(x))(1-\eta(N^{-1}x))| + \sum_{\beta_1+\beta_2=\beta, \beta_2\neq 0} C'_{\beta_1,\beta_2} ||\varphi||_{\alpha,\beta_1} N^{-|\beta_2|} \leq$$

$$|x^\alpha \partial^\beta \varphi(x)| \cdot C_\xi \cdot N^{-1}|x| + \sum_{\beta_1+\beta_2=\beta, \beta_2\neq 0} C'_{\beta_1,\beta_2} ||\varphi||_{\alpha,\beta_1} N^{-|\beta_2|},$$

meaning

$$||\varphi - \varphi_N||_{\alpha,\beta} \leq ||\varphi||_{\alpha+1,\beta} \cdot C_\xi \cdot N^{-1} + \sum_{\beta_1+\beta_2=\beta, \beta_2\neq 0} C'_{\beta_1,\beta_2} ||\varphi||_{\alpha,\beta_1} N^{-|\beta_2|}.$$

RHS $\to 0$ as $N \to \infty$ because it's a finite sum and each term goes to 0. $\square$

12 Fourier Theory

1. **Fourier series:** $L^p(\mathbb{T}^n) :=$ measurable functions f on $\mathbb{R}^n$ which are periodic of period 1 in all directions, i.e. $f(x+k) = f(x)$ for all $k \in \mathbb{Z}^n$.

$$||f||_{L^p(\mathbb{T}^n)} = \left(\int_{[0,1]^n} |f|^p\right)^{\frac{1}{p}} = \left(\int_{[-1/2,1/2]^n} |f|^p\right)^{\frac{1}{p}}.$$

All functions are identifiable with functions on $[0,1)^n$.
Define $e_\xi(x) := e^{2\pi i \xi \cdot x}$ where $x, \xi \in \mathbb{R}^n$.
Observe: if $\xi \in \mathbb{Z}^n$, then $e_\xi(x)$ is 1-periodic in every direction.
Q: If $f \in L^p(\mathbb{T}^n)$, can we write $f = \sum_{k \in \mathbb{Z}^d} c_k e_k(x)$?
On $\mathbb{T}^n$, $\{e_k\}_{k \in \mathbb{Z}^d}$ are orthonormal.

$$\int_{[0,1]^n} e_k(x)\overline{e_\ell(x)} = \int_{[0,1]^n} e^{2\pi i (k-\ell)\cdot x} dx = \begin{cases} 1, & k = \ell \\ 0, & k \neq \ell \end{cases}$$

because if m_j is a nonzero component of $k - \ell$,

$$\int_0^1 e^{2\pi i m_j x_j} dx_j = \left[\frac{e^{2\pi i m_j x_j}}{2\pi i m_j}\right]_0^1 = \frac{e^{2\pi i m_j} - e^0}{2\pi i m_j} = 0.$$

In $L^2(\mathbb{T}^n)$, $\{e_k\}_{k \in \mathbb{Z}^d}$ must be an orthonormal basis of its span.
Q1: What is the span?
Q2: What about $p \neq 2$?

2. **Fourier transform:** If $f \in L^1(\mathbb{R}^n)$,

$$\hat{f}(\xi) := \int_{\mathbb{R}^n} e^{-2\pi i x \cdot \xi} f(x) dx.$$

We think of this as a version of the Fourier series with periodicity $\to \infty$.
Q: What does f say about $\hat{f}$? Can properties be deduced from $\hat{f}$?

12.1 Basic properties of Fourier Transform

Lemma 7 (Riemann Lebesgue Lemma). *If $f \in L^1(\mathbb{R}^n)$ then $\hat{f}$ is continuous, vanishes at ∞, and $||\hat{f}||_\infty \leq ||f||_1$. If $f \in L^1(\mathbb{T}^n)$ then $\hat{f}$ vanishes at ∞ and $||\hat{f}||_{\ell^\infty(\mathbb{Z}^n)} \leq ||f||_{L^1(\mathbb{T}^n)}$.*

Proof. $||\hat{f}||_\infty \leq ||f||_1$ since

$$|\hat{f}(\xi)| \leq \int |f(x) e^{-2\pi i x \cdot \xi}| dx = \int |f|.$$

To show $\xi \mapsto \int e^{-2\pi i x \cdot \xi} f(x) dx$ is continuous, we use Dominated Convergence. If $\xi_n \to \xi$ then $e^{-2\pi i \xi_n \cdot x} \to e^{-2\pi i \xi \cdot x}$, and $|e^{2\pi i x \cdot \xi_n} f(x)| \leq |f(x)| \in L^1$.

So

$$\lim_{n\to\infty} \hat{f}(\xi_n) = \lim_{n\to\infty} \int_{\mathbb{R}^n} e^{-2\pi i x\cdot\xi_n} f(x)dx = \int_{\mathbb{R}^n} e^{-2\pi i x\cdot\xi} f(x)dx = \hat{f}(\xi)$$

and hence $\xi \mapsto \hat{f}(\xi)$ is continuous.

To see why $\hat{f} \to 0$ at ∞, there are two approaches. We could use the fact that Schwartz functions are dense in L^1 and prove $\hat{f} \to 0$ when f is Schwartz. Alternatively, we can use a clever exploitation of cancellation:

$$\hat{f}(\xi) := \int e^{2\pi i x\cdot\xi} f(x)dx \to 0 \text{ as } |\xi| \to \infty$$

Use: $\tau_h f \to f$ in the L^1 norm as $h \to 0$.
Using the substitution $x \mapsto x - \frac{\xi}{2|\xi|^2}$,

$$\hat{f}(\xi) = \int e^{2\pi i(x - \frac{\xi}{2|\xi|^2})\cdot\xi} f(x - \frac{\xi}{2|\xi|^2})dx$$

and $e^{2\pi i x\cdot\xi} e^{-2\pi i \frac{\xi\cdot\xi}{2|\xi|^2}} = e^{2\pi i x\cdot\xi} \cdot e^{-\pi i} = -e^{2\pi i x\cdot\xi}$ so

$$\hat{f}(\xi) = -\int e^{2\pi i x\cdot\xi} f(x - \frac{\xi}{2|\xi|^2})dx.$$

If we take the average of our two definitions for $\hat{f}(\xi)$,

$$\hat{f}(\xi) = \frac{1}{2}\int e^{2\pi i x\cdot\xi}\left[f(x) - f\left(x - \frac{\xi}{2|\xi|^2}\right)\right]dx \implies$$

$$|\hat{f}(\xi)| \le ||f - \tau_{(\frac{\xi}{2|\xi|^2})} f||_1.$$

As $|\xi| \to 0$, $|\frac{\xi}{2|\xi|^2}| = \frac{1}{2|\xi|} \to 0$. Because $\tau_h f \to f$ in L^1 as $h \to 0$,

$$||f - \tau_{(\frac{\xi}{2|\xi|^2})} f||_1 \to 0 \text{ as } |\xi| \to \infty.$$

So $|\hat{f}(\xi)| \to 0$ as $|\xi| \to \infty$.

$\square$

Consequence: continuous and vanishing at $\infty \implies$ uniformly continuous.

We denote by $C_0(\mathbb{R}^n)$ the set of continuous functions vanishing at ∞ (= the closure of the space of continuous functions of compact support in the uniform norm).

12.2 Symmetries of Fourier Transform

Recall:

$$\tau_h f(x) := f(x-h), \quad h \in \mathbb{R}^n$$
$$e_\xi(x) := e^{2\pi i x\cdot\xi} \quad (\xi \in \mathbb{R}^n)$$
$$\hat{f}(\xi) = \int e^{-2\pi i x\cdot\xi} f(x)dx$$

Given $A \in \mathbb{R}^{n\times n}$ invertible, define $D_A f(x) := f(A^{-1}x)$.

Proposition 22. *Let $f \in L^1(\mathbb{R}^n)$.*

1. $(\tau_h f)^\wedge = e_{-h}\hat{f}$
2. $(e_h f)^\wedge = \tau_h \hat{f}$
3. $(D_A f)^\wedge = |\det A| \cdot D_{(A^T)^{-1}}\hat{f}$

On $\mathbb{T}^n$: these 3 properties remain true, provided:

- *For τ_h, $h \in \mathbb{T}^n$.*
- *For e_h, $h \in \mathbb{Z}^n$ i.e. $\theta \in \mathbb{T}^n$, $k \in \mathbb{Z}^n$, $\theta\cdot k$ is well-defined modulo integers, and $e^{2\pi i(k\cdot\theta)}$ is well-defined.*
- *For A, entries are integers and* $\det A = \pm 1$.

Example 9.

$$(\tau_h f)^\wedge = \int e^{-2\pi i x\cdot\xi} f(x-h)dx = \int e^{-2\pi i (x+h)\cdot\xi} f(x)dx = e^{-2\pi i h\cdot\xi}\int e^{-2\pi i x\cdot\xi} f(x)dx = e_{-h}(\xi)\hat{f}(\xi).$$

and

$$(e_h f)^\wedge(\xi) = \int e^{-2\pi i x\cdot\xi} e^{2\pi i h\cdot x} f(x)dx = \int e^{-2\pi i x\cdot(\xi-h)} f(x)dx.$$

Others are similar.

Corollaries: There are two "infinitesimal" symmetries which relate differentiation and multiplication by monomials. They arise as limits of the above formulas.

For example, using $(\tau_h f)^\wedge = e_{-h}\hat{f}$, fix $e_j \in \mathbb{R}^n$ standard coordinate vector in direction j. Then:

$$\frac{f - \tau_{he_j} f}{h}(x) = \frac{f(x) - f(x-he_j)}{h} \to \frac{\partial f}{\partial x_j} \text{ as } h \to 0.$$

Suppose $f \in L^1(\mathbb{R}^n)$.

$$\left(\frac{f - \tau_{he_j} f}{h}\right)^\wedge = \int e^{-2\pi i x\cdot\xi}\left(\frac{f(x) - f(x-he_j)}{h}\right)dx.$$

Assume f has the property that

$$\frac{f(x)-f(x-he_j)}{h} \to \frac{\partial f}{\partial x_j}$$

weakly as $h \to 0$. Then

$$\int e^{-2\pi i x\cdot\xi}\left(\frac{f(x)-f(x-he_j)}{h}\right)dx \to \int e^{-2\pi i x\cdot\xi}\frac{\partial f}{\partial x_j}(x)dx = \left(\frac{\partial f}{\partial x_j}\right)^{\wedge}(\xi).$$

On the other hand,

$$\left(\frac{f-\tau_{he_j}f}{h}\right)^{\wedge}(\xi) \to \left(\frac{\partial f}{\partial x_j}\right)^{\wedge}(\xi)$$

and also

$$\left(\frac{f-\tau_{he_j}f}{h}\right)^{\wedge}(\xi) = \frac{\hat{f}(\xi)-e^{-2\pi i h e_j\cdot\xi}\hat{f}(\xi)}{h} = \hat{f}(\xi)\frac{1-e^{-2\pi i h e_j\cdot\xi}}{h} = \hat{f}(\xi)\left(\frac{1-e^{-2\pi i h\xi_j}}{h}\right)$$

and $\frac{1-e^{-2\pi i h\xi_j}}{h} \to 2\pi i\xi_j$ as $h \to \infty$ (recognize this as limit definition of derivative).

Conclude:

Proposition 23. *If $f \in L^1(\mathbb{R}^n)$ and $\frac{f-\tau_{he_j}f}{h} \to \frac{\partial f}{\partial x_j} \in L^1$ weakly, then*

$$\left(\frac{\partial f}{\partial x_j}\right)^{\wedge}(\xi) = 2\pi i\xi_j\hat{f}(\xi).$$

Note: If $f \in C^1(\mathbb{R}^n)$, then if $\frac{f-\tau_{he_j}f}{h}$ converges to something weakly, since $\frac{f-\tau_{he_j}f}{h} = \frac{f(x)-f(x-he_j)}{h} = \frac{\partial f}{\partial x_j}(x+h^*e_j)$ where $|h*|<|h|$, this value converges pointwise to the typical derivative.

Proposition 24. *If $f \in L^1(\mathbb{R}^n)$ and $x_jf \in L^1(\mathbb{R}^n)$ then $\frac{\partial\hat{f}}{\partial\xi_j}(\xi)$ exists at every point in the classical sense and satisfies*

$$\frac{\partial\hat{f}}{\partial\xi_j}(\xi) = (-2\pi i x_j f)^{\wedge}(\xi).$$

Proof.

$$\frac{\partial\hat{f}}{\partial\xi_j}(\xi) = \frac{\hat{f}(\xi+he_j)-\hat{f}(\xi)}{h} = \frac{1}{h}\int(e^{-2\pi i x\cdot(\xi+he_j)} - e^{-2\pi i x\cdot\xi})f(x)dx = \int e^{-2\pi i x\cdot\xi}\frac{e^{-2\pi i x_j h}-1}{h}f(x)dx.$$

Take the pointwise limit as $h \to 0$:

1. $\frac{e^{-2\pi i x_j h}-1}{h} \to -2\pi i x_j$ as $h \to 0$ (recognize limit definition of derivative)

2. To use DCT, need uniform bound on

$$\left|\frac{1}{h}\left(e^{-2\pi i x_j \cdot h}-1\right)\right| = \left|\frac{1}{h}\int_0^h \frac{\partial}{\partial t}(e^{-2\pi i x_j \cdot t})dt\right| = \left|\frac{1}{h}\int_0^h -2\pi i x_j e^{-2\pi i x_j \cdot t}dt\right| \leq$$

$$|2\pi x|\left|\frac{1}{h}\int_0^h e^{-2\pi i x_j \cdot t}dt\right| \leq |2\pi x|\frac{1}{h}\int_0^h |e^{-2\pi i x_j \cdot t}|dt \leq |2\pi x| \cdot \frac{1}{h} \cdot h \leq |2\pi x|.$$

So

$$\left|\frac{e^{-2\pi i x_j \cdot h}-1}{h} f(x)\right| \leq 2\pi|x||f(x)|$$

and $|x||f(x)|$ is assumed to be in L^1, so we can use it for DCT.

Then

$$\frac{\partial \hat{f}}{\partial \xi_j}(\xi) = \int e^{-2\pi i x \cdot \xi}\frac{e^{-2\pi i x_j h}-1}{h} f(x)dx \to \int e^{-2\pi i x \cdot \xi}(-2\pi i x_j f(x))dx = (-2\pi i x_j f)\hat{}(\xi).$$

□

Corollary 9. *If $f \in C^k(\mathbb{R}^n)$ and $\partial^\alpha f \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$, then*

$$(\partial^\alpha f)^\wedge(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$$

for all $|\alpha| \leq k$.

Proof. If $f \in C^1$ and $f, \partial_j \in L^1$, then

$$\frac{f(x)-f(x-he_j)}{h} \to \partial_j f$$

weakly as $h \to 0$. We would like to show it converges in L^1.

Integrate the derivative again: $\phi(x,t) := f(x-the_j), \quad t \in \mathbb{R}$.

$$\phi(x,1)-\phi(x,0) = f(x-he_j)-f(x) \implies \phi(x,1)-\phi(x,0) = \int_0^1 \frac{\partial}{\partial t}\phi(x,t)dt = \int_0^1 -h\frac{\partial f}{\partial x_j}(x-the_j)dt.$$

Conclusion:

$$\frac{f(x)-f(x-he_j)}{h} = \int_0^1 \frac{\partial f}{\partial x_j}(x-the_j)dt.$$

We want to show the LHS converges weakly. The term in the RHS is the classical partial derivative, which we assumed is in L^1.

$$\left\|\frac{f(x)-f(x-he_j)}{h} - \frac{\partial f}{\partial x_j}\right\|_1 = \left\|\int_0^1 \left(\frac{\partial f}{\partial x_j}(x-the_j) - \frac{\partial f}{\partial x_j}(x)\right)dt\right\|_1 = \left\|\int_0^1 \tau_{the_j}\frac{\partial f}{\partial x_j} - \frac{\partial f}{\partial x_j}dt\right\|_1$$

and then by Minkowski,

$$\left|\left|\int_0^1 \tau_{the_j}\frac{\partial f}{\partial x_j} - \frac{\partial f}{\partial x_j} dt\right|\right|_1 \leq \int_0^1 \left|\left|\tau_{the_j}\frac{\partial f}{\partial x_j} - \frac{\partial f}{\partial x_j}\right|\right|_1 dt$$

and

$$\left|\left|\tau_{the_j}\frac{\partial f}{\partial x_j} - \frac{\partial f}{\partial x_j}\right|\right|_1 \to 0 \text{ as } h \to 0.$$

Now if $f \in C^1(\mathbb{R}^n)$ and $f, \frac{\partial f}{\partial x_j} \in L^1(\mathbb{R}^n)$, then difference quotients converge in the L^1 norm, so also converge weakly.

So we are allowed to replace the symbol ∂f with the limit of fractions inside our Fourier integral, thus the result holds. □

Corollary 10. *The Fourier transform maps $\mathcal{S}$ to itself continuously.*

Proof. $f \in \mathcal{S} \implies \partial^\alpha(x^\beta f) \in L^1(\mathbb{R}^n)$, continuous, etc.
The result above $\implies$

$$(\partial^\alpha(x^\beta f))^\wedge(\xi) = (2\pi i\xi)^\alpha (x^\beta f)^\wedge(\xi) = (2\pi i\xi)^\alpha (\frac{i\partial}{2\pi})^\beta \hat{f}(\xi).$$

Note that the L^∞ norm of the RHS is a Schwartz norm of $\hat{f}$. So

$$||\hat{f}||_{\alpha,\beta} \leq \pm(2\pi)^{|\beta|-|\alpha|}||\partial^\alpha(x^\beta f)||_1$$

and we claim that this is controlled by a Schwartz seminorm too.

$$|f(x)| \leq ||f||_\infty \implies$$

$$|x|^{n+1}|f(x)| \leq C\sum ||x^\beta f||_\infty \implies$$

$$(1+|x|^{n+1})|f(x)| \leq C \cdot (\text{Schwartz seminorms}),$$

so

$$|f(x)| \leq C\frac{(\text{Schwartz seminorms})}{1+|x|^{n+1}} \implies$$

$$||f||_1 \leq C\int \frac{dx}{1+|x|^{n+1}} \cdot \text{Schwartz seminorms} < \infty.$$

Conclude: $\hat{f} \in \mathcal{S}$ and $f \mapsto \hat{f}$ is continuous in the Schwartz topology. □

Proposition 25. *If $f, g \in L^1$, then $(f * g)^\wedge(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.*

Proof.

$$(f*g)^\wedge(\xi) = \int e^{-2\pi i x\cdot\xi}\int f(x-y)g(y)dydx.$$

Use Fubini:

$$e^{-2\pi i x\cdot\xi}f(x-y)g(y) \in L^1(\mathbb{R}^n)$$

so

$$(f{*}g)^\wedge(\xi) = \int\int e^{-2\pi i x\cdot\xi} f(x{-}y)g(y)dxdy = \int g(y)\left(\int e^{-2\pi i x\cdot\xi} f(x-y)dx\right)dy =$$

$$\int g(y)e^{-2\pi i y\ \cdot\xi}\left(\int f(x)e^{-2\pi i x\cdot\xi}dx\right)dy = \hat{f}(\xi)\hat{g}(\xi).$$

□

Proposition 26 (Multiplication Formula)**.** *If $f, g \in L^1$, then*

$$\int \hat{f}(\xi)g(\xi)d\xi = \int f(x)\hat{g}(x)dx.$$

To prove this: substitute the definition and apply Fubini.

Summary: for Fourier transform,

- translation = modulation (multiplication by complex exponential)
- differentiation = multiplication by monomials
- $(f * g)^\wedge = \hat{f}\hat{g}$
- $\int \hat{f}g = \int f\hat{g}, \quad f, g \in L^1(\mathbb{R}^n)$ (Multiplication Formula)

Note: By the multiplication formula, $\hat{f}$ interpreted as a linear functional on L^1 is equivalent to f as a linear functional on L^∞ acting on $\hat{g}$. i.e. $\hat{f}(g) = f(\hat{g})$.

We will later generalize this: If D = tempered distribution, $\varphi \in \mathcal{S}$, we'll define $\hat{D}(\varphi) := D(\hat{\varphi})$: allows us to define $\hat{D}$ of any tempered distribution and agrees with the usual definition when $D \in L^1(\mathbb{R}^n)$.

12.3 Inverting the Fourier Transform

Idea: if we let

$$f(x) = \int e^{2\pi i x\cdot\xi}\hat{f}(\xi)d\xi$$

(so we have a + in the exponential instead of −), this could work as an inverse to the Fourier transform, but makes no formal sense in most cases because $\hat{f} \notin L^1(\mathbb{R}^n)$ in general.

Proposition 27. *Let $f(x) = e^{-\pi|x|^2}$ (on $\mathbb{R}^n$). Then $\hat{f}(\xi) = e^{-\pi|\xi|^2}$.*

Proof.

$$\hat{f}(\xi) = \int e^{-2\pi i x\cdot\xi}e^{-\pi|x|^2}dx = \int e^{-\pi(|x|^2+2ix\cdot\xi)}dx = \int e^{-\pi((x+i\xi)(x+i\xi)+|\xi|^2)}dx =$$

$$e^{-\pi|\xi|^2}\int e^{-\pi(x+i\xi)(x+i\xi)}dx = e^{-\pi|\xi|^2}\prod_{j=1}^{n}\int e^{-\pi(x_j+i\xi_j)^2}dx_j.$$

First step: prove $\xi_j \mapsto \int e^{-\pi(x_j+i\xi_j)^2}dx_j$ is constant. Not a proof: change of variables $x_j \mapsto x_j - i\xi_j$ (not allowed in multivariable context)
Approach #1: Take $\xi_j \mapsto \int e^{-\pi(x_j+i\xi_j)^2}dx_j$, differentiate with respect to ξ_j, show that what you get equals 0.

$$\frac{\partial}{\partial\xi}e^{-\pi(x+i\xi)^2} = i\frac{\partial}{\partial x}e^{-\pi(x+i\xi)^2}$$

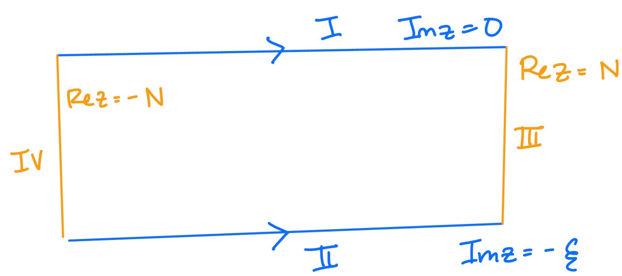
and

$$\frac{\partial}{\partial\xi}\int e^{-\pi(x+i\xi)^2}dx = i\int\frac{\partial}{\partial x}e^{-\pi(x+i\xi)^2}dx = 0$$

since

$$\left[-2\pi i e^{-\pi(x+i\xi)^2}\right]_{-\infty}^{\infty} = 0 - 0 = 0.$$

Approach #2: Treat as complex contour integral. Let $f(z) = e^{-\pi(z+i\xi)^2}$.



Notice that

$$\int_{(II)} f(z)dz = \int_{-\infty}^{\infty} e^{\pi x^2}dx = \text{ independent of } \xi.$$

Want to show: $\int_{(I)} f(z)dz = \int_{(II)} f(z)dz$ because this will show that the integral over path (I) is also independent of ξ.

If $R = [-N,N]\times[-\xi,0], \xi > 0$, then $\int_{\partial R} f dz = 0$ because f has no poles (Cauchy's Theorem).
Then it suffices to show

$$\int_{(III)} f(z)dz, \int_{(IV)} f(z)dz \to 0$$

as $N\to\infty$. If $z = N + iy$,

$$(z+i\xi)^2 = (N + i(y+\xi))^2 = N^2 + 2i(y+\xi)N - (y+\xi)^2$$

so

$$e^{-\pi(N+iy+i\xi)^2} = e^{-\pi N^2} e^{-2\pi i(y+i\xi)\cdot N} e^{\pi(y+\xi)^2}$$

where on the RHS, the first factor is small, the second factor is magnitude 1, and the third factor is bounded as a function of y.

Second step: Since we just proved that the integral does not depend on ξ, we can plug in 0 for ξ in the integral:

$$f = e^{-\pi|x|^2} \implies \hat{f}(\xi) = e^{-\pi|\xi|^2} \int e^{-\pi|x|^2} dx$$

and

$$\left(\int e^{-\pi x^2} dx\right)^2 = \int e^{-\pi x^2} e^{-\pi y^2} dxdy = \int_0^{2\pi} \int_0^\infty e^{-\pi r^2} r dr d\theta = 2\pi \int_0^\infty e^{-\pi r^2} r dr = 1.$$

$\square$

Corollary 11 (Fourier Inversion)**.** *Let*

$$F_{x,\varepsilon}(\xi) = e^{2\pi i x\cdot\xi} e^{-\pi\varepsilon|\xi|^2}$$

for $\varepsilon > 0$.
Then as a function of ξ,

$$F_{x,\varepsilon} = (\textit{modulation of dilation of } f)^\wedge =$$

$$\textit{translation(dilation } f)^\wedge = \textit{ translation of inverse dilation of } \hat{f}$$

and if $f \in L^1$ *and* $\hat{f} \in L^1$,

$$\lim_{\varepsilon\to 0^+} \int F_{x,\varepsilon} \hat{f}(\xi) d\xi = f(x).$$

Proof. By the definition of the Fourier transform:

$$\widehat{F}_{x,\varepsilon}(y) = \int_{\mathbb{R}^n} F_{x,\varepsilon}(\xi) e^{-2\pi i y\cdot\xi}\, d\xi$$

Substituting the definition $F_{x,\varepsilon}(\xi) = e^{2\pi i x\cdot\xi} e^{-\pi\varepsilon|\xi|^2}$:

$$\widehat{F}_{x,\varepsilon}(y) = \int_{\mathbb{R}^n} e^{-\pi\varepsilon|\xi|^2} e^{-2\pi i(y-x)\cdot\xi}\, d\xi$$

By a change of variables,

$$\widehat{F}_{x,\varepsilon}(y) = \varepsilon^{-n/2} \int_{\mathbb{R}^n} e^{-\pi|u|^2} e^{-2\pi i\left(\frac{y-x}{\sqrt{\varepsilon}}\right)\cdot u}\, du$$

Then using our previous result,

$$\widehat{F}_{x,\varepsilon}(y) = \varepsilon^{-n/2} e^{-\pi\left|\frac{y-x}{\sqrt{\varepsilon}}\right|^2} = \varepsilon^{-n/2} e^{-\pi\varepsilon^{-1}|x-y|^2} \implies \hat{F}_{x,z}(y) = \varepsilon^{\frac{-n}{2}} e^{-\pi\varepsilon^{-1}|x-y|^2}.$$

Plugging this into the multiplication formula:

$$\int \hat{f}(\xi) e^{2\pi i x\cdot\xi} e^{-\pi\varepsilon|\xi|^2} d\xi = \int \varepsilon^{-\frac{n}{2}} e^{-\pi\varepsilon^{-1}|x-y|^2} f(y) dy$$

LHS: As $\varepsilon \to 0^+$, $e^{-\pi\varepsilon|\xi|^2} \to 1$.
RHS: $T_\varepsilon(x) := (f * \varepsilon^{-\frac{n}{2}} e^{-\pi\varepsilon^{-1}|x|^2})(x)$.
Study the limit of the RHS: we know $\varepsilon^{-\frac{n}{2}} e^{-\pi\varepsilon^{-1}|x|^2}$ becomes peaked at origin, and that for all $\varepsilon > 0$,

$$\int \varepsilon^{-\frac{n}{2}} e^{-\pi\varepsilon^{-1}} |x|^2 dx = 1.$$

So we have a radially decreasing function with unit mass, meaning

$$\sup_{\varepsilon>0} |T_\varepsilon f(x)| \leq 1 \cdot Mf(x)$$

Recall: the Hardy-Littlewood Maximal function satisfies a weak (1,1) inequality, is bounded on L^∞, is bounded on L^p for $1 < p < \infty$, so we can use it as a dominating function.

We also know that $\lim_{\varepsilon\to 0} T_\varepsilon f(x) \to f(x)$ pointwise a.e. and in norm, because: Mf is weak 1-1, so we can approximate $f \in L^1$ by a continuous function and show we get convergence everywhere for continuous functions. Putting these together and using DCT,

$$\lim_{\varepsilon\to 0^+} \int e^{2\pi i x\cdot\xi} \hat{f}(\xi) e^{-\pi\varepsilon|\xi|^2} d\xi = f(x)$$

pointwise a.e. and in norm. □

This tells us: given $\hat{f}$, you can recover f.

Example 10. *If $f, \hat{f} \in L^1(\mathbb{R}^n)$, then*

$$\int e^{2\pi i x\cdot\xi} \hat{f}(\xi) dx = f(x) \text{ a.e.}$$

So for $g \in L^1(\mathbb{R}^n)$, we define

$$g^\vee(x) := \int e^{2\pi i x\cdot\xi} g(\xi) d\xi.$$

Then the above example says if $f, \hat{f} \in L^1$, then $f = (\hat{f})^\vee = (f^\vee)^\wedge$ a.e. Also note that the Riemann Lebesgue Lemma $\implies$ the Fourier transform of an L^1 function, so this shows that f is equal a.e. to a continuous function.

Corollary 12. *The Fourier transform FT: $\mathcal{S} \to \mathcal{S}$ bijectively.*

Proof. $\varphi \in \mathcal{S} \implies$ we can take $\hat{\varphi}$. Recall: if $g \in L^1(\mathbb{R})$, then

$$g^{\vee}(x) = \int e^{2\pi i x\cdot\xi} g(\xi) d\xi$$

so $\varphi(x) = (\hat{\varphi})^{\vee}(x) = (\hat{\varphi})^{\wedge}(-x)$. Then $\hat{\varphi} = 0 \implies \varphi = 0$.

Further, given $\varphi \in \mathcal{S}$, if $\psi := \varphi^{\vee}$, then $\hat{\psi} = \varphi$, so FT is onto. $\square$

12.4 L^2 Theory of the Fourier Transform

Theorem 25 (Plancherel's Theorem)**.** *If $f \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$, then $\hat{f} \in L^2(\mathbb{R}^n)$, and $||\hat{f}||_2 = ||f||_2$.*

Proof.

$$\int e^{-\pi\varepsilon|\xi|^2} |\hat{f}(\xi)|^2 d\xi$$

always makes sense when $\varepsilon > 0$, because $f \in L^1 \implies \hat{f} \in L^{\infty}$. Then we may write

$$\int e^{-\pi\varepsilon|\xi|^2} |\hat{f}(\xi)|^2 d\xi = \int e^{-\pi\varepsilon|\xi|^2} \hat{f}(\xi)\overline{\hat{f}(\xi)} d\xi = \int e^{-\pi\varepsilon|\xi|^2} \int e^{-2\pi i x\cdot\xi} f(x)dx \overline{\int e^{-2\pi i y\cdot\xi} f(y)dy} d\xi =$$

$$\int e^{-\pi\varepsilon|\xi|^2} \int e^{-2\pi i x\cdot\xi} f(x)dx \int e^{2\pi i y\cdot\xi}\overline{f(y)}dyd\xi = \int\int f(x)\overline{f(y)} \int e^{-\pi\varepsilon|\xi|^2} e^{2\pi i(y-x)\cdot\xi} d\xi dx dy$$

and using our previous result

$$= \int\int f(x)\overline{f(y)} \int \varepsilon^{-\frac{n}{2}} e^{-\pi\varepsilon^{-1}|x-y|^2} dxdy = \int T_\varepsilon f(y)\overline{f(y)}dy$$

where $T_\varepsilon(x) := (f * \varepsilon^{-\frac{n}{2}} e^{-\pi\varepsilon^{-1}|x|^2})(x)$ is same as above and $T_\varepsilon f \in L^1$.

$$\lim_{\varepsilon\to 0} \int T_\varepsilon f(y)\overline{f(y)}dy = \int f(y)\overline{f(y)}dy = \int |f|^2.$$

But also by Monotone Convergence

$$\lim_{\varepsilon\to 0} \int e^{-\pi\varepsilon|\xi|^2} |\hat{f}(\xi)|^2 d\xi = \int |\hat{f}|^2$$

and this is true whether or not this value is finite. Thus $\int |\hat{f}|^2 = \int |f|^2 \implies ||\hat{f}||_2 = ||f||_2$ and both are finite. $\square$

Using this we can show that FT extends uniquely from $L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$ to an isometry of L^2 with itself:

Pick $f \in L^2(\mathbb{R}^n)$, $\{f_n\}_{n=1}^\infty \subseteq L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$, chosen so that $f_n \to f$ in the L^2 norm. Then $f_n - f_m \to 0$ as $n, m \to \infty$.

$$||\hat{f}_n - \hat{f}_m||_2 = ||f_n - f_m||_2 \text{ for all } n, m,$$

so $\{\hat{f}_n\}$ is Cauchy in L^2, converges to, say $\hat{f}$.

$$||\hat{f}||_2 = \lim_{n\to\infty} ||\hat{f}_n||_2 = \lim_{n\to\infty} ||f_n||_2 = ||f||_2.$$

Need to check: this is well-defined, i.e. two different sequences give the same answer. Suppose $f_n \to f$ and $g_n \to f$. Then

$$f_n - g_n \to 0 \text{ in } L^2 \implies \hat{f}_n - \hat{g}_n \to 0 \text{ in } L^2 \implies \lim_{n\to\infty} \hat{f}_n = \lim_{n\to\infty} \hat{g}_n \text{ in } L^2.$$

Corollary 13 (Hausdorff-Young)**.** $||\hat{f}||_\infty \leq ||f||_1$ *and* $||\hat{f}||_2 = ||f||_2$*. In fact, if* $f \in L^p(\mathbb{R}^n)$*,* $1 \leq p \leq 2$ *and* p' *is dual to* p*, then* $\hat{f} \in L^{p'}(\mathbb{R}^n)$ *and* $||\hat{f}||_{p'} \leq ||f||_p$*.*

12.5 Fourier Series and Fourier Theory of the Torus

Recall: In $L^p(\mathbb{T}^n)$, functions are periodic on $\mathbb{R}^n$ with respect to $\mathbb{Z}^n$, i.e. $f(x) = f(x+k)$ for a.e. $x \in \mathbb{R}^n$ when $k \in \mathbb{Z}^n$.

$$L^p(\mathbb{T}^n) = \{f \text{ as above } \mid \left(\int_{[0,1]^n} |f|^p\right)^{\frac{1}{p}} < \infty\}.$$

Note that we could integrate over any fundamental domain; i.e. E measurable and $\sum_{k\in\mathbb{Z}^n} \chi_E(x+k) = 1$ a.e.

$$||f||_{L^p(\mathbb{T}^n)} := ||f\chi_{[0,1)^n}||_{L^p(\mathbb{R}^n)}.$$

Fourier coefficients:

$$\hat{f}(k) = \int_{[0,1]^n} f(x) e^{-2\pi i k\cdot x} dx.$$

Question: when is $f = \sum_{k\in\mathbb{Z}^n} \hat{f}(k) e^{2\pi i k\cdot x}$ and in what sense does the sum give back f?

Observation 1: Given f which is 1-periodic in each direction, let $F = f\chi_{[0,1)^n}$.

$$\hat{f}(k) = k\text{-th Fourier coefficient}$$

equals

$$\hat{F}(k) = \text{ Fourier transform at } \xi = k.$$

By the Riemann Lebesgue Lemma, $\hat{F} \in C_0(\mathbb{R}^n)$ ($\hat{F}$ is continuous and vanishes at ∞) and $||\hat{F}||_\infty \leq ||F||_1 \implies |\hat{f}(k)| \to 0$ as $|k| \to \infty$ and $||\hat{f}||_{\ell^\infty(\mathbb{Z}^n)} \leq ||f||_{L^1(\mathbb{T}^n)}$.

Observation 2: $\{e^{2\pi i k\cdot x}\}_{k\in\mathbb{Z}^n}$ is an orthonormal system in $L^2(\mathbb{T}^n)$, so by Bessel's inequality,

$$||\hat{f}||_{\ell^2(\mathbb{Z}^n)} \leq ||f||_{L^2(\mathbb{T}^n)}.$$

By Hausdorff-Young, $f \in L^p(\mathbb{T}^n)$ for $1 \leq p \leq 2 \implies \hat{f} \in \ell^{p'}(\mathbb{Z}^n)$ and $||\hat{f}||_{\ell^{p'}(\mathbb{Z}^n)} \leq ||f||_{L^p(\mathbb{T}^n)}$.

Note: If $f \in L^p(\mathbb{T}^n)$ for $p > 2$, we know $L^p(\mathbb{T}^n) \subseteq L^2(\mathbb{T}^n), ||f||_2 \leq ||f||_p$ (because this is counting measure). So if $p > 2, ||\hat{f}||_2 \leq ||f||_2 \leq ||f||_p$.

To recover f from $\hat{f}$, we think in terms of convolution.

Proposition 28. *Suppose $\{c_k\}_{k\in\mathbb{Z}^n} \in \ell^1(\mathbb{Z}^n)$. If $\varphi(x) = \sum_{k\in\mathbb{Z}^n} c_k e^{2\pi i k\cdot x}$, then φ is continuous, $\hat{\varphi}(k) = c_k$ for each k, and $f \in L^1(\mathbb{T}^n) \implies$*

$$f * \varphi(x) = \sum_{k\in\mathbb{Z}^n} \hat{f}(k) c_k e^{2\pi i k\cdot x}.$$

Proof. The sum for φ is uniformly convergent, so the limit is continuous and passes through integrals. Thus

$$\hat{\varphi}(k) = \int_{[0,1]^n} e^{-2\pi i k\cdot x}\varphi(x)dx = \sum_{\ell\in\mathbb{Z}^n}\int_{[0,1]^n} e^{-2\pi i k\cdot x}e^{2\pi i \ell\cdot x}c_\ell dx = \sum_{\ell\in\mathbb{Z}^n} c_\ell \int_{[0,1]^n} e^{-2\pi i(\ell-k)\cdot x}dx.$$

Observe that

$$\int_{[0,1]^n} e^{-2\pi i(\ell-k)\cdot x}dx = \begin{cases} 1, & \ell = k \\ 0, & \ell \neq k \end{cases}$$

so $\hat{\varphi}(k) = c_k$. Moreover,

$$f * \varphi(x) = \int_{[0,1]^n} f(y)\varphi(x-y)dy$$

and $f(y) \in L^1$ and $\varphi(x-y)$ converges uniformly so by dominated convergence with dominating function $|f(y)| \cdot ||c||_{\ell^1(\mathbb{Z}^n)}$,

$$\int_{[0,1]^n} f(y)\varphi(x-y)dy = \sum_{k\in\mathbb{Z}^n}\int_{[0,1]^n} f(y)c_k e^{2\pi i(x-y)\cdot k}dy$$

$$= \sum_{k\in\mathbb{Z}^n} c_k e^{2\pi i x\cdot k}\int_{[0,1]^n} f(y)e^{-2\pi i y\cdot k}dy = \sum_{k\in\mathbb{Z}^n}\hat{f}(k)c_k e^{2\pi i k\cdot x}$$

for a.e. $x \in \mathbb{T}^n$. We can also show this convergence is uniform, so $f * \varphi$ must be continuous modulo a.e. and is the uniform limit of the series

$$\sum_{k\in\mathbb{Z}^n}\hat{f}(k)c_k e^{2\pi i k\cdot x}.$$

$\square$

Note. *We can bootstrap this argument:*

$$(f * \varphi)^\wedge(k) = \hat{f}(k)c_k.$$

So if $\{c_k\} \in \ell^1$*, then* $f * \varphi$ *is continuous,* $(f * \varphi)^\wedge(k) = \hat{f}(k)c_k$*, and* $f * \varphi$ *is uniform (and pointwise) limit of the sum of its Fourier series.*

Goal: imagine a sequence of φ concentrating around the origin, i.e. tending to a $\mathcal{S}$ function. Study the formula in the limit.

Let $\varphi = C^\infty$ function of compact support, nonnegative, $\int_{\mathbb{R}^n} \varphi dx = 1$.

Let $\varphi_\varepsilon(x) = \varepsilon^{-n}\varphi(\varepsilon^{-1}x)$. Let

$$\Phi_\varepsilon(x) := \sum_{k\in\mathbb{Z}^n} \varphi_\varepsilon(x-k).$$

This is now a periodic function.

$$\hat{\Phi}_\varepsilon(x) = \int_{[0,1]^n} e^{2\pi ik\cdot x} \sum_{\ell\in\mathbb{Z}^n} \varphi_\varepsilon(x-\ell)dx = \sum_{\ell\in\mathbb{Z}^n} \int_{[0,1]^n} e^{-2\pi ik\cdot x}\varphi_\varepsilon(x-\ell)dx = \sum_{\ell\in\mathbb{Z}^n} \int_{[0,1]^n} e^{-2\pi ik\cdot(x+\ell)}\varphi_\varepsilon(x)dx.$$

Since $e^{-2\pi ik\cdot\ell} = 1$ and $\varphi_\varepsilon(x) \in L^1$, by dominated convergence and the fact that $\sum_{\ell\in\mathbb{Z}^n} \chi_{\ell+[0,1]^n} = 1$ a.e.,

$$\sum_{\ell\in\mathbb{Z}^n} \int_{[0,1]^n} e^{-2\pi ik\cdot(x+\ell)}\varphi_\varepsilon(x)dx = \int e^{-2\pi ik\cdot x}\varphi_\varepsilon(x)dx = \hat{\varphi}_\varepsilon(k) = \hat{\varphi}(\varepsilon k)$$

(last equality by the change of variables $x \mapsto \varepsilon x$)

Note: $\varphi \in C^\infty$, compactly supported $\implies \varphi \in \mathcal{S} \implies \hat{\varphi} \in \mathcal{S} \implies$ rapid decay $\implies \{\hat{\varphi}(\varepsilon k)\}_{k\in\mathbb{Z}^n} \in \ell^1(\mathbb{Z}^n)$. So

$$f \in L^1(\mathbb{T}^n) \implies (f * \Phi_\varepsilon)^\wedge(\xi) = \hat{f}(\xi)\hat{\Phi}_\varepsilon(\xi) \implies$$

$$f * \Phi_\varepsilon(x) = \sum_{k\in\mathbb{Z}^n} \left((f * \Phi_\varepsilon(x))^\wedge\right) e^{2\pi ik\cdot x} = \sum_{k\in\mathbb{Z}^n} \hat{f}(k)\hat{\varphi}(\varepsilon k)e^{2\pi ik\cdot x}$$

where the convergence is uniform and pointwise, $\hat{f}(k)$ is the Fourier series, and $\hat{\varphi}(\varepsilon k)$ is the Fourier transform.

$$f * \Phi_\varepsilon(x) = \int_{[-\frac{1}{2},\frac{1}{2})^n} f(x-y)\Phi_\varepsilon(y)dy = \int_{[-\frac{1}{2},\frac{1}{2})^n} f(x-y)\varphi_\varepsilon(y)dy$$

when ε is sufficiently small.

$$\int_{[-\frac{1}{2},\frac{1}{2})^n} f(x-y)\varphi_\varepsilon(y)dy = \int f(x-y)\varphi_\varepsilon(y)dy = \int f(y)\varphi_\varepsilon(x-y)dy = (f*\varphi_\varepsilon)(x).$$

If we restrict x to live in $[0,1)^n$, $f * \Phi_\varepsilon(x) \to f$ pointwise a.e. (Lebesgue Differentiation Theorem).
Minkowski Integral Inequality $\implies$ convergence in $L^p([0,1]^n)$:

$$\begin{aligned} \|f * \Phi_\varepsilon - f\|_p &= \left\| \int_{\mathbb{T}^n} [f(x-y) - f(x)]\Phi_\varepsilon(y)\, dy \right\|_p \\ &\leq \int_{\mathbb{T}^n} \|f(x-y) - f(x)\|_p |\Phi_\varepsilon(y)|\, dy \end{aligned}$$

Conclude: as $\varepsilon \to 0$, $f * \Phi_\varepsilon(x) \to f$ a.e. and in L^p for $1 \leq p < \infty$. Also

$$\sum_{k \in \mathbb{Z}^n} \hat{f}(k)\hat{\varphi}(\varepsilon k)e^{2\pi i k\cdot x} \to f, \quad \text{provided that } \varphi(\varepsilon k) \to 1 \text{ as } \varepsilon \to 0^+, \hat{\varphi}(0) = 1, \text{ and } \hat{\varphi} \in C_0(\mathbb{R}^n).$$

Corollary 14 (Riesz-Fischer Theorem)**.**

$$\{e^{2\pi i k\cdot x}\}_{k \in \mathbb{Z}^n}$$

are an orthonormal basis of $L^2(\mathbb{T}^n)$.

Proof. If $\overline{\operatorname{span}\{e^{2\pi i k\cdot x}\}_{k\in\mathbb{Z}^n}} \neq L^2(\mathbb{T}^n)$, there exists g so that $g \perp \overline{\operatorname{span}\{e^{2\pi i k\cdot x}\}_{k\in\mathbb{Z}^n}}$.
So $\langle g, e^{2\pi i k\cdot x}\rangle = 0$ for all k. By definition this is $\hat{g}(k)$.

Recall that the previous result said $\sum_{k\in\mathbb{Z}^n} \hat{g}(k)\hat{\varphi}(\varepsilon k)e^{2\pi i k\cdot x} \to g(x)$ (provided that $\varphi(\varepsilon k) \to 1$ as $\varepsilon \to 0^+, \hat{\varphi}(0) = 1$, and $\hat{\varphi} \in C_0(\mathbb{R}^n)$), so this forces $g = 0$. $\square$

Corollary 15. $||\hat{f}||_{\ell^2(\mathbb{Z}^n)} = ||f||_{L^2(\mathbb{T}^n)}$.

13 Tempered Distributions

Recall:

$\mathcal{S} =$ Schwartz class $= \{\varphi \text{ on } \mathbb{R}^n \mid ||x^\alpha \partial^\beta \varphi||_\infty < \infty \text{ for all } \alpha, \beta \text{ multi-indices}\}$.

A **tempered distribution** is a continuous linear functional on $\mathcal{S}$.
If $\ell : \mathcal{S} \to \mathbb{C}$, then $\ell \in \mathcal{S}' \iff \ell$ is linear and $|\ell(\varphi)| \leq C \sum_{k=1}^N ||\varphi||_{\alpha_k,\beta_k}$.

"Key" kind of tempered distribution:

$$\ell(\varphi) = \int f\varphi \text{ for } f \in L^1(\mathbb{R}^n).$$

(Proof: $|\ell(\varphi)| \leq ||f||_1 ||\varphi||_{0,0}$.)

A tempered distribution ℓ which admits an $f \in L^1(\mathbb{R}^n)$ such that $\ell(\varphi) = \int f\varphi$ for all $\varphi \in \mathcal{S}$ is said to be a **function**.

To define operations on tempered distributions, we first work out the rule for functions.

Example 11. *$\tau_h f(x) = f(x-h)$, $f \in L^1(\mathbb{R}^n)$. We have*

$$\int \tau_h f(x)\varphi(x)dx = \int f(x-h)\varphi(x)dx = \int f(x)\varphi(x+h)dx = \int f(x)\tau_{-h}\varphi(x).$$

So should have:

$$D \in \mathcal{S}' \implies (\tau_h D)(\varphi) := D(\tau_{-h}\varphi).$$

All of our usual operations get extended. Key: define it so that it agrees with what is already true about functions.

Derivatives: Assume f is a well-behaved function.

$$\int \partial^\alpha f \varphi(x)dx = \int f(x)(-\partial)^\alpha \varphi(x)dx = (-1)^{|\alpha|}\int f(x)\partial^\alpha \varphi(x)dx.$$

Definition 20. *If D is a tempered distribution, we define*

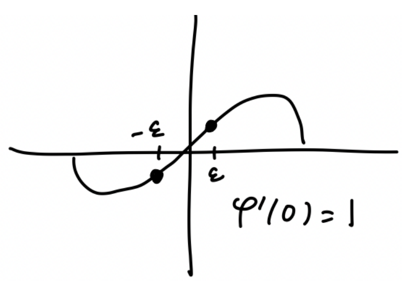
$$\partial^\alpha D(\varphi) := (-1)^{|\alpha|} D(\partial^\alpha \varphi).$$

So all tempered distributions have derivatives of all orders, as tempered distributions.

Example 12. *Let $f(\varphi) = \varphi(0)$. Then $|f(\varphi)| = |\varphi(0)| \leq ||\varphi||_{0,0}$ so $f \in \mathcal{S}'$. f is the Dirac δ -function. Informally: if $\varphi(0) = \int f\varphi$, what properties would f have to have to make this true? This would force f to have total mass 1 and be nonzero only at the origin.*

$$(\tau_h f)(\varphi) = f(\tau_{-h}(\varphi)) = (\tau_{-h}\varphi)(0) = \varphi(h).$$

On $\mathbb{R}$, if $f = \delta$, what is δ'?
$f'(\varphi) = -f(\varphi') = -\varphi'(0)$. Again: informally, if $\int f'\varphi = -\varphi'(0)$, what would f' have to look like for this to be true? It would have to live at the origin and have "mass" zero. What we get is a dipole: two charges of equal and opposite charge infinitesimally close to one another.



$$(\delta_\varepsilon - \delta_{-\varepsilon})(\varphi) = \varphi(\varepsilon) - \varphi(-\varepsilon) = \varphi'(0)\varepsilon - \varphi'(0)(-\varepsilon) + o(\varepsilon) = 2\varphi'(0)\varepsilon + o(\varepsilon).$$

$$\frac{\delta_\varepsilon - \delta_{-\varepsilon}}{-2\varepsilon}(\varphi) = -\varphi'(0) + o(1).$$

$$\delta' = \lim_{\varepsilon \to 0}\left(\frac{\delta_\varepsilon - \delta_{-\varepsilon}}{-2\varepsilon}\right).$$

Typically: we first compute some distribution and then ask if it's a function.

Example 13. *$\hat{D} := D(\hat{\varphi})$, definition motivated by the multiplication formula:*

$$\int f\hat{\varphi} = \int f\hat{\varphi} \text{ when } f, \varphi \in L^1.$$

If D is a function in L^1, then $\hat{D}$ is also a function (in L^∞).

Automatic formulas:

1. $\partial^\beta \partial^\alpha D = \partial^\alpha \partial^\beta D$.

$$\partial^\beta \partial^\alpha D(\varphi) = (-1)^{|\beta|}(\partial^\alpha D)(\partial^\beta \varphi) = (-1)^{|\beta|+|\alpha|} D(\partial^\alpha \partial^\beta D).$$

2. $\partial^\alpha \hat{F} = ((-2\pi i\xi)^\alpha F)^\wedge$ in the sense of distributions.
   $(x^\alpha D(\varphi) := D(x^\alpha \varphi))$
3. $(2\pi i\xi)^\alpha \hat{F} = (\partial^\alpha F)^\wedge$
4. $\tau_h \hat{F} = (e^{2\pi i x\cdot h} F)^\wedge$
5. $e^{-2\pi i h\cdot\xi} \hat{F} = (\tau_h F)^\wedge$

13.1 Common Example of Tempered Distributions

Translation invariant operators: $T : L^p(\mathbb{R}^n) \to L^q(\mathbb{R}^n)$ such that $\tau_h T = T\tau_h$. Will show: Every such T admits a tempered distribution D called the **Schwartz kernel** such that $T\varphi = D * \varphi$ for all $\varphi \in \mathcal{S}$.

Definition 21. *Given $D \in \mathcal{S}'$, $\varphi \in \mathcal{S}$, define $(D * \varphi)(x) := D(\tau_x \tilde{\varphi})$, where $\tilde{\varphi}(y) = \varphi(-y)$.*

Motivation: if D = function, $D(\varphi) = \int f\varphi$,

$$D(\tau_x \tilde{\varphi}) = \int f(y)(\tau_x \tilde{\varphi})(y)dy = \int f(y)\tilde{\varphi}(y-x)dy = \int f(y)\varphi(x-y)dy.$$

If g is a smooth function that tends to 0 as $|x| \to \infty$, then

$$||g||_{L^\infty(\mathbb{R}^n)} \leq \left|\left|\frac{\partial^n}{\partial x_1 \ldots \partial x_n} g\right|\right|_{L^1(\mathbb{R}^n)},$$

assuming $\frac{\partial^n}{\partial x_1 \ldots \partial x_n} g \in L^1(\mathbb{R}^n)$. This is a multi-dimensional analogue of FTC.

Bootstrapping:

Proposition 29 (Sobolev Embedding Inequality)**.** *If $g \in L^1$ and g has weak derivatives in $L^1(\mathbb{R}^n)$ of all orders $|\alpha| \leq n$, then g is a.e. equal to a continuous function, it's bounded, and*

$$||g||_{L^\infty(\mathbb{R}^n)} \leq \left|\left|\frac{\partial^n}{\partial x_1 \ldots \partial x_n} g\right|\right|_{L^1(\mathbb{R}^n)}.$$

Proof. Let φ be C^∞, radial, compactly supported, nonnegative, $\int \varphi dx = 1$ on $\mathbb{R}^d$. Define $\varphi_\varepsilon(x) = \varepsilon^{-n}\varphi(\varepsilon^{-1}x)$. So $\int \varphi_\varepsilon dx = 1$. Let g be as above, and consider $g * \varphi_\varepsilon(x)$.

1. This is a smooth function

2. It has derivatives which "agree" with the weak derivatives:

$$\partial^\alpha(g*\varphi_\varepsilon(x)) = g*\partial^\alpha\varphi_\varepsilon(x) = \int g(y)\partial_x^\alpha\varphi_\varepsilon(x-y)dy = \int g(y)(-\partial_y)^\alpha\varphi_\varepsilon(x-y)dy$$

   where the last equality is pointwise exactly equality.

$$\int g(y)(-\partial_y)^\alpha\varphi_\varepsilon(x-y)dy = \int(-\partial_y^\alpha g)(y)\varphi_\varepsilon(x-y)dy = (\partial^\alpha g) * \varphi_\varepsilon(x)$$

   where the first equality is using the definition of weak derivative. Conclude

$$\partial^\alpha(g * \varphi_\varepsilon(x)) = (\partial^\alpha g) * \varphi_\varepsilon(x)$$

   where the LHS is the usual derivative and the RHS is the weak derivative. Then

$$||g * \varphi_\varepsilon||_\infty \leq ||\partial^\alpha g * \varphi_\varepsilon||_1 \leq ||\partial^\alpha g||_1||\varphi_\varepsilon||_1 = ||\partial^\alpha g||_1.$$

Using the facts:

1. $\partial^\alpha(g * \varphi_\varepsilon(x)) \to \partial^\alpha g$ in the L^1 norm, i.e. $\{\partial^\alpha(g * \varphi_\varepsilon)\}$ is Cauchy in $L^1 \implies \{g * \varphi_\varepsilon\}$ is Cauchy in L^∞ since

$$||g * \varphi_{\varepsilon_1} - g * \varphi_{\varepsilon_2}||_\infty \leq ||\partial^\alpha g * \varphi_{\varepsilon_1} - \partial^\alpha g * \varphi_{\varepsilon_2}||_1 \to 0$$

   as $\varepsilon_1, \varepsilon_2 \to 0^+$.

2. So $g * \varphi_\varepsilon$ converges uniformly to some continuous function $\implies$ g is equivalent modulo a.e. to a bounded continuous function, and

$$||g||_\infty \leq ||\partial^{(1,\ldots,1)}g||_1.$$

□

Corollary 16. *Suppose F is locally integrable and has weak derivatives of order $|\alpha| \leq n$ which are also locally integrable. Then there exists $C > 0$ such that for all $R > 0$, F is continous on $B_R(0)$, and*

$$||F||_{L^\infty(B_R(0))} \leq C \sum_{\alpha\leq(1,\ldots,1)} R^{-(n-|\alpha|)}||\partial^\alpha F||_{L^1(B_{2R}(0))}.$$

Proof. Let η be C^∞ such that $\eta \equiv 1$ on $B_1(0)$ and $\eta \equiv 0$ outside $B_2(0)$. Apply the previous result to $F(x)\eta(R^{-1}x)$. Then

$$||F(x)\eta(R^{-1}x)||_\infty \leq ||\partial^{(1,\ldots,1)}(F\eta(R^{-1}(\cdot)))||_1$$

To finish the proof, expand the RHS using product rule, and use

$$||F||_{L^\infty(B_R(0))} \leq ||\partial^{(1,\ldots,1)}(F\eta(R^{-1}\cdot))||_1.$$

□

Now consider $T\varphi$ where T is translation-invariant and $\varphi \in \mathcal{S}$. We assume that $T\varphi \in L^q$ for some q.

Proposition 30. *If $\psi \in \mathcal{S}$, e_j= standard basis vector in $\mathbb{R}^n$, then*

$$\frac{\psi - \tau_{he_j}\psi}{h} \to \partial_j \psi$$

in the Schwartz topology.

Proof. By Fundamental Theorem of Calculus,

$$\Psi_h(x) := \frac{\psi(x) - \psi(x - he_j)}{h} = -\frac{1}{h}\int_0^1 \frac{d}{d\theta}\psi(x - \theta he_j)d\theta$$

$$= -\frac{1}{h}\int_0^1 \partial_j\psi(x - \theta he_j)(-h)d\theta = \int_0^1 \partial_j\psi(x - \theta he_j)d\theta.$$

Want to show:

$$||\Psi_h - \partial_j\psi||_{\alpha,\beta} \to 0$$

as $h \to 0$ for all multi-indices α, β.

$$\Psi_h(x) - \partial_j\psi = \int_0^1 (\partial_j\psi(x - \theta he_j) - \partial_j\psi(x))d\theta \implies$$

$$\partial^\beta \left(\Psi_h(x) - \partial_j\psi\right) = \int_0^1 \left(\partial^{\beta+e_j}\psi(x - \theta he_j) - \partial^{\beta+e_j}\psi(x)\right) d\theta$$

and by Mean Value Theorem,

$$|\partial^\beta \Psi_h(x) - \partial_j\psi(x)| \leq \int_0^1 |\partial^{\beta+2e_j}\psi(x + \xi he_j)||\theta h|d\theta$$

some $\xi \in [0, \theta]$ (so $|\xi| \leq 1$).

$$\int_0^1 |\partial^{\beta+2e_j}\psi(x + \xi he_j)||\theta h|d\theta \leq \frac{1}{2}|h| \sup_{|t|\leq|h|} |\partial^{\beta+2e_j}\psi(x + te_j)|.$$

$$|x^\alpha||\partial^\beta\Psi_h(x) - \partial_j\psi(x)| \leq \frac{1}{2}\sup_{|t|\leq|h|} |x^\alpha||\partial^{\beta+2e_j}\psi(x + te_j)|$$

so we can write

$$x^\alpha = (x + te_j - te_j)^\alpha = \sum_{\alpha'\leq\alpha}\binom{\alpha}{\alpha'}(x + te_j)^{\alpha'}(-te_j)^{\alpha-\alpha'}$$

and then

$$|x^\alpha||\partial^\beta\Psi_h(x) - \partial_j\psi(x)| \leq \frac{1}{2}|h| \sup_{|t|\leq|h|}\sum_{\alpha'\leq\alpha}\binom{\alpha}{\alpha'}|-te_j|^{\alpha-\alpha'}|x+te_j|^{\alpha'}|\partial^{\beta+2e_j}\psi(x+te_j)| \leq$$

$$\frac{1}{2}|h| \sup_{|t|\leq|h|} \sum_{\alpha'\leq\alpha} \binom{\alpha}{\alpha'} |h|^{|\alpha-\alpha'|} ||\psi||_{\alpha',\beta+2e_j}.$$

Conclude:

$$||\Psi_h - \partial_j\psi||_{\alpha,\beta} \leq \frac{1}{2}|h| \sup_{|t|\leq|h|} \sum_{\alpha'\leq\alpha} \binom{\alpha}{\alpha'} |h|^{|\alpha-\alpha'|} ||\psi||_{\alpha',\beta+2e_j}$$

and

$$\sum_{\alpha'\leq\alpha} \binom{\alpha}{\alpha'} |h|^{|\alpha-\alpha'|} ||\psi||_{\alpha',\beta+2e_j}$$

is constant and finite, so the RHS $\to 0$ as $|h| \to 0$. And this is true for any fixed α, β. □

Theorem 26. *If $T : L^p(\mathbb{R}^n) \to L^q(\mathbb{R}^n)$ is a translation-invariant operator such that $\tau_h T = T\tau_h$, then T admits a tempered distribution D (called the **Schwartz kernel**) such that $T\varphi = D * \varphi$ for all $\varphi \in \mathcal{S}$.*

Proof. Let $\psi \in \mathcal{S}$. Consider $T\psi$. We know that $T\psi \in L^q(\mathbb{R}^n)$ for some given q. So $T\psi \in L^1_{\text{loc}}(\mathbb{R}^n)$.

$$\frac{T\psi - \tau_{he_j}T\psi}{h} = T\left(\frac{\psi - \tau_{he_j}\psi}{h}\right)$$

and

$$\frac{\psi - \tau_{he_j}\psi}{h} \to \partial_j$$

in $\mathcal{S}$, so does in the L^p norm too. Thus the LHS converges to $T(\partial_j\psi)$ in the L^q norm.

Using the fact that $\frac{\eta-\tau_{he_j}\eta}{h}$ converges uniformly,

$$\int(-\partial_j\eta)T\psi = \lim_{h\to 0}\int -\left(\frac{\eta - \tau_{he_j}\eta}{h}\right)(T\psi) = \lim_{h\to 0}\int \eta\left(\frac{T\psi - \tau_{-he_j}T\psi}{h}\right) = \int \eta T\partial_j\psi.$$

So $T\psi$ has weak derivatives of all orders. In particular, $T\psi$ is continuous. Moreover,

$$||T\psi||_{B_1(0)} \leq C \sum_{\alpha\leq(1,\ldots,1)} ||\partial^\alpha T\psi||_{L^1(B_2(0))} \leq C \sum_{\alpha\leq(1,\ldots,1)} ||T(\partial^\alpha\psi)||_{L^1(B_2(0))} \leq C \sum_{\alpha\leq(1,\ldots,1)} ||\partial^\alpha\psi||_{L^P(\mathbb{R}^n)}$$

by continuity of T

$$\leq C\sum_{k=1}^N ||\psi||_{\alpha_k,\beta_k} \implies |T\psi(0)| \leq C\sum_{k=1}^N ||\psi||_{\alpha_k,\beta_k}$$

$\implies \psi \mapsto T\psi(0)$ is a tempered distribution, i.e. $T\psi(0) = D(\psi)$ some $D \in \mathcal{S}'$. So

$$T\psi(x) = T\psi(0-(-x)) = (\tau_{-x}T\psi)(0) = T(\tau_{-x}\psi)(0) = D(\tau_{-x}\psi).$$

Recall: we defined convolution so that $(D * \varphi)(x) := D(\tau_x \tilde{\varphi})$, where $\tilde{\varphi}(y) = \varphi(-y)$. So what we have currently is not quite how we defined convolution, but

$$T\psi = K * \psi \text{ where } K = \tilde{D}, \text{ i.e. } K(\psi) = D(\tilde{\psi})$$

So we showed:

$$T\psi = D(\tau_{-x}\psi) = \tilde{K}(\tau_{-x}\psi) = K(\widetilde{\tau_{-x}\psi}) = K(\tau_x \tilde{\psi}).$$

where now the RHS is how we defined convolution. $\square$